

Chapter 4.1

Use of packagings, including intermediate bulk containers (IBCs) and large packagings

4.1.0 Definitions

Effectively closed: liquid-tight closure.

Hermetically sealed: vapour-tight closure.

Securely closed: so closed that dry contents cannot escape during normal handling; the minimum provisions for any closure.

4.1.1 General provisions for the packing of dangerous goods in packagings, including IBCs and large packagings

Note: For the packing of goods of classes 2, 6.2 and 7, the general provisions of this section only apply as indicated in 4.1.8.2 (class 6.2, UN 2814 and UN 2900), 4.1.9.1.5 (class 7) and in the applicable packing instructions of 4.1.4 (P201 and LP02 for class 2 and P620, P621, P622, IBC620, LP621 and LP622 for class 6.2).

4.1.1.1 Dangerous goods shall be packed in good quality packagings, including IBCs and large packagings, which shall be strong enough to withstand the shocks and loadings normally encountered during transport, including trans-shipment between cargo transport units and between cargo transport units and warehouses as well as any removal from a pallet or overpack for subsequent manual or mechanical handling. Packagings, including IBCs and large packagings, shall be constructed and closed so as to prevent any loss of contents when prepared for transport which may be caused under normal conditions of transport, by vibration, or by changes in temperature, humidity or pressure (resulting from altitude, for example). Packagings, including IBCs and large packagings, shall be closed in accordance with the information provided by the manufacturer. No dangerous residue shall adhere to the outside of packages, IBCs and large packagings during transport. These provisions apply, as appropriate, to new, reused, reconditioned or remanufactured packagings, and to new, reused, repaired or remanufactured IBCs, and to new, reused or remanufactured large packagings.

4.1.1.2 Parts of packagings, including IBCs and large packagings, which are in direct contact with dangerous goods:

- .1 shall not be affected or significantly weakened by those dangerous goods; and
- .2 shall not cause a dangerous effect, such as catalysing a reaction or reacting with the dangerous goods;
- .3 shall not allow permeation of the dangerous goods that could constitute a danger under normal conditions of transport.

Where necessary, they shall be provided with a suitable inner coating or treatment.

4.1.1.3 Unless otherwise provided elsewhere in this Code, each packaging, including IBCs and large packagings, except inner packagings, shall conform to a design type successfully tested in accordance with the provisions of 6.1.5, 6.3.5, 6.5.6 or 6.6.5, as applicable. However, IBCs manufactured before 1 January 2011 and conforming to a design type which has not passed the vibration test of 6.5.6.13 or which was not required to meet the criteria of 6.5.6.9.5.4 at the time it was subjected to the drop test may still be used.

4.1.1.3.1 Packagings, including IBCs and large packagings, may conform to one or more than one successfully tested design type and may bear more than one mark.

4.1.1.4 When filling packagings, including IBCs and large packagings, with liquids,* sufficient ullage (outage) shall be left to ensure that neither leakage nor permanent distortion of the packaging occurs as a result of an expansion of the liquid caused by temperatures likely to occur during transport. Unless specific provisions

* With respect to ullage limits only, the provisions applicable for packagings for solid substances may be used if the viscous substance has an outflow time via a DIN-cup with a 4 mm diameter outlet exceeding 10 minutes at 20°C (corresponding to an outflow time via a Ford cup 4 of more than 690 seconds at 20°C, or to a viscosity of more than 2,680 centistokes at 20°C).

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are prescribed, liquids shall not completely fill a packaging at a temperature of 55°C. However, sufficient ullage shall be left in an IBC to ensure that at the mean bulk temperature of 50°C it is not filled to more than 98% of its water capacity.*

- 4.1.1.4.1 For air transport, packagings intended to contain liquids shall also be capable of withstanding a pressure differential without leakage as specified in the international regulations for air transport.
- 4.1.1.5 Inner packagings shall be packed in an outer packaging in such a way that, under normal conditions of transport, they cannot break, be punctured or leak their contents into the outer packaging. Inner packagings containing liquids shall be packaged with their closures upward and placed within outer packagings consistent with the orientation marks prescribed in 5.2.1.7.1 of this Code. Inner packagings that are liable to break or be punctured easily, such as those made of glass, porcelain or stoneware or of certain plastics materials, etc., shall be secured in outer packagings with suitable cushioning material. Any leakage of the contents shall not substantially impair the protective properties of the cushioning material or of the outer packaging.
- 4.1.1.5.1 Where an outer packaging of a combination packaging or a large packaging has been successfully tested with different types of inner packagings, a variety of such different inner packagings may also be assembled in this outer packaging or large packagings. In addition, provided an equivalent level of performance is maintained, the following variations in inner packagings are allowed without further testing of the package:
- .1 Inner packagings of equivalent or smaller size may be used provided:
 - the inner packagings are of similar design to the tested inner packagings (such as shape – round, rectangular, etc.);
 - the material of construction of inner packagings (glass, plastics, metal, etc.) offers resistance to impact and stacking forces equal to or greater than that of the originally tested inner packaging;
 - the inner packagings have the same or smaller openings and the closure is of similar design (such as screw cap, friction lid, etc.);
 - sufficient additional cushioning material is used to take up void spaces and to prevent significant movement of the inner packagings;
 - inner packagings are oriented within the outer packaging in the same manner as in the tested package; and
 - .2 A lesser number of the tested inner packagings or of the alternative types of inner packagings identified in .1 above may be used, provided sufficient cushioning is added to fill the void space(s) and to prevent significant movement of the inner packagings.
- 4.1.1.5.2 Use of supplementary packagings within an outer packaging (e.g. an intermediate packaging or a receptacle inside a required inner packaging) additional to what is required by the packing instructions is authorized provided all relevant requirements are met, including those of 4.1.1.3, and, if appropriate, suitable cushioning is used to prevent movement within the packaging.
- 4.1.1.5.3 Cushioning and absorbent material shall be inert and suited to the nature of the contents.
- 4.1.1.5.4 The nature and the thickness of the outer packagings shall be such that friction during transport does not generate any heating likely to alter dangerously the chemical stability of the contents.
- 4.1.1.6 Dangerous goods shall not be packed together in the same outer packaging, or in large packagings, with dangerous or other goods if they react dangerously with each other and cause:
- .1 combustion and/or evolution of considerable heat;
 - .2 evolution of flammable, toxic or asphyxiant gases;
 - .3 the formation of corrosive substances; or
 - .4 the formation of unstable substances.
- 4.1.1.7 The closures of packagings containing wetted or diluted substances shall be such that the percentage of liquid (water, solvent or phlegmatizer) does not fall below the prescribed limits during transport.
- 4.1.1.7.1 Where two or more closure systems are fitted in series on an IBC, that nearest to the substance being transported shall be closed first.

* For a differing temperature, the maximum degree of filling may be determined as follows:

$$\text{Degree of filling} = \frac{98}{1 + \alpha (50 - t_F)} \text{ per cent of the capacity of the IBC.}$$

In this formula α represents the mean coefficient of cubic expansion of the liquid substance between 15°C and 50°C; that is to say, for a maximum rise in the temperature of 35°C, “ α ” is calculated according to the formula:

$$\alpha = \frac{d_{15} - d_{50}}{35 \times d_{50}}$$

where d_{15} and d_{50} are the relative densities of the liquid at 15°C and 50°C and t_F is the mean temperature of the liquid at the time of filling.

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- 4.1.1.7.2** Unless otherwise specified in the Dangerous Goods List, packages containing substances which:
- .1 evolve flammable gases or vapour;
 - .2 may become explosive if allowed to dry;
 - .3 evolve toxic gases or vapour;
 - .4 evolve corrosive gases or vapour; or
 - .5 may react dangerously with the atmosphere
- should be hermetically sealed.
- 4.1.1.8** Where pressure may develop in a package by the emission of gas from the contents (as a result of temperature increase or other causes), the packaging or IBC may be fitted with a vent provided that the gas emitted will not cause danger on account of its toxicity, its flammability, the quantity released, etc.
- A venting device shall be fitted if dangerous overpressure may develop due to normal decomposition of substances. The vent shall be so designed that, when the packaging or IBC is in the attitude in which it is intended to be transported, leakages of liquid and the penetration of foreign substances are prevented under normal conditions of transport.
- 4.1.1.8.1** Liquids may only be filled into inner packagings which have an appropriate resistance to internal pressure that may be developed under normal conditions of transport.
- 4.1.1.9** New, remanufactured or re-used packagings, including IBCs and large packagings, or reconditioned packagings and repaired or routinely maintained IBCs shall be capable of passing the tests prescribed in 6.1.5, 6.3.5, 6.5.6 or 6.6.5, as applicable. Before being filled and handed over for transport, every packaging, including IBCs and large packagings, shall be inspected to ensure that it is free from corrosion, contamination or other damage and every IBC shall be inspected with regard to the proper functioning of any service equipment. Any packaging which shows signs of reduced strength as compared with the approved design type shall no longer be used or shall be so reconditioned that it is able to withstand the design type tests. Any IBC which shows signs of reduced strength as compared with the tested design type shall no longer be used or shall be so repaired or routinely maintained that it is able to withstand the design type tests.
- 4.1.1.10** Liquids shall be filled only into packagings, including IBCs, which have an appropriate resistance to the internal pressure that may develop under normal conditions of transport. As the vapour pressure of low-boiling-point liquids is usually high, the strength of receptacles for these liquids shall be sufficient to withstand, with an ample factor of safety, the internal pressure likely to be generated. Packagings and IBCs marked with the hydraulic test pressure prescribed in 6.1.3.1(d) and 6.5.2.2.1, respectively, shall be filled only with a liquid having a vapour pressure:
- .1 such that the total gauge pressure in the packaging or IBC (i.e. the vapour pressure of the filling substance plus the partial pressure of air or other inert gases, less 100 kPa) at 55°C, determined on the basis of a maximum degree of filling in accordance with 4.1.1.4 and a filling temperature of 15°C, will not exceed two thirds of the marked test pressure; or
 - .2 at 50°C, less than four sevenths of the sum of the marked test pressure plus 100 kPa; or
 - .3 at 55°C, less than two thirds of the sum of the marked test pressure plus 100 kPa.
- IBCs intended for the transport of liquids shall not be used to carry liquids having a vapour pressure of more than 110 kPa (1.1 bar) at 50°C or 130 kPa (1.3 bar) at 55°C.

**Examples of required marked test pressures for packagings, including IBCs,
calculated as in 4.1.1.10.3**

UN No.	Name	Class	Packing group	V_{p55} (kPa)	$V_{p55} \times 1.5$ (kPa)	$(V_{p55} \times 1.5) \text{ minus } 100$ (kPa)	Required minimum test pressure (gauge) under 6.1.5.5.4.3 (kPa)	Minimum test pressure (gauge) to be marked on the packaging (kPa)
2056	Tetrahydrofuran	3	II	70	105	5	100	100
2247	<i>n</i> -Decane	3	III	1.4	2.1	-97.9	100	100
1593	Dichloromethane	6.1	III	164	246	146	146	150
1155	Diethyl ether	3	I	199	299	199	199	250

Note 1: For pure liquids, the vapour pressure at 55°C (V_{p55}) can often be obtained from scientific tables.

Note 2: The table refers to the use of 4.1.1.10.3 only, which means that the marked test pressure shall exceed 1.5 times the vapour pressure at 55°C less 100 kPa. When, for example, the test pressure for *n*-decane is determined according to 6.1.5.5.4.1, the minimum marked test pressure may be lower.

Note 3: For diethyl ether, the required minimum test pressure under 6.1.5.5.5 is 250 kPa.

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- 4.1.1.11** Empty packagings, including IBCs and large packagings, that have contained a dangerous substance shall be treated in the same manner as is required by this Code for a filled packaging, unless adequate measures have been taken to nullify any hazard.
- 4.1.1.12** Every packaging as specified in chapter 6.1 intended to contain liquids shall successfully undergo a suitable leakproofness test. This test is part of a quality assurance programme as stipulated in 6.1.1.3 which shows the capability of meeting the appropriate test level indicated in 6.1.5.4.4:
- .1 before it is first used for transport;
 - .2 after remanufacturing or reconditioning of any packaging, before it is re-used for transport.
- For this test, the packaging need not have its closures fitted. The inner receptacle of a composite packaging may be tested without the outer packaging, provided the test results are not affected. This test is not necessary for inner packagings of combination packagings or large packagings.
- 4.1.1.13** Packagings, including IBCs, used for solids which may become liquid at temperatures likely to be encountered during transport shall also be capable of containing the substance in the liquid state.
- 4.1.1.14** Packagings, including IBCs, used for powdery or granular substances shall be sift-proof or shall be provided with a liner.
- 4.1.1.15** For plastics drums and jerricans, rigid plastics IBCs and composite IBCs with plastics inner receptacles, unless otherwise approved by the competent authority, the period of use permitted for the transport of dangerous substances shall be five years from the date of manufacture of the receptacles, except where a shorter period of use is prescribed because of the nature of the substance to be transported.
- Note:** For composite IBCs the period of use refers to the date of manufacture of the inner receptacle.
- 4.1.1.16** Where ice is used as a coolant it shall not affect the integrity of the packaging.
- 4.1.1.17** **Explosives, self-reactive substances and organic peroxides**
- Unless specific provision to the contrary is made in this Code, the packagings, including IBCs and large packagings, used for goods of class 1, self-reactive substances of class 4.1 and organic peroxides of class 5.2 shall comply with the provisions for the medium danger group (packing group II).
- 4.1.1.18** **Use of salvage packagings and large salvage packagings**
- 4.1.1.18.1** Damaged, defective, leaking or non-conforming packages, or dangerous goods that have spilled or leaked may be transported in salvage packagings mentioned in 6.1.5.1.11 and 6.6.5.1.9. This does not prevent the use of a larger size packaging or large packaging of appropriate type and performance level and under the conditions of 4.1.1.18.2 and 4.1.1.18.3.
- 4.1.1.18.2** Appropriate measures shall be taken to prevent excessive movement of the damaged or leaking packages within a salvage packaging. When the salvage packaging contains liquids, sufficient inert absorbent material shall be added to eliminate the presence of free liquid.
- 4.1.1.18.3** Appropriate measures shall be taken to ensure there is no dangerous build-up of pressure.
- 4.1.1.19** **Use of salvage pressure receptacles**
- 4.1.1.19.1** In the case of damaged, defective, leaking or non-conforming pressure receptacles, salvage pressure receptacles according to 6.2.3 may be used.
- Note:** A salvage pressure receptacle may be used as an overpack in accordance with 5.1.2. When used as an overpack, marks shall be in accordance with 5.1.2.1 instead of 5.2.1.3.
- 4.1.1.19.2** Pressure receptacles shall be placed in salvage pressure receptacles of suitable size. More than one pressure receptacle may be placed in the same salvage pressure receptacle only if the contents are known and do not react dangerously with each other (see 4.1.1.6). In this case the total sum of water capacities of the placed pressure receptacles shall not exceed 3,000 L. Measures shall be taken to prevent movement of the pressure receptacles within the salvage pressure receptacle, e.g. by partitioning, securing or cushioning.
- 4.1.1.19.3** A pressure receptacle may only be placed in a salvage pressure receptacle if:
- .1 the salvage pressure receptacle is in accordance with 6.2.3.5 and a copy of the approval certificate is available;
 - .2 parts of the salvage pressure receptacle which are, or are likely to be in direct contact with the dangerous goods will not be affected or weakened by those dangerous goods and will not cause a dangerous effect (e.g. catalysing reaction or reacting with the dangerous goods); and
 - .3 the contents of the contained pressure receptacle(s) is limited in pressure and volume so that if totally discharged into the salvage pressure receptacle, the pressure in the salvage pressure receptacle at 65°C will not exceed the test pressure of the salvage pressure receptacle (for gases, see packing instruction

in P200 (3) 4.1.4.1). The reduction of the useable water capacity of the salvage pressure receptacle, e.g. by any contained equipment and cushioning, shall be taken into account.

- 4.1.1.19.4 The proper shipping name, the UN number preceded by the letters “UN” and label(s) as required for packages in chapter 5.2 applicable to the dangerous goods inside the contained pressure receptacle(s) shall be applied to the salvage pressure receptacle for transport.
- 4.1.1.19.5 Salvage pressure receptacles shall be cleaned, purged and visually inspected internally and externally after each use. They shall be periodically inspected and tested in accordance with 6.2.1.6 at least once every five years.
- 4.1.1.20 During transport, packagings, including IBCs and large packagings, shall be securely fastened to or contained within the cargo transport unit, so that lateral or longitudinal movement or impact is prevented and adequate external support is provided.
- 4.1.1.21 Except as provided in 4.1.1.18 and 4.1.1.19, packagings including large packagings and IBCs shall not be filled or discharged while they remain on board.

4.1.2 Additional general provisions for the use of IBCs

- 4.1.2.1 When IBCs are used for the transport of liquids with a flashpoint of 60°C (closed cup) or lower, or of powders liable to dust explosion, measures shall be taken to prevent a dangerous electrostatic discharge.
- 4.1.2.2 Every metal, rigid plastics and composite IBC shall be inspected and tested, as relevant, in accordance with 6.5.4.4 or 6.5.4.5:
 - before it is put into service;
 - thereafter at intervals not exceeding two and a half and five years, as appropriate; and
 - after the repair or remanufacture, before it is re-used for transport.

An IBC shall not be filled and offered for transport after the date of expiry of the last periodic test or inspection. However, an IBC filled prior to the date of expiry of the last periodic test or inspection may be transported for a period not to exceed three months beyond the date of expiry of the last periodic test or inspection. In addition, an IBC may be transported after the date of expiry of the last periodic test or inspection:

- .1 after emptying but before cleaning, for purposes of performing the required test or inspection prior to refilling; and
 - .2 unless otherwise approved by the competent authority, for a period not to exceed six months beyond the date of expiry of the last periodic test or inspection in order to allow the return of dangerous goods or residues for proper disposal or recycling. Reference to this exemption shall be entered in the transport document.
- 4.1.2.3 IBCs of type 31HZ2 when transporting liquids shall be filled to at least 80% of the volume of the outer casing and shall be transported in closed cargo transport units.
 - 4.1.2.4 Except for routine maintenance of metal, rigid plastics, composite and flexible IBCs performed by the owner of the IBC, whose State and name or authorized symbol is durably marked on the IBC, the party performing routine maintenance shall durably mark the IBC near the manufacturer’s UN design type mark to show:
 - .1 the State in which the routine maintenance was carried out; and
 - .2 the name or authorized symbol of the party performing the routine maintenance.

4.1.3 General provisions concerning packing instructions

- 4.1.3.1 Packing instructions applicable to dangerous goods of classes 1 to 9 are specified in 4.1.4. They are subdivided in three subsections depending on the type of packagings to which they apply:
 - subsection 4.1.4.1 for packagings other than IBCs and large packagings; these packing instructions are designated by an alphanumeric code comprising the letter “P”;
 - subsection 4.1.4.2 for IBCs; these are designated by an alphanumeric code comprising the letters “IBC”;
 - subsection 4.1.4.3 for large packagings; these are designated by an alphanumeric code comprising the letters “LP”.

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Generally, packing instructions specify that the general provisions of 4.1.1, 4.1.2 and/or 4.1.3, as appropriate, are applicable. They may also require compliance with the special provisions of 4.1.5, 4.1.6, 4.1.7, 4.1.8 or 4.1.9 when appropriate. Special packing provisions may also be specified in the packing instruction for individual substances or articles. They are also designated by an alphanumeric code comprising the letters:

- “PP” for packagings other than IBCs and large packagings
- “B” for IBCs
- “L” for large packagings.

Unless otherwise specified, each packaging shall conform to the applicable provisions of part 6. Generally, packing instructions do not provide guidance on compatibility and the user shall not select a packaging without checking that the substance is compatible with the packaging material selected (such as, most fluorides are unsuitable for glass receptacles). Where glass receptacles are permitted in the packing instructions, porcelain, earthenware and stoneware packagings are also allowed.

4.1.3.2 Column 8 of the Dangerous Goods List shows for each article or substance the packing instruction(s) that shall be used. Column 9 indicates the special packing provisions applicable to specific substances or articles.

4.1.3.3 Each packing instruction shows, where applicable, the acceptable single and combination packagings. For combination packagings, the acceptable outer packagings, inner packagings and, when applicable, the maximum quantity permitted in each inner or outer packaging are shown. *Maximum net mass* and *maximum capacity* are as defined in 1.2.1. Where packagings which need not meet the requirements of 4.1.1.3 (e.g. crates, pallets, etc.) are authorized in a packing instruction or the special provisions named in the Dangerous Goods List, these packagings are not subject to the mass or volume limits generally applicable to packagings conforming to the requirements of chapter 6.1, unless otherwise indicated in the relevant packing instruction or special provision.

4.1.3.4 The following packagings shall not be used when the substances being transported are liable to become liquid during transport:

Packagings

Drums: 1D and 1G

Boxes: 4C1, 4C2, 4D, 4F, 4G and 4H1

Bags: 5L1, 5L2, 5L3, 5H1, 5H2, 5H3, 5H4, 5M1 and 5M2

Composite: 6HC, 6HD1, 6HD2, 6HG1, 6HG2, 6PC, 6PD1, 6PD2, 6PG1, 6PG2 and 6PH1

Large packagings

Flexible plastics: 51H (outer packaging)

IBCs

For substances of packing group I:

All types of IBCs

For substances of packing groups II and III:

Wooden: 11C, 11D and 11F

Fibreboard: 11G

Flexible: 13H1, 13H2, 13H3, 13H4, 13H5, 13L1, 13L2, 13L3, 13L4, 13M1 and 13M2

Composite: 11HZ2 and 21HZ2

4.1.3.5 Where the packing instructions in this chapter authorize the use of a particular type of packaging (such as 4G; 1A2), packagings bearing the same packaging identification code followed by the letters “V”, “U” or “W” marked in accordance with the provisions of part 6 (such as “4GV”, “4GU” or “4GW”, “1A2V”, “1A2U” or “1A2W”) may also be used under the same conditions and limitations applicable to the use of that type of packaging according to the relevant packing instructions. For example, a combination packaging marked with the packaging code “4GV” may be used whenever a combination packaging marked “4G” is authorized, provided the provisions in the relevant packing instruction regarding types of inner packagings and quantity limitations are respected.

4.1.3.6 **Pressure receptacles for liquids and solids**

4.1.3.6.1 Unless otherwise indicated in this Code, pressure receptacles conforming to:

- .1 the applicable requirements of chapter 6.2; or

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- .2 the National or International standards on the design, construction, testing, manufacturing and inspection, as applied by the country in which the pressure receptacles are manufactured, provided that the provisions of 4.1.3.6 and 6.2.3.3 are met,

are authorized for the transport of any liquid or solid substance other than explosives, thermally unstable substances, organic peroxides, self-reactive substances, substances where significant pressure may develop by evolution of chemical reaction and radioactive material (unless permitted in 4.1.9).

This subsection is not applicable to the substances mentioned in 4.1.4.1, packing instruction P200, table 3.

- 4.1.3.6.2 Every design type of pressure receptacle shall be approved by the competent authority of the country of manufacture or as indicated in chapter 6.2.

- 4.1.3.6.3 Unless otherwise indicated, pressure receptacles having a minimum test pressure of 0.6 MPa shall be used.

- 4.1.3.6.4 Unless otherwise indicated, pressure receptacles may be provided with an emergency pressure relief device designed to avoid bursting in case of overfill or fire accidents.

Pressure receptacle valves shall be designed and constructed in such a way that they are inherently able to withstand damage without release of the contents or shall be protected from damage which could cause inadvertent release of the contents of the pressure receptacle, by one of the methods as given in 4.1.6.1.8.1 to 4.1.6.1.8.5.

- 4.1.3.6.5 The level of filling shall not exceed 95% of the capacity of the pressure receptacle at 50°C. Sufficient ullage (outage) shall be left to ensure that the pressure receptacle will not be liquid-full at a temperature of 55°C.

- 4.1.3.6.6 Unless otherwise indicated, pressure receptacles shall be subjected to a periodic inspection and test every five years. The periodic inspection shall include an external examination, an internal examination or alternative method as approved by the competent authority, a pressure test or equivalent effective non-destructive testing with the agreement of the competent authority, including an inspection of all accessories (e.g. tightness of valves, emergency relief valves or fusible elements). Pressure receptacles shall not be filled after they become due for periodic inspection and test but may be transported after the expiry of the time limit. Pressure receptacle repairs shall meet the requirements of 4.1.6.1.11.

- 4.1.3.6.7 Prior to filling, the filler shall perform an inspection of the pressure receptacle and ensure that the pressure receptacle is authorized for the substances to be transported and that the provisions of this Code have been met. Shut-off valves shall be closed after filling and remain closed during transport. The consignor shall verify that the closures and equipment are not leaking.

- 4.1.3.6.8 Refillable pressure receptacles shall not be filled with a substance different from that previously contained unless the necessary operations for change of service have been performed.

- 4.1.3.6.9 Marking of pressure receptacles for liquids and solids according to 4.1.3.6 (not conforming to the requirements of chapter 6.2) shall be in accordance with the requirements of the competent authority of the country of manufacturing.

- 4.1.3.7 Packagings, including IBCs and large packagings, not specifically authorized in the applicable packing instruction shall not be used for the transport of a substance or article unless specifically approved by the competent authority and provided:

- .1 the alternative packaging complies with the general provisions of this chapter;
- .2 when the packing instruction indicated in the Dangerous Goods List so specifies, the alternative packaging meets the provisions of part 6;
- .3 the competent authority determines that the alternative packaging provides at least the same level of safety as if the substance were packed in accordance with a method specified in the particular packing instruction indicated in the Dangerous Goods List; and
- .4 a copy of the competent authority approval accompanies each consignment or the transport document includes an indication that alternative packaging was approved by the competent authority.

Note: The competent authorities granting such approvals shall take action to amend the Code to include the provisions covered by the approval as appropriate.

- 4.1.3.8 **Unpackaged articles other than class 1 articles**

- 4.1.3.8.1 Where large and robust articles cannot be packaged in accordance with the requirements of chapter 6.1 or 6.6 and they have to be transported empty, uncleaned and unpackaged, the competent authority may approve such transport. In doing so, the competent authority shall take into account that:

- .1 Large and robust articles shall be strong enough to withstand the shocks and loadings normally encountered during transport, including trans-shipment between cargo transport units and between cargo transport units and warehouses, as well as any removal from a pallet for subsequent manual or mechanical handling.

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- .2 All closures and openings shall be sealed so that there can be no loss of contents which might be caused under normal conditions of transport, by vibration, or by changes in temperature, humidity or pressure (resulting from altitude, for example). No dangerous residue shall adhere to the outside of the large and robust articles.
- .3 Parts of large and robust articles, which are in direct contact with dangerous goods:
 - .1 shall not be affected or significantly weakened by those dangerous goods; and
 - .2 shall not cause a dangerous effect, e.g. catalysing a reaction or reacting with the dangerous goods.
- .4 Large and robust articles containing liquids shall be stowed and secured to ensure that neither leakage nor permanent distortion of the article occurs during transport.
- .5 They shall be fixed in cradles or crates or other handling devices in such a way that they will not become loose during normal conditions of transport.

4.1.3.8.2 Unpackaged articles approved by the competent authority in accordance with the provisions of 4.1.3.8.1 shall be subject to the consignment procedures of part 5. In addition the consignor of such articles shall ensure that a copy of any such approval is transported with the large and robust articles.

Note: A large and robust article may include flexible fuel containment systems, military equipment, machinery or equipment containing dangerous goods above the limited quantity thresholds.

4.1.3.9 Where, in 4.1.3.6 and in the individual packing instructions, cylinders and other pressure receptacles for gases are authorized for the transport of any liquid or solid substance, use is also authorized of cylinders and pressure receptacles of a kind normally used for gases which conform to the requirements of the competent authority of the country in which the cylinder or pressure receptacle is filled. Valves shall be suitably protected. Pressure receptacles with capacities of 1 L or less shall be packed in outer packagings constructed of suitable material of adequate strength and design in relation to the capacity of the packaging and its intended use and secured or cushioned so as to prevent significant movement within the outer packaging during normal conditions of transport.

4.1.4 List of packing instructions

4.1.4.1 Packing instructions concerning the use of packagings (except IBCs and large packagings)

P001 PACKING INSTRUCTION (LIQUIDS) P001				
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.				
Combination packagings		Maximum capacity/net mass (see 4.1.3.3)		
Inner packagings	Outer packagings	Packing group I	Packing group II	Packing group III
Glass 10 L Plastics 30 L Metal 40 L	Drums			
	steel (1A1, 1A2)	75 kg	400 kg	400 kg
	aluminium (1B1, 1B2)	75 kg	400 kg	400 kg
	other metal (1N1, 1N2)	75 kg	400 kg	400 kg
	plastics (1H1, 1H2)	75 kg	400 kg	400 kg
	plywood (1D)	75 kg	400 kg	400 kg
	fibre (1G)	75 kg	400 kg	400 kg
	Boxes			
	steel (4A)	75 kg	400 kg	400 kg
	aluminium (4B)	75 kg	400 kg	400 kg
	other metal (4N)	75 kg	400 kg	400 kg
	natural wood (4C1, 4C2)	75 kg	400 kg	400 kg
	plywood (4D)	75 kg	400 kg	400 kg
	reconstituted wood (4F)	75 kg	400 kg	400 kg
	fibreboard (4G)	75 kg	400 kg	400 kg
	expanded plastics (4H1)	40 kg	60 kg	60 kg
	solid plastics (4H2)	75 kg	400 kg	400 kg
	Jerricans			
	steel (3A1, 3A2)	60 kg	120 kg	120 kg
	aluminium (3B1, 3B2)	60 kg	120 kg	120 kg
	plastics (3H1, 3H2)	30 kg	120 kg	120 kg
Single packagings				
Drums				
steel, non-removable head (1A1)		250 L	450 L	450 L
steel, removable head (1A2)		prohibited	250 L	250 L
aluminium, non-removable head (1B1)		250 L	450 L	450 L
aluminium, removable head (1B2)		prohibited	250 L	250 L
other metal, non-removable head (1N1)		250 L	450 L	450 L
other metal, removable head (1N2)		prohibited	250 L	250 L
plastics, non-removable head (1H1)		250 L*	450 L	450 L
plastics, removable head (1H2)		prohibited	250 L	250 L
Jerricans				
steel, non-removable head (3A1)		60 L	60 L	60 L
steel, removable head (3A2)		prohibited	60 L	60 L
aluminium, non-removable head (3B1)		60 L	60 L	60 L
aluminium, removable head (3B2)		prohibited	60 L	60 L
plastics, non-removable head (3H1)		60 L*	60 L	60 L
plastics, removable head (3H2)		prohibited	60 L	60 L
Composite packagings				
Plastics receptacle in steel, aluminium or plastics drum (6HA1, 6HB1, 6HH1)		250 L	250 L	250 L
Plastics receptacle in fibre or plywood drum (6HG1, 6HD1)		120 L*	250 L	250 L
Plastics receptacle in steel or aluminium crate or box or plastics receptacle in wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)		60 L*	60 L	60 L
Glass receptacle in steel, aluminium, fibre, plywood, expanded or solid plastics drum (6PA1, 6PB1, 6PG1, 6PD1, 6PH1 or 6PH2) or in a steel, aluminium, wood or fibreboard box or in a wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 or 6PD2)		60 L	60 L	60 L
Pressure receptacles, provided that the general provisions of 4.1.3.6 are met				

* Not permitted for class 3, packing group I.

Part 4 – Packing and tank provisions

P001	PACKING INSTRUCTION (LIQUIDS) (continued)	P001
Special packing provisions:		
PP1	For UN Nos. 1133, 1210, 1263 and 1866 and for adhesives, printing inks, printing ink related materials, paints, paint related materials and resin solutions which are assigned to UN 3082, metal or plastics packagings for substances of packing groups II and III in quantities of 5 L or less per packaging are not required to meet the performance tests in chapter 6.1 when transported:	
	(a) in palletized loads, a pallet box or a unit load device, such as individual packagings placed or stacked and secured by strapping, shrink- or stretch-wrapping or other suitable means to a pallet. For sea transport, the palletized loads, pallet boxes or unit load devices shall be firmly packed and secured in closed cargo transport units. On roll-on/roll-off ships the unit loads may be carried in vehicles other than closed vehicles provided they are securely fenced to the full height of the cargo carried; or	
	(b) as an inner packaging of a combination packaging with a maximum net mass of 40 kg.	
PP2	For UN 3065, wooden barrels with a maximum capacity of 250 L and which do not meet the provisions of chapter 6.1 may be used.	
PP4	For UN 1774, packagings shall meet the packing group II performance level.	
PP5	For UN 1204, packagings shall be so constructed that explosion is not possible by reason of increased internal pressure. Gas cylinders and gas receptacles shall not be used for these substances.	
PP10	For UN 1791, for packing group II, the packaging shall be vented.	
PP31	For UN Nos. 1131, 1553, 1693, 1694, 1699, 1701, 2478, 2604, 2785, 3148, 3183, 3184, 3185, 3186, 3187, 3188, 3398 (PG II and III), 3399 (PG II and III), 3413 and 3414, packagings shall be hermetically sealed.	
PP33	For UN 1308, for packing groups I and II, only combination packagings with a maximum gross mass of 75 kg are allowed.	
PP81	For UN 1790 with more than 60% but not more than 85% hydrogen fluoride and UN 2031 with more than 55% nitric acid, the permitted use of plastics drums and jerricans as single packagings shall be two years from their date of manufacture	
PP93	For UN Nos. 3532 and 3534, packagings shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the packagings in the event of loss of stabilization.	

P002		PACKING INSTRUCTION (SOLIDS)			P002
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.					
Combination packagings		Maximum net mass (see 4.1.3.3)			
Inner packagings		Outer packagings	Packing group I	Packing group II	Packing group III
Glass	10 kg	Drums			
Plastics ¹	30 kg	steel (1A1, 1A2)	125 kg	400 kg	400 kg
Metal	40 kg	aluminium (1B1, 1B2)	125 kg	400 kg	400 kg
Paper ^{1, 2, 3}	50 kg	other metal (1N1, 1N2)	125 kg	400 kg	400 kg
Fibre ^{1, 2, 3}	50 kg	plastics (1H1, 1H2)	125 kg	400 kg	400 kg
		plywood (1D)	125 kg	400 kg	400 kg
		fibre (1G)	125 kg	400 kg	400 kg
		Boxes			
		steel (4A)	125 kg	400 kg	400 kg
		aluminium (4B)	125 kg	400 kg	400 kg
		other metal (4N)	125 kg	400 kg	400 kg
		natural wood (4C1)	125 kg	400 kg	400 kg
		natural wood with sift-proof walls (4C2)	250 kg	400 kg	400 kg
		plywood (4D)	125 kg	400 kg	400 kg
		reconstituted wood (4F)	125 kg	400 kg	400 kg
		fibreboard (4G)	75 kg	400 kg	400 kg
		expanded plastics (4H1)	40 kg	60 kg	60 kg
		solid plastics (4H2)	125 kg	400 kg	400 kg
		Jerricans			
		steel (3A1, 3A2)	75 kg	120 kg	120 kg
		aluminium (3B1, 3B2)	75 kg	120 kg	120 kg
		plastics (3H1, 3H2)	75 kg	120 kg	120 kg
Single packagings					
Drums					
steel (1A1 or 1A2 ⁴)			400 kg	400 kg	400 kg
aluminium (1B1 or 1B2 ⁴)			400 kg	400 kg	400 kg
metal, other than steel or aluminium (1N1 or 1N2 ⁴)			400 kg	400 kg	400 kg
plastics (1H1 or 1H2 ⁴)			400 kg	400 kg	400 kg
fibre (1G ⁵)			400 kg	400 kg	400 kg
plywood (1D ⁵)			400 kg	400 kg	400 kg
⁴ These packagings shall not be used for substances of packing group I that may become liquid during transport (see 4.1.3.4).					
⁵ These packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).					

¹ These inner packagings shall be sift-proof.² These inner packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).³ Paper and fibre inner packagings shall not be used for substances of packing group I.

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P002	PACKING INSTRUCTION (SOLIDS) (continued)			P002
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.				
		Maximum net mass (see 4.1.3.3)		
Single packagings (continued)		Packing group I	Packing group II	Packing group III
Jerricans				
steel (3A1 or 3A2 ⁴)		120 kg	120 kg	120 kg
aluminium (3B1 or 3B2 ⁴)		120 kg	120 kg	120 kg
plastics (3H1 or 3H2 ⁴)		120 kg	120 kg	120 kg
Boxes				
steel (4A) ⁵		Not allowed	400 kg	400 kg
aluminium (4B) ⁵		Not allowed	400 kg	400 kg
other metal (4N) ⁵		Not allowed	400 kg	400 kg
natural wood (4C1) ⁵		Not allowed	400 kg	400 kg
natural wood with sift-proof walls (4C2) ⁵		Not allowed	400 kg	400 kg
plywood (4D) ⁵		Not allowed	400 kg	400 kg
reconstituted wood (4F) ⁵		Not allowed	400 kg	400 kg
fibreboard (4G) ⁵		Not allowed	400 kg	400 kg
solid plastics (4H2) ⁵		Not allowed	400 kg	400 kg
Bags				
bags (5H3, 5H4, 5L3, 5M2) ⁵		Not allowed	50 kg	50 kg
Composite packagings				
Plastics receptacle in steel, aluminium, plywood, fibre or plastics drum (6HA1, 6HB1, 6HG1 ⁵ , 6HD1 ⁵ , or 6HH1)		400 kg	400 kg	400 kg
Plastics receptacle in steel or aluminium crate or box, wooden box, plywood box, fibreboard box or solid plastics box (6HA2, 6HB2, 6HC, 6HD2 ⁵ , 6HG2 ⁵ or 6HH2)		75 kg	75 kg	75 kg
Glass receptacle in steel, aluminium, plywood or fibre drum (6PA1, 6PB1, 6PD1 ⁵ or 6PG1 ⁵) or in steel, aluminium, wood, or fibreboard box or in wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 ⁵ or 6PD2 ⁵) or in expanded or solid plastics packaging (6PH2 or 6PH1 ⁵)		75 kg	75 kg	75 kg
⁴ These packagings shall not be used for substances of packing group I that may become liquid during transport (see 4.1.3.4).				
⁵ These packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).				
Pressure receptacles, provided that the general provisions of 4.1.3.6 are met.				
Special packing provisions:				
PP7	For UN 2000, celluloid may be transported unpacked on pallets, wrapped in plastic film and secured by appropriate means, such as steel bands, as a single commodity in closed cargo transport units. Each pallet shall not exceed 1,000 kg.			
PP8	For UN 2002, packagings shall be so constructed that explosion is not possible by reason of increased internal pressure. Gas cylinders and gas receptacles shall not be used for these substances.			
PP9	For UN Nos. 3175, 3243 and 3244, packagings shall conform to a design type that has passed a leakproofness test at the packing group II performance level. For UN 3175 the leakproofness test is not required when the liquids are fully absorbed in solid material contained in sealed bags.			
PP11	For UN 1309, packing group III, and UN 1362, 5H1, 5L1 and 5M1 bags are allowed if they are overpacked in plastic bags and are wrapped in shrink or stretch wrap on pallets.			
PP12	For UN Nos. 2213 and 3077, 5H1, 5L1 and 5M1 bags are allowed when transported in closed cargo transport units.			
PP13	For articles classified under UN 2870, only combination packagings meeting the packing group I performance level are authorized.			
PP14	For UN Nos. 2211, 2698 and 3314, packagings are not required to meet the performance tests in chapter 6.1.			
PP15	For UN Nos. 1324 and 2623, packagings shall meet the packing group III performance level.			
PP20	For UN 2217, any sift-proof, tearproof receptacle may be used.			
PP30	For UN 2471, paper or fibre inner packagings are not permitted.			
PP31	For UN Nos. 1362, 1463, 1565, 1575, 1626, 1680, 1689, 1698, 1868, 1889, 1932, 2471, 2545, 2546, 2881, 3048, 3088, 3170, 3174, 3181, 3182, 3189, 3190, 3205, 3206, 3341, 3342, 3448, 3449 and 3450, packagings shall be hermetically sealed.			
PP34	For UN 2969 (as whole beans), 5H1, 5L1 and 5M1 bags are permitted.			
PP37	For UN Nos. 2590 and 2212, 5M1 bags are permitted. All bags of any type shall be transported in closed cargo transport units or be placed in closed rigid overpacks.			
PP38	For UN 1309, bags are permitted only in closed cargo transport units or as unit loads.			
PP84	For UN 1057, rigid outer packagings meeting the packing group II performance level shall be used. The packagings shall be designed and constructed and arranged to prevent movement, inadvertent ignition of the devices or inadvertent release of flammable gas or liquid.			
PP85	For UN Nos. 1748, 2208, 2880, 3485, 3486 and 3487, bags are not allowed.			
PP92	For UN Nos. 3531 and 3533, packagings shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the packagings in the event of loss of stabilization.			
PP100	For UN numbers 1309, 1323, 1333, 1376, 1435, 1449, 1457, 1472, 1476, 1483, 1509, 1516, 1567, 1869, 2210, 2858, 2878, 2968, 3089, 3096 and 3125, flexible, fibreboard or wooden packagings shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.			

Part 4 – Packing and tank provisions

P003	PACKING INSTRUCTION	P003
	Dangerous goods shall be placed in suitable outer packagings. The packagings shall meet the provisions of 4.1.1.1, 4.1.1.2, 4.1.1.4, 4.1.1.8 and 4.1.3 and be so designed that they meet the construction provisions of 6.1.4. Outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use, shall be used. Where this packing instruction is used for the transport of articles or inner packagings of combination packagings, the packaging shall be designed and constructed to prevent inadvertent discharge of articles during normal conditions of transport.	
	Special packing provisions:	
	PP16 For UN 2800, batteries shall be protected from short circuit within the packagings.	
	PP17 For UN 2037, packages shall not exceed 55 kg net mass for fibreboard packagings or 125 kg net mass for other packagings.	
	PP18 For UN 1845, packagings shall be designed and constructed to permit the release of carbon dioxide gas to prevent a build-up of pressure that could rupture the packagings.	
	PP19 For UN Nos. 1327, 1364, 1365, 1856 and 3360, transport as bales is authorized.	
	PP20 For UN Nos. 1363, 1386, 1408 and 2793, any sift-proof, tearproof receptacle may be used.	
	PP32 UN Nos. 2857 and 3358 and robust articles consigned under UN 3164 may be transported unpackaged, in crates or in appropriate overpacks.	
	Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).	
△	PP90 For UN Nos. 3506 and 3554, sealed inner liners or bags of strong leakproof and puncture resistant material impervious to mercury or gallium, as appropriate, which will prevent escape of the substance from the package irrespective of the position of the package shall be used.	
	PP91 For UN 1044, large fire extinguishers may also be transported unpackaged provided that the requirements of 4.1.3.8.1.1 to 4.1.3.8.1.5 are met, the valves are protected by one of the methods in accordance with 4.1.6.1.8.1 to 4.1.6.1.8.4 and other equipment mounted on the fire extinguisher is protected to prevent accidental activation. For the purpose of this special packing provision, “large fire extinguishers” means fire extinguishers as described in subparagraphs .3 to .5 of special provision 225 of chapter 3.3.	
	PP96 For UN 2037 waste gas cartridges transported in accordance with special provision 327, the packagings shall be adequately ventilated to prevent the creation of dangerous atmospheres and the build-up of pressure.	
	PP100 For UN Nos. 1408 and 2793 flexible, fibreboard or wooden packagings shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.	

P004	PACKING INSTRUCTION	P004
	This instruction applies to UN Nos. 3473, 3476, 3477, 3478 and 3479.	
	The following packagings are authorized:	
	(1) For fuel cell cartridges, provided that the general provisions of 4.1.1.1, 4.1.1.2, 4.1.1.3, 4.1.1.6 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group II performance level.	
	(2) For fuel cell cartridges packed with equipment: strong outer packagings which meet the general provisions of 4.1.1.1, 4.1.1.2, 4.1.1.6 and 4.1.3. When fuel cell cartridges are packed with equipment, they shall be packed in inner packagings or placed in the outer packaging with cushioning material or divider(s) so that the fuel cell cartridges are protected against damage that may be caused by the movement or placement of the contents within the outer packaging. The equipment shall be secured against movement within the outer packaging. For the purpose of this packing instruction, “equipment” means apparatus requiring the fuel cell cartridges with which it is packed for its operation.	
	(3) For fuel cell cartridges contained in equipment: strong outer packagings which meet the general provisions of 4.1.1.1, 4.1.1.2, 4.1.1.6 and 4.1.3. Large robust equipment (see 4.1.3.8) containing fuel cell cartridges may be transported unpackaged. For fuel cell cartridges contained in equipment, the entire system shall be protected against short circuit and inadvertent operation. Note: The packagings authorized in (2) and (3) may exceed a net mass of 400 kg (see 4.1.3.3).	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P005	PACKING INSTRUCTION	P005
This instruction applies to UN Nos. 3528, 3529 and 3530.		
If the engine or machinery is constructed and designed so that the means of containment containing the dangerous goods affords adequate protection, an outer packaging is not required. Dangerous goods in engines or machinery shall otherwise be packed in outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use, and meeting the applicable requirements of 4.1.1.1, or they shall be fixed in such a way that they will not become loose during normal conditions of transport, e.g. in cradles or crates or other handling devices. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3). In addition, the manner in which means of containment are contained within the engine or machinery, shall be such that under normal conditions of transport, damage to the means of containment containing the dangerous goods is prevented; and in the event of damage to the means of containment containing liquid dangerous goods, no leakage of the dangerous goods from the engine or machinery is possible (a leakproof liner may be used to satisfy this requirement). Means of containment containing dangerous goods shall be so installed, secured or cushioned as to prevent their breakage or leakage and so as to control their movement within the engine or machinery during normal conditions of transport. Cushioning material shall not react dangerously with the content of the means of containment. Any leakage of the contents shall not substantially impair the protective properties of the cushioning material. Additional requirement: Other dangerous goods (e.g. batteries, fire extinguishers, compressed gas accumulators or safety devices) required for the functioning or safe operation of the engine or machinery shall be securely mounted in the engine or machine.		
P006	PACKING INSTRUCTION	P006
This instruction applies to UN Nos. 3537, 3538, 3540, 3541, 3546, 3547 and 3548.		
(1) The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group II performance level. (2) In addition, for robust articles the following packagings are authorized: Strong outer packagings constructed of suitable material and of adequate strength and design in relation to the packaging capacity and its intended use. The packagings shall meet the provisions of 4.1.1.1, 4.1.1.2, 4.1.1.8 and 4.1.3 in order to achieve a level of protection that is at least equivalent to that provided by chapter 6.1. Articles may be transported unpackaged or on pallets when the dangerous goods are afforded equivalent protection by the article in which they are contained. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3). (3) Additionally, the following conditions shall be met: (a) Receptacles within articles containing liquids or solids shall be constructed of suitable materials and secured in the article in such a way that, under normal conditions of transport, they cannot break, be punctured or leak their contents into the article itself or the outer packaging; (b) Receptacles containing liquids with closures shall be packed with their closures correctly oriented. The receptacles shall in addition conform to the internal pressure test provisions of 6.1.5.5; (c) Receptacles that are liable to break or be punctured easily, such as those made of glass, porcelain or stoneware or of certain plastics materials shall be properly secured. Any leakage of the contents shall not substantially impair the protective properties of the article or of the outer packaging; (d) Receptacles within articles containing gases shall meet the requirements of section 4.1.6 and chapter 6.2 as appropriate or be capable of providing an equivalent level of protection to packing instructions P200 or P208; (e) Where there is no receptacle within the article, the article shall fully enclose the dangerous substances and prevent their release under normal conditions of transport. (4) Articles shall be packed to prevent movement and inadvertent operation during normal conditions of transport. (5) Articles containing pre-production prototype lithium cells or batteries when these prototypes are transported for testing or production runs of not more than 100 lithium cells or batteries that are of a type that have not met the testing requirements of the <i>Manual of Tests and Criteria</i>, part III, subsection 38.3 shall in addition meet the following: (a) Packagings shall conform to the requirements in paragraph (1) of this packing instruction. (b) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the article within the package that may lead to damage and a dangerous condition during transport. When cushioning material is used to meet this requirement it shall be non-combustible and electrically non-conductive. (c) Non-combustibility of the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured. (d) The article may be transported unpackaged under conditions specified by the competent authority. Additional conditions that may be considered in the approval process include, but are not limited to: (i) the article shall be strong enough to withstand the shocks and loadings normally encountered during transport, including trans-shipment between cargo transport units and between cargo transport units and warehouses as well as any removal from a pallet for subsequent manual or mechanical handling; and (ii) the article shall be fixed in cradles or crates or other handling devices in such a way that it will not become loose during normal conditions of transport.		

Part 4 – Packing and tank provisions

P010 PACKING INSTRUCTION		P010
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.		
Combination packagings		Maximum net mass (see 4.1.3.3)
Inner packagings	Outer packagings	
Glass 1 L Steel 40 L	Drums	
	steel (1A1, 1A2)	400 kg
	plastics (1H1, 1H2)	400 kg
	plywood (1D)	400 kg
	fibre (1G)	400 kg
	Boxes	
	steel (4A)	400 kg
	natural wood (4C1, 4C2)	400 kg
	plywood (4D)	400 kg
	reconstituted wood (4F)	400 kg
	fibreboard (4G)	400 kg
	expanded plastics (4H1)	60 kg
	solid plastics (4H2)	400 kg
Single packagings		Maximum capacity (see 4.1.3.3)
Drums		
steel, non-removable head (1A1)		450 L
Jerricans		
steel, non-removable head (3A1)		60 L
Composite packagings		
plastics receptacle in steel drums (6HA1)		250 L
Steel pressure receptacles provided that the general provisions of 4.1.3.6 are met.		

P099	PACKING INSTRUCTION	P099
Only packagings which are approved for these goods by the competent authority may be used (see 4.1.3.7). A copy of the competent authority approval shall accompany each consignment or the transport document shall include an indication that the packaging was approved by the competent authority.		

P101	PACKING INSTRUCTION	P101
Only packagings which are approved by the competent authority may be used. The distinguishing sign used on vehicles in international road traffic* of the country for which the authority acts shall be marked on the transport documents as follows: “Packaging approved by the competent authority of ...”		

* Distinguishing sign of the State of registration used on motor vehicles and trailers in international road traffic, e.g. in accordance with the Geneva Convention on Road Traffic of 1949 or the Vienna Convention on Road Traffic of 1968.

P110(a)	PACKING INSTRUCTION	P110(a)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics textile, plastic coated or lined rubber textile, rubberized textile	Bags plastics textile, plastic coated or lined rubber textile, rubberized	Drums steel (1A1, 1A2) metal, other than steel or aluminium (1N1, 1N2) plastics (1H1, 1H2)
Receptacles wood	Receptacles plastics metal wood	
Additional provisions:		
1 The intermediate packagings shall be filled with water-saturated material such as an anti-freeze solution or wetted cushioning.		
2 Outer packagings shall be filled with water-saturated material such as an anti-freeze solution or wetted cushioning. Outer packagings shall be constructed and sealed to prevent evaporation of the wetting solution, except for UN 0224 when transported dry.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P110(b) PACKING INSTRUCTION P110(b)		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Receptacles metal wood rubber, conductive plastics, conductive Bags rubber, conductive plastics, conductive	Dividing partitions metal wood plastics fibreboard	Boxes natural wood, sift-proof wall (4C2) plywood (4D) reconstituted wood (4F)
Special packing provision: PP42 For UN Nos. 0074, 0113, 0114, 0129, 0130, 0135 and 0224, the following conditions shall be met: .1 inner packagings shall not contain more than 50 g of explosive substance (quantity corresponding to dry substance); .2 compartments between dividing partitions shall not contain more than one inner packaging, firmly fitted; and .3 the outer packaging may be partitioned into up to 25 compartments.		

P111 PACKING INSTRUCTION P111		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper, waterproofed plastics textile, rubberized Sheets plastics textile, rubberized Receptacles wood	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provision: PP43 For UN 0159, inner packagings are not required when metal (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2) or plastics (1H1 or 1H2) drums are used as outer packagings.		

Part 4 – Packing and tank provisions

P112(a)	PACKING INSTRUCTION (Solid wetted, 1.1D)		P112(a)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags paper, multiwall, water-resistant plastics textile textile, rubberized woven plastics Receptacles metal plastics wood	Bags plastics textile, plastic coated or lined Receptacles metal plastics wood	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Additional provision: Intermediate packagings are not required if leakproof removable head drums are used as the outer packaging.			
Special packing provisions: PP26 For UN Nos. 0004, 0076, 0078, 0154, 0219 and 0394, packagings shall be lead-free. PP45 For UN Nos. 0072 and 0226, intermediate packagings are not required.			

P112(b)	PACKING INSTRUCTION (Solid dry, other than powder 1.1D)		P112(b)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags paper, kraft paper, multiwall, water-resistant plastics textile textile, rubberized woven plastics	Bags (for UN 0150 only) plastics textile, plastic coated or lined	Bags woven plastics, sift-proof (5H2) woven plastics, water-resistant (5H3) plastics, film (5H4) textile, sift-proof (5L2) textile, water-resistant (5L3) paper, multiwall, water-resistant (5M2) Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Special packing provisions: PP26 For UN Nos. 0004, 0076, 0078, 0154, 0216, 0219 and 0386, packagings shall be lead-free. PP46 For UN 0209, bags, sift-proof (5H2) are recommended for flake or prilled TNT in the dry state and a maximum net mass of 30 kg. PP47 For UN 0222, inner packagings are not required when the outer packaging is a bag.			

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P112(c)	PACKING INSTRUCTION (Solid dry powder 1.1D)		P112(c)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags paper, multiwall, water-resistant plastics woven plastics Receptacles fibreboard metal plastics wood	Bags paper, multiwall, water-resistant with inner lining plastics Receptacles metal plastics wood	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Additional provisions: 1 Inner packagings are not required if drums are used as the outer packaging. 2 The packaging shall be sift-proof.			
Special packing provisions: PP26 For UN Nos. 0004, 0076, 0078, 0154, 0216, 0219 and 0386, packagings shall be lead-free. PP46 For UN 0209, bags, sift-proof (5H2) are recommended for flake or prilled TNT in the dry state and a maximum net mass of 30 kg. PP48 For UN 0504, metal packagings shall not be used. Packagings of other material with a small amount of metal, for example metal closures or other metal fittings such as those mentioned in 6.1.4, are not considered metal packagings.			

P113	PACKING INSTRUCTION		P113
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags paper plastics textile, rubberized Receptacles fibreboard metal plastics wood	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Additional provision: The packaging shall be sift-proof.			
Special packing provisions: PP49 For UN Nos. 0094 and 0305, no more than 50 g of substance shall be packed in an inner packaging. PP50 For UN 0027, inner packagings are not necessary when drums are used as the outer packaging. PP51 For UN 0028, paper kraft or waxed paper sheets may be used as inner packagings.			

Part 4 – Packing and tank provisions

P114(a)	PACKING INSTRUCTION (Solid wetted)		P114(a)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags plastics textile woven plastics Receptacles metal plastics wood	Bags plastics textile, plastic coated or lined Receptacles metal plastics Dividing partitions wood	Boxes steel (4A) metal, other than steel or aluminium (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Additional provision: Intermediate packagings are not required if leakproof removable head drums are used as the outer packaging.			
Special packing provisions: PP26 For UN Nos. 0077, 0132, 0234, 0235 and 0236, packagings shall be lead-free. PP43 For UN 0342, inner packagings are not required when metal (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2) or plastics (1H1 or 1H2) drums are used as outer packagings.			

P114(b)	PACKING INSTRUCTION (Solid dry)		P114(b)
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Bags paper, kraft plastics textile, sift-proof woven plastics, sift-proof Receptacles fibreboard metal paper plastics woven plastics, sift-proof wood	<i>Not necessary</i>	Boxes natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)	
Special packing provisions: PP26 For UN Nos. 0077, 0132, 0234, 0235 and 0236, packagings shall be lead-free. PP48 For UN Nos. 0508 and 0509, metal packagings shall not be used. Packagings of other material with a small amount of metal, for example metal closures or other metal fittings such as those mentioned in 6.1.4, are not considered metal packagings. PP50 For UN Nos. 0160, 0161 and 0508, inner packagings are not necessary when drums are used as the outer packaging. PP52 For UN Nos. 0160 and 0161, when metal drums (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2) are used as the outer packaging, metal packagings shall be so constructed that the risk of explosion, by reason of increase in internal pressure from internal or external causes, is prevented.			

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P115 PACKING INSTRUCTION P115		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Receptacles plastics wood	Bags plastics in metal receptacles Drums metal Receptacles wood	Boxes natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provisions: PP45 For UN 0144, intermediate packagings are not required. PP53 For UN Nos. 0075, 0143, 0495 and 0497, when boxes are used as the outer packaging, inner packagings shall have taped screw-cap closures and be not more than 5 L capacity each. Inner packagings shall be surrounded with non-combustible absorbent cushioning materials. The amount of absorbent cushioning material shall be sufficient to absorb the liquid contents. Metal receptacles shall be cushioned from each other. Net mass of propellant is limited to 30 kg for each package when outer packagings are boxes. PP54 For UN Nos. 0075, 0143, 0495 and 0497, when drums are used as the outer packaging and when intermediate packagings are drums, they shall be surrounded with non-combustible cushioning material in a quantity sufficient to absorb the liquid contents. A composite packaging consisting of a plastics receptacle in a metal drum may be used instead of the inner and intermediate packagings. The net volume of propellant in each package shall not exceed 120 L. PP55 For UN 0144, absorbent cushioning material shall be inserted. PP56 For UN 0144, metal receptacles may be used as inner packagings. PP57 For UN Nos. 0075, 0143, 0495 and 0497, bags shall be used as intermediate packagings when boxes are used as outer packagings. PP58 For UN Nos. 0075, 0143, 0495 and 0497, drums shall be used as intermediate packagings when drums are used as outer packagings. PP59 For UN 0144, fibreboard boxes (4G) may be used as outer packagings. PP60 For UN 0144, aluminium drums (1B1 or 1B2) and metal, other than steel or aluminium, drums (1N1 or 1N2) shall not be used.		

Part 4 – Packing and tank provisions

P116 PACKING INSTRUCTION P116		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper, water- and oil-resistant plastics textile, plastic coated or lined woven plastics, sift-proof Receptacles fibreboard, water-resistant metal plastics wood, sift-proof Sheets paper, water-resistant paper, waxed plastics	<i>Not necessary</i>	Bags woven plastics (5H1, 5H2, 5H3) paper, multiwall, water-resistant (5M2) plastics, film (5H4) textile, sift-proof (5L2) textile, water-resistant (5L3) Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) fibre (1G) plastics (1H1, 1H2) plywood (1D) Jerricans steel (3A1, 3A2) plastics (3H1, 3H2)
Special packing provisions: PP61 For UN Nos. 0082, 0241, 0331 and 0332, inner packagings are not required if leakproof removable head drums are used as the outer packaging. PP62 For UN Nos. 0082, 0241, 0331 and 0332, inner packagings are not required when the explosive is contained in a material impervious to liquid. PP63 For UN 0081, inner packagings are not required when contained in rigid plastic which is impervious to nitric esters. PP64 For UN 0331, inner packagings are not required when bags (5H2, 5H3 or 5H4) are used as outer packagings. PP65 [Deleted] PP66 For UN 0081, bags shall not be used as outer packagings.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P130 PACKING INSTRUCTION P130		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
<i>Not necessary</i>	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provision: PP67 The following applies to UN Nos. 0006, 0009, 0010, 0015, 0016, 0018, 0019, 0034, 0035, 0038, 0039, 0048, 0056, 0137, 0138, 0168, 0169, 0171, 0181, 0182, 0183, 0186, 0221, 0243, 0244, 0245, 0246, 0254, 0280, 0281, 0286, 0287, 0297, 0299, 0300, 0301, 0303, 0321, 0328, 0329, 0344, 0345, 0346, 0347, 0362, 0363, 0370, 0412, 0424, 0425, 0434, 0435, 0436, 0437, 0438, 0451, 0488, 0502 and 0510: large and robust explosives articles, normally intended for military use, without their means of initiation or with their means of initiation containing at least two effective protective features, may be transported unpackaged. When such articles have propelling charges or are self-propelled, their ignition systems shall be protected against stimuli encountered during normal conditions of transport. A negative result in Test Series 4 on an unpackaged article indicates that the article can be considered for transport unpackaged. Such unpackaged articles may be fixed to cradles or contained in crates or other suitable handling devices. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).		

P131 PACKING INSTRUCTION P131		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper plastics Receptacles fibreboard metal plastics wood Reels	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provision: PP68 For UN Nos. 0029, 0267 and 0455, bags and reels shall not be used as inner packagings.		

Part 4 – Packing and tank provisions

P132(a)	PACKING INSTRUCTION		P132(a)
(Articles consisting of closed metal, plastics or fibreboard casings that contain a detonating explosive, or consisting of plastics-bonded detonating explosives)			
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Not necessary	Not necessary	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2)	

P132(b)	PACKING INSTRUCTION		P132(b)
(Articles without closed casings)			
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Receptacles fibreboard metal plastics wood Sheets paper plastics	Not necessary	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2)	

P133	PACKING INSTRUCTION		P133
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.			
Inner packagings	Intermediate packagings	Outer packagings	
Receptacles fibreboard metal plastics wood Trays, fitted with dividing partitions fibreboard plastics wood	Receptacles fibreboard metal plastics wood	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2)	
Additional provision: Receptacles are only required as intermediate packagings when the inner packagings are trays.			
Special packing provision: PP69 For UN Nos. 0043, 0212, 0225, 0268 and 0306, trays shall not be used as inner packagings.			

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P134 PACKING INSTRUCTION P134		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags water-resistant Receptacles fibreboard metal plastics wood Sheets fibreboard, corrugated Tubes fibreboard	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)

P135 PACKING INSTRUCTION P135		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper plastics Receptacles fibreboard metal plastics wood Sheets paper plastics	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)

P136 PACKING INSTRUCTION P136		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics textile Boxes fibreboard plastics wood Dividing partitions in the outer packagings	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)

Part 4 – Packing and tank provisions

P137 PACKING INSTRUCTION P137		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics Boxes fibreboard wood Tubes fibreboard metal plastics Dividing partitions in the outer packagings	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provision: PP70 For UN Nos. 0059, 0439, 0440 and 0441, when the shaped charges are packed singly, the conical cavity shall face downwards and the package shall be marked as illustrated in figures in 5.2.1.7.1. When the shaped charges are packed in pairs, the conical cavities shall face inwards to minimize the jetting effect in the event of accidental initiation.		

P138 PACKING INSTRUCTION P138		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Additional provision: If the ends of the articles are sealed, inner packagings are not necessary.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P139 PACKING INSTRUCTION P139		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics Receptacles fibreboard metal plastics wood Reels Sheets paper plastics	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provisions: PP71 For UN Nos. 0065, 0102, 0104, 0289 and 0290, the ends of the detonating cord shall be sealed; for example, by a plug firmly fixed so that the explosive cannot escape. The ends of flexible detonating cord shall be fastened securely. PP72 For UN Nos. 0065 and 0289, inner packagings are not required when they are in coils.		

P140 PACKING INSTRUCTION P140		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags plastics Reels Sheets paper, kraft plastics Receptacles wood	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Special packing provisions: PP73 For UN 0105, no inner packagings are required if the ends are sealed. PP74 For UN 0101, the packaging shall be sift-proof except when the fuse is covered by a paper tube and both ends of the tube are covered with removable caps. PP75 For UN 0101, steel, aluminium or other metal boxes or drums shall not be used.		

Part 4 – Packing and tank provisions

P141 PACKING INSTRUCTION P141		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Receptacles fibreboard metal plastics wood Trays, fitted with dividing partitions plastics wood Dividing partitions in the outer packagings	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)

P142 PACKING INSTRUCTION P142		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper plastics Receptacles fibreboard metal plastics wood Sheets paper Trays, fitted with dividing partitions plastics	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P143 PACKING INSTRUCTION P143		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags paper, kraft plastics textile textile, rubberized Receptacles fibreboard metal plastics wood Trays, fitted with dividing partitions plastics wood	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary (4C1) natural wood, sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plywood (1D) fibre (1G) plastics (1H1, 1H2)
Additional provision: Instead of the above inner and outer packagings, composite packagings (6HH2) (plastics receptacle with outer solid box) may be used.		
Special packing provision: PP76 For UN Nos. 0271, 0272, 0415 and 0491, when metal packagings are used, metal packagings shall be so constructed that the risk of explosion, by reason of increase in internal pressure from internal or external causes, is prevented.		

P144 PACKING INSTRUCTION P144		
The following packagings are authorized, provided the general packing provisions of 4.1.1, 4.1.3 and special packing provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Receptacles fibreboard metal plastics wood Dividing partitions in the outer packagings	<i>Not necessary</i>	Boxes steel (4A) aluminium (4B) other metal (4N) natural wood, ordinary with metal liner (4C1) plywood (4D) with metal liner reconstituted wood with metal liner (4F) plastics, expanded (4H1) plastics, solid (4H2) Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plastics (1H1, 1H2)
Special packing provision: PP77 For UN Nos. 0248 and 0249, packagings shall be protected against the ingress of water. When water-activated contrivances are transported unpackaged, they shall be provided with at least two independent protective features which prevent the ingress of water. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).		

Part 4 – Packing and tank provisions

P200	PACKING INSTRUCTION	P200
	For pressure receptacles, the general packing provisions of 4.1.6.1 shall be met. In addition, for MEGCs, the general requirements of 4.2.4 shall be met. Cylinders, tubes, pressure drums, bundles of cylinders constructed as specified in 6.2 and MEGCs constructed as specified in 6.7.5 are authorized for the transport of a specific substance when specified in the following tables. For some substances, the special packing provisions may prohibit a particular type of cylinder, tube, pressure drum or bundle of cylinders. (1) Pressure receptacles containing toxic substances with an LC_{50} less than or equal to 200 mL/m³ (ppm) as specified in the table shall not be equipped with any pressure relief device. Pressure relief devices shall be fitted on pressure receptacles used for the transport of UN 1013 carbon dioxide and UN 1070 nitrous oxide. Other pressure receptacles shall be fitted with a pressure relief device if specified by the competent authority of the country of use. The type of pressure relief device, the set-to-discharge pressure and relief capacity of pressure relief devices, if required, shall be specified by the competent authority of the country of use. (2) The following three tables cover compressed gases (Table 1), liquefied and dissolved gases (Table 2) and substances not in Class 2 (Table 3). They provide: (a) The UN number, proper shipping name and classification of the substance; (b) The LC_{50} for toxic substances; (c) The types of pressure receptacles authorized for the substance, shown by the letter "X"; (d) The maximum test period for periodic inspection of the pressure receptacles. Note: For pressure receptacles which make use of composite materials, the maximum test period shall be 5 years. The test period may be extended to that specified in Tables 1 and 2 (i.e. up to 10 years), if approved by the competent authority of the country of use. (e) The minimum test pressure of the pressure receptacles; (f) The maximum working pressure of the pressure receptacles for compressed gases (where no value is given, the working pressure shall not exceed two thirds of the test pressure) or the maximum filling ratio(s) dependent on the test pressure(s) for liquefied and dissolved gases; (g) Special packing provisions that are specific to a substance. (3) In no case shall pressure receptacles be filled in excess of the limit permitted in the following requirements. (a) For compressed gases, the working pressure shall be not more than two thirds of the test pressure of the pressure receptacles. Restrictions to this upper limit on working pressure are imposed by special packing provision "o" in (5) below. In no case shall the internal pressure at 65°C exceed the test pressure. (b) For high pressure liquefied gases, the filling ratio shall be such that the settled pressure at 65°C does not exceed the test pressure of the pressure receptacles. The use of test pressures and filling ratios other than those in the table is permitted, except where (5), special packing provision "o" applies, provided that: (i) the criterion of (5), special packing provision "r" is met when applicable; or (ii) the above criterion is met in all other cases. For high pressure liquefied gases and gas mixtures for which relevant data are not available, the maximum filling ratio (FR) shall be determined as follows: $FR = 8.5 \times 10^{-4} \times d_g \times P_h$ where FR = maximum filling ratio d_g = gas density (at 15°C, 1 bar) (in g/L) P_h = minimum test pressure (in bar) If the density of the gas is unknown, the maximum filling ratio shall be determined as follows: $FR = \frac{P_h \times MM \times 10^{-3}}{R \times 338}$ where FR = maximum filling ratio P_h = minimum test pressure (in bar) MM = molecular mass (in g/mol) R = 8.31451×10^{-2} bar·L/mol·K (gas constant) For gas mixtures, the average molecular mass is to be taken, taking into account the volumetric concentrations of the various components. (c) For low pressure liquefied gases, the maximum mass of contents per litre of water capacity shall equal 0.95 times the density of the liquid phase at 50°C; in addition, the liquid phase shall not fill the pressure receptacle at any temperature up to 60°C. The test pressure of the pressure receptacle shall be at least equal to the vapour pressure (absolute) of the liquid at 65°C, minus 100 kPa (1 bar). For low pressure liquefied gases and gas mixtures for which relevant data are not available, the maximum filling ratio shall be determined as follows: $FR = (0.0032 \times BP - 0.24) \times d_l$ where FR = maximum filling ratio BP = boiling point (in kelvin) d_l = density of the liquid at boiling point (in kg/L) (d) For UN 1001, acetylene, dissolved, and UN 3374 acetylene, solvent free, see (5), special packing provision "p".	

P200	PACKING INSTRUCTION (continued)	P200										
(e)	For liquefied gases charged with compressed gases, both components – the liquefied gas and the compressed gas – have to be taken into consideration in the calculation of the internal pressure in the pressure receptacle. The maximum mass of contents per litre of water capacity shall not exceed 0.95 times the density of the liquid phase at 50°C; in addition, the liquid phase shall not completely fill the pressure receptacle at any temperature up to 60°C. When filled, the internal pressure at 65°C shall not exceed the test pressure of the pressure receptacles. The vapour pressures and volumetric expansions of all substances in the pressure receptacles shall be considered. When experimental data is not available, the following steps shall be carried out: (i) calculation of the vapour pressure of the liquefied gas and of the partial pressure of the compressed gas at 15°C (filling temperature); (ii) calculation of the volumetric expansion of the liquid phase resulting from the heating from 15°C to 65°C and calculation of the remaining volume for the gaseous phase; (iii) calculation of the partial pressure of the compressed gas at 65°C considering the volumetric expansion of the liquid phase; Note: The compressibility factor of the compressed gas at 15°C and 65°C shall be considered. (iv) calculation of the vapour pressure of the liquefied gas at 65°C; (v) the total pressure is the sum of the vapour pressure of the liquefied gas and the partial pressure of the compressed gas at 65°C; (vi) consideration of the solubility of the compressed gas at 65°C in the liquid phase. The test pressure of the pressure receptacle shall not be less than the calculated total pressure minus 100 kPa (1bar). If the solubility of the compressed gas in the liquid phase is not known for the calculation, the test pressure can be calculated without taking the gas solubility (subparagraph (vi)) into account.											
(4)	The filling of pressure receptacles shall be carried out by qualified staff using appropriate equipment and procedures. The procedures should include checks of: (a) the conformity of receptacles and accessories with the provisions of this Code; (b) their compatibility with the product to be transported; (c) the absence of damage which might affect safety; (d) compliance with the degree or pressure of filling, as appropriate; (e) marks and identification. These requirements are deemed to be met if the following standards are applied: <table><tr><td>ISO 10691:2004</td><td>Gas cylinders – Refillable welded steel cylinders for liquefied petroleum gas (LPG) – Procedures for checking before, during and after filling.</td></tr><tr><td>ISO 11372:2011</td><td>Gas cylinders – Acetylene cylinders – Filling conditions and filling inspection</td></tr><tr><td>ISO 11755:2005</td><td>Gas cylinders – Cylinder bundles for compressed and liquefied gases (excluding acetylene) – Inspection at time of filling</td></tr><tr><td>ISO 13088:2011 + Amd 1:2020</td><td>Gas cylinders – Acetylene cylinder bundles – Filling conditions and filling inspection</td></tr><tr><td>ISO 24431:2016</td><td>Gas cylinders – Seamless, welded and composite cylinders for compressed and liquefied gases (excluding acetylene) – Inspection at time of filling</td></tr></table>	ISO 10691:2004	Gas cylinders – Refillable welded steel cylinders for liquefied petroleum gas (LPG) – Procedures for checking before, during and after filling.	ISO 11372:2011	Gas cylinders – Acetylene cylinders – Filling conditions and filling inspection	ISO 11755:2005	Gas cylinders – Cylinder bundles for compressed and liquefied gases (excluding acetylene) – Inspection at time of filling	ISO 13088:2011 + Amd 1:2020	Gas cylinders – Acetylene cylinder bundles – Filling conditions and filling inspection	ISO 24431:2016	Gas cylinders – Seamless, welded and composite cylinders for compressed and liquefied gases (excluding acetylene) – Inspection at time of filling	
ISO 10691:2004	Gas cylinders – Refillable welded steel cylinders for liquefied petroleum gas (LPG) – Procedures for checking before, during and after filling.											
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ISO 13088:2011 + Amd 1:2020	Gas cylinders – Acetylene cylinder bundles – Filling conditions and filling inspection											
ISO 24431:2016	Gas cylinders – Seamless, welded and composite cylinders for compressed and liquefied gases (excluding acetylene) – Inspection at time of filling											
(5)	Special packing provisions: <i>Material compatibility</i> - a: Aluminium alloy pressure receptacles shall not be used. - b: Copper valves shall not be used. - c: Metal parts in contact with the contents shall not contain more than 65% copper. - d: When steel pressure receptacles or composite pressure receptacles with steel liners are used, only those bearing the "H" mark in accordance with 6.2.2.7.4 (p) are permitted. <i>Requirements for toxic substances with an LC₅₀ less than or equal to 200 mL/m³ (ppm)</i> - k: Valve outlets shall be fitted with pressure-retaining gas-tight plugs or caps having threads that match those of the valve outlets. Each cylinder within a bundle shall be fitted with an individual valve that shall be closed during transport. After filling, the manifold shall be evacuated, purged and plugged. Bundles containing UN 1045 fluorine, compressed, may be constructed with isolation valves on groups of cylinders not exceeding 150 L total water capacity instead of isolation valves on every cylinder. Cylinders and individual cylinders in a bundle shall have a test pressure greater than or equal to 200 bar and a minimum wall thickness of 3.5 mm for aluminium alloy or 2 mm for steel. Individual cylinders not complying with this requirement shall be transported in a rigid outer packaging that will adequately protect the cylinder and its fittings and meeting the packing group I performance level. Pressure drums shall have a minimum wall thickness as specified by the competent authority. Pressure receptacles shall not be fitted with a pressure relief device. Cylinders and individual cylinders in a bundle shall be limited to a maximum water capacity of 85 L. Each valve shall be capable of withstanding the test pressure of the pressure receptacle and be connected directly to the pressure receptacle by either a taper thread or other means which meets the requirements of ISO 10692-2:2001.											

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Part 4 – Packing and tank provisions

P200	PACKING INSTRUCTION (continued)	P200
	Each valve shall either be of the packless type with non-perforated diaphragm, or be of a type which prevents leakage through or past the packing. Each pressure receptacle shall be tested for leakage after filling. <i>Gas specific provisions</i> l: UN 1040 ethylene oxide may also be packed in hermetically sealed glass or metal inner packagings suitably cushioned in fibreboard, wooden or metal boxes meeting the packing group I performance level. The maximum quantity permitted in any glass inner packaging is 30 g, and the maximum quantity permitted in any metal inner packaging is 200 g. After filling, each inner packaging shall be determined to be leaktight by placing the inner packaging in a hot water bath at a temperature, and for a period of time, sufficient to ensure that an internal pressure equal to the vapour pressure of ethylene oxide at 55°C is achieved. The maximum net mass in any outer packaging shall not exceed 2.5 kg. m: Pressure receptacles shall be filled to a working pressure not exceeding 5 bar. n: Cylinders and individual cylinders in a bundle shall contain not more than 5 kg of the gas. When bundles containing UN 1045 fluorine, compressed are divided into groups of cylinders in accordance with special packing provision "k" each group shall contain not more than 5 kg of the gas. o: In no case shall the working pressure or filling ratio shown in the table be exceeded. p: For UN 1001 acetylene, dissolved and UN 3374 acetylene, solvent free: cylinders shall be filled with a homogeneous monolithic porous material; the working pressure and the quantity of acetylene shall not exceed the values prescribed in the approval or in ISO 3807-1:2000, ISO 3807-2:2000 or ISO 3807:2013, as applicable. For UN 1001 acetylene, dissolved: cylinders shall contain a quantity of acetone or suitable solvent as specified in the approval (see ISO 3807-1:2000, ISO 3807-2:2000 or ISO 3807:2013, as applicable); cylinders fitted with pressure relief devices or manifolded together shall be transported vertically. The test pressure of 52 bar applies only to cylinders fitted with a fusible plug. q: Valve outlets of pressure receptacles for pyrophoric gases or flammable mixtures of gases containing more than 1% of pyrophoric compounds shall be fitted with gas-tight plugs or caps. When these pressure receptacles are manifolded in a bundle, each of the pressure receptacles shall be fitted with an individual valve that shall be closed during transport, and the outlet of the manifold valve shall be fitted with a pressure-retaining gas-tight plug or cap. Gas-tight plugs or caps shall have threads that match those of the valve outlets. r: The filling ratio of this gas shall be limited such that, if complete decomposition occurs, the pressure does not exceed two thirds of the test pressure of the pressure receptacle. ra: This gas may also be packed in capsules under the following conditions: (i) The mass of gas shall not exceed 150 g per capsule; (ii) The capsules shall be free from faults liable to impair the strength; (iii) The leakproofness of the closure shall be ensured by an additional device (cap, crown, seal, binding, etc.) capable of preventing any leakage of the closure during transport; (iv) The capsules shall be placed in an outer packaging of sufficient strength. A package shall not weigh more than 75 kg. s: Aluminium alloy pressure receptacles shall be: (a) equipped only with brass or stainless steel valves; and (b) cleaned in accordance with ISO 11621:1997 and not contaminated with oil. t: (a) The wall thickness of pressure receptacles shall be not less than 3 mm. (b) Prior to transport, it shall be ensured that the pressure has not risen due to potential hydrogen generation. <i>Periodic inspection</i> u: The interval between periodic tests may be extended to 10 years for aluminium alloy pressure receptacles when the alloy of the pressure receptacle has been subjected to stress corrosion testing as specified in ISO 7866:2012 + Cor 1:2014. v: The interval between periodic inspections for steel cylinders may be extended to 15 years if approved by the competent authority of the country of use. <i>Requirements for N.O.S. descriptions and for mixtures</i> z: The construction materials of the pressure receptacles and their accessories shall be compatible with the contents and shall not react to form harmful or dangerous compounds therewith. The test pressure and filling ratio shall be calculated in accordance with the relevant requirements of (3). Toxic substances with an LC₅₀ less than or equal to 200 mL/m³ shall not be transported in tubes, pressure drums or MEGCs and shall meet the requirements of special packing provision "k". However, UN 1975 nitric oxide and dinitrogen tetroxide mixtures may be transported in pressure drums. For pressure receptacles containing pyrophoric gases or flammable mixtures of gases containing more than 1% pyrophoric compounds, the requirements of special packing provision "q" shall be met. The necessary steps shall be taken to prevent dangerous reactions (i.e. polymerization or decomposition) during transport. If necessary, stabilization or addition of an inhibitor shall be required. Mixtures containing UN 1911 diborane shall be filled to a pressure such that, if complete decomposition of the diborane occurs, two thirds of the test pressure of the pressure receptacle shall not be exceeded. Mixtures containing UN 2192 germane, other than mixtures of up to 35% germane in hydrogen or nitrogen or up to 28% germane in helium or argon, shall be filled to a pressure such that, if complete decomposition of the germane occurs, two thirds of the test pressure of the pressure receptacle shall not be exceeded.	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

△	Mixtures of fluorine and nitrogen with a fluorine concentration below 35% by volume may be filled in pressure receptacles up to a maximum allowable working pressure for which the partial pressure of fluorine does not exceed 31 bar (absolute). $\text{working pressure (bar)} < \frac{31}{x_f} - 1$ in which x_f = fluorine concentration in % by volume/100.
△	Mixtures of fluorine and inert gases with a fluorine concentration below 35% by volume may be filled in pressure receptacles up to a maximum allowable working pressure for which the partial pressure of fluorine does not exceed 31 bar (absolute), additionally taking the coefficient of nitrogen equivalency in accordance with ISO 10156:2017 into account when calculating the partial pressure. $\text{working pressure (bar)} < \frac{31}{x_f} (x_f + K_k \times x_k) - 1$ in which x_f = fluorine concentration in % by volume/100; K_k = coefficient of equivalency of an inert gas relative to nitrogen (coefficient of nitrogen equivalency); and x_k = inert gas concentration in % by volume/100. However, the working pressure for mixtures of fluorine and inert gases shall not exceed 200 bar. The minimum test pressure of pressure receptacles for mixtures of fluorine and inert gases equals 1.5 times the working pressure or 200 bar, with the greater value to be applied.

Part 4 – Packing and tank provisions

P200		PACKING INSTRUCTION (continued)												P200	
Table 1: COMPRESSED GASES															
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ , mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar*	Maximum working pressure, bar*	Special packing provisions		
1002	AIR, COMPRESSED	2.2			X	X	X	X	X	10					
1006	ARGON, COMPRESSED	2.2			X	X	X	X	X	10					
1016	CARBON MONOXIDE, COMPRESSED	2.3	2.1	3,760	X	X	X	X	X	5			u		
1023	COAL GAS, COMPRESSED	2.3	2.1		X	X	X	X	X	5					
1045	FLUORINE, COMPRESSED	2.3	5.1, 8	185	X			X		5	200	30	a, k, n, o		
1046	HELIUM, COMPRESSED	2.2			X	X	X	X	X	10					
1049	HYDROGEN, COMPRESSED	2.1			X	X	X	X	X	10			d		
1056	KRYPTON, COMPRESSED	2.2			X	X	X	X	X	10					
1065	NEON, COMPRESSED	2.2			X	X	X	X	X	10					
1066	NITROGEN, COMPRESSED	2.2			X	X	X	X	X	10					
1071	OIL GAS, COMPRESSED	2.3	2.1		X	X	X	X	X	5					
1072	OXYGEN, COMPRESSED	2.2	5.1		X	X	X	X		10			s		
1612	HEXAETHYL TETRAPHOSPHATE AND COMPRESSED GAS MIXTURE	2.3			X	X	X	X		5			z		
1660	NITRIC OXIDE, COMPRESSED	2.3	5.1, 8	115	X			X		5	225	33	k, o		
1953	COMPRESSED GAS, TOXIC, FLAMMABLE, N.O.S.	2.3	2.1	≤ 5,000	X	X	X	X	X	5			z		
1954	COMPRESSED GAS, FLAMMABLE, N.O.S	2.1			X	X	X	X	X	10			z		
1955	COMPRESSED GAS, TOXIC, N.O.S.	2.3		≤ 5,000	X	X	X	X	X	5			z		
1956	COMPRESSED GAS, N.O.S.	2.2			X	X	X	X	X	10			z		
1957	DEUTERIUM, COMPRESSED	2.1			X	X	X	X	X	10			d		
1964	HYDROCARBON GAS MIXTURE, COMPRESSED, N.O.S.	2.1			X	X	X	X	X	10			z		
1971	METHANE, COMPRESSED or NATURAL GAS, COMPRESSED with high methane content	2.1			X	X	X	X	X	10					
2034	HYDROGEN AND METHANE MIXTURE, COMPRESSED	2.1			X	X	X	X	X	10			d		
2190	OXYGEN DIFLUORIDE, COMPRESSED	2.3	5.1, 8	2.6	X			X		5	200	30	a, k, n, o		
3156	COMPRESSED GAS, OXIDIZING, N.O.S.	2.2	5.1		X	X	X	X	X	10			z		
3303	COMPRESSED GAS, TOXIC, OXIDIZING, N.O.S.	2.3	5.1	≤ 5,000	X	X	X	X	X	5			z		
3304	COMPRESSED GAS, TOXIC, CORROSIVE, N.O.S.	2.3	8	≤ 5,000	X	X	X	X	X	5			z		
3305	COMPRESSED GAS, TOXIC, FLAMMABLE, CORROSIVE, N.O.S.	2.3	2.1, 8	≤ 5,000	X	X	X	X	X	5			z		
3306	COMPRESSED GAS, TOXIC, OXIDIZING, CORROSIVE, N.O.S.	2.3	5.1, 8	≤ 5,000	X	X	X	X	X	5			z		

* Where the entries are blank, the maximum working pressure shall not exceed two thirds of the test pressure.

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P200		PACKING INSTRUCTION (continued)												P200
Table 2: LIQUEFIED GASES AND DISSOLVED GASES														
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ , mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions	
1001	ACETYLENE, DISSOLVED	2.1			X			X		10	60 52		c, p c, p	
1005	AMMONIA, ANHYDROUS	2.3	8	4,000	X	X	X	X	X	5	29	0.54	b	
1008	BORON TRIFLUORIDE	2.3	8	864	X	X	X	X	X	5	225 300	0.715 0.86	a a	
1009	BROMOTRIFLUOROMETHANE (REFRIGERANT GAS R 13B1)	2.2			X	X	X	X	X	10	42 120 250	1.13 1.44 1.60		
1010	BUTADIENES, STABILIZED (1,2-butadiene), or	2.1			X	X	X	X	X	10	10	0.59		
1010	BUTADIENES, STABILIZED (1,3-butadiene), or	2.1			X	X	X	X	X	10	10	0.55		
1010	BUTADIENES AND HYDROCARBON MIXTURE, STABILIZED with more than 20% butadienes	2.1			X	X	X	X	X	10			v, z	
1011	BUTANE	2.1			X	X	X	X	X	10	10	0.52	v	
1012	BUTYLENE (butylenes mixture) or	2.1			X	X	X	X	X	10	10	0.50	z	
1012	BUTYLENE (1-butylene) or	2.1			X	X	X	X	X	10	10	0.53		
1012	BUTYLENE (cis-2-butylene) or	2.1			X	X	X	X	X	10	10	0.55		
1012	BUTYLENE (trans-2-butylene)	2.1			X	X	X	X	X	10	10	0.54		
1013	CARBON DIOXIDE	2.2			X	X	X	X	X	10	190 250	0.68 0.76		
1017	CHLORINE	2.3	5.1, 8	293	X	X	X	X	X	5	22	1.25	a	
1018	CHLORODIFLUOROMETHANE (REFRIGERANT GAS R 22)	2.2			X	X	X	X	X	10	27	1.03		
1020	CHLOROPENTAFLUOROETHANE (REFRIGERANT GAS R 115)	2.2			X	X	X	X	X	10	25	1.05		
1021	1-CHLORO-1,2,2,2-TETRAFLUOROETHANE (REFRIGERANT GAS R 124)	2.2			X	X	X	X	X	10	11	1.20		
1022	CHLOROTRIFLUOROMETHANE (REFRIGERANT GAS R 13)	2.2			X	X	X	X	X	10	100 120 190 250	0.83 0.90 1.04 1.11		
1026	CYANOGEN	2.3	2.1	350	X	X	X	X	X	5	100	0.70	u	
1027	CYCLOPROPANE	2.1			X	X	X	X	X	10	18	0.55		
1028	DICHLORODIFLUOROMETHANE (REFRIGERANT GAS R 12)	2.2			X	X	X	X	X	10	16	1.15		
1029	DICHLOROFLUOROMETHANE (REFRIGERANT GAS R 21)	2.2			X	X	X	X	X	10	10	1.23		
1030	1,1-DIFLUOROETHANE (REFRIGERANT GAS R 152a)	2.1			X	X	X	X	X	10	16	0.79		
1032	DIMETHYLAMINE, ANHYDROUS	2.1			X	X	X	X	X	10	10	0.59	b	
1033	DIMETHYL ETHER	2.1			X	X	X	X	X	10	18	0.58		
1035	ETHANE	2.1			X	X	X	X	X	10	95 120 300	0.25 0.30 0.40		
1036	ETHYLAMINE	2.1			X	X	X	X	X	10	10	0.61	b	
1037	ETHYL CHLORIDE	2.1			X	X	X	X	X	10	10	0.80	a, ra	
1039	ETHYL METHYL ETHER	2.1			X	X	X	X	X	10	10	0.64		
1040	ETHYLENE OXIDE or ETHYLENE OXIDE WITH NITROGEN up to a total pressure of 1 MPa (10 bar) at 50°C	2.3	2.1	2,900	X	X	X	X	X	5	15	0.78	l	

Part 4 – Packing and tank provisions

P200		PACKING INSTRUCTION (continued)											P200
Table 2: LIQUEFIED GASES AND DISSOLVED GASES (continued)													
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ , mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions
1041	ETHYLENE OXIDE AND CARBON DIOXIDE MIXTURE with more than 9% ethylene oxide but not more than 87%	2.1			X	X	X	X	X	10	190 250	0.66 0.75	
1043	FERTILIZER AMMONIATING SOLUTION with free ammonia	2.2			X		X	X		5			b, z
1048	HYDROGEN BROMIDE, ANHYDROUS	2.3	8	2,860	X	X	X	X	X	5	60	1.51	a, d
1050	HYDROGEN CHLORIDE, ANHYDROUS	2.3	8	2,810	X	X	X	X	X	5	100	0.30	a, d
											120	0.56	a, d
											150	0.67	a, d
											200	0.74	a, d
1053	HYDROGEN SULPHIDE	2.3	2.1	712	X	X	X	X	X	5	48	0.67	d, u
1055	ISOBUTYLENE	2.1			X	X	X	X	X	10	10	0.52	
1058	LIQUEFIED GASES, non-flammable, charged with nitrogen, carbon dioxide or air	2.2			X	X	X	X	X	10			z
1060	METHYLACETYLENE AND PROPADIENE MIXTURE, STABILIZED or	2.1			X	X	X	X	X	10			c, z
1060	METHYLACETYLENE AND PROPADIENE MIXTURE, STABILIZED (Propadiene with 1% to 4% methylacetylene)	2.1			X	X	X	X	X	10	22	0.52	c
1061	METHYLAMINE, ANHYDROUS	2.1			X	X	X	X	X	10	13	0.58	b
1062	METHYL BROMIDE with not more than 2% chloropicrin	2.3		850	X	X	X	X	X	5	10	1.51	a
1063	METHYL CHLORIDE (REFRIGERANT GAS R 40)	2.1			X	X	X	X	X	10	17	0.81	a
1064	METHYL MERCAPTAN	2.3	2.1	1,350	X	X	X	X	X	5	10	0.78	d, u
1067	DINITROGEN TETROXIDE (NITROGEN DIOXIDE)	2.3	5.1, 8	115	X		X	X		5	10	1.30	k
1069	NITROSYL CHLORIDE	2.3	8	35	X			X		5	13	1.10	k
1070	NITROUS OXIDE	2.2	5.1		X	X	X	X	X	10	180	0.68	
											225	0.74	
											250	0.75	
1075	PETROLEUM GASES, LIQUEFIED	2.1			X	X	X	X	X	10			v, z
1076	PHOSGENE	2.3	8	5	X		X	X		5	20	1.23	k, a
1077	PROPYLENE	2.1			X	X	X	X	X	10	27	0.43	
1078	REFRIGERANT GAS, N.O.S.	2.2			X	X	X	X	X	10			z
1079	SULPHUR DIOXIDE	2.3	8	2,520	X	X	X	X	X	5	12	1.23	
1080	SULPHUR HEXAFLUORIDE	2.2			X	X	X	X	X	10	70	1.06	
											140	1.34	
											160	1.38	
1081	TETRAFLUOROETHYLENE, STABILIZED	2.1			X	X	X	X	X	10	200		m, o
1082	TRIFLUOROCHLOROETHYLENE, STABILIZED (REFRIGERANT GAS R 1113)	2.3	2.1	2,000	X	X	X	X	X	5	19	1.13	u
1083	TRIMETHYLAMINE, ANHYDROUS	2.1			X	X	X	X	X	10	10	0.56	b
1085	VINYL BROMIDE, STABILIZED	2.1			X	X	X	X	X	10	10	1.37	a
1086	VINYL CHLORIDE, STABILIZED	2.1			X	X	X	X	X	10	12	0.81	a
1087	VINYL METHYL ETHER, STABILIZED	2.1			X	X	X	X	X	10	10	0.67	
1581	CHLOROPICRIN AND METHYL BROMIDE MIXTURE with more than 2% chloropicrin	2.3		850	X	X	X	X	X	5	10	1.51	a
1582	CHLOROPICRIN AND METHYL CHLORIDE MIXTURE	2.3			X	X	X	X	X	5	17	0.81	a
1589	CYANOGEN CHLORIDE, STABILIZED	2.3	8	80	X			X		5	20	1.03	k
1741	BORON TRICHLORIDE	2.3	8	2,541	X	X	X	X	X	5	10	1.19	a

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P200		PACKING INSTRUCTION (continued)											P200	
Table 2: LIQUEFIED GASES AND DISSOLVED GASES (continued)														
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ , mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions	
1749	CHLORINE TRIFLUORIDE	2.3	5.1, 8	299	X	X	X	X	X	5	30	1.40	a	
1858	HEXAFLUOROPROPYLENE (REFRIGERANT GAS R 1216)	2.2			X	X	X	X	X	10	22	1.11		
1859	SILICON TETRAFLUORIDE	2.3	8	922	X	X	X	X	X	5	200 300	0.74 1.10	a a	
1860	VINYL FLUORIDE, STABILIZED	2.1			X	X	X	X	X	10	250	0.64	a	
1911	DIBORANE	2.3	2.1	80	X			X		5	250	0.07	d, k, o	
1912	METHYL CHLORIDE AND METHYLENE CHLORIDE MIXTURE	2.1			X	X	X	X	X	10	17	0.81	a	
1952	ETHYLENE OXIDE AND CARBON DIOXIDE MIXTURE with not more than 9% ethylene oxide	2.2			X	X	X	X	X	10	190 250	0.66 0.75		
1958	1,2-DICHLORO-1,1,2,2-TETRAFLUOROETHANE (REFRIGERANT GAS R 114)	2.2			X	X	X	X	X	10	10	1.30		
1959	1,1-DIFLUOROETHYLENE (REFRIGERANT GAS R 1132a)	2.1			X	X	X	X	X	10	250	0.77		
1962	ETHYLENE	2.1			X	X	X	X	X	10	225 300	0.34 0.38		
1965	HYDROCARBON GAS MIXTURE, LIQUEFIED, N.O.S.	2.1			X	X	X	X	X	10			v, z	
1967	INSECTICIDE GAS, TOXIC, N.O.S.	2.3			X	X	X	X	X	5			z	
1968	INSECTICIDE GAS, N.O.S.	2.2			X	X	X	X	X	10			z	
1969	ISOBUTANE	2.1			X	X	X	X	X	10	10	0.49	v	
1973	CHLORODIFLUOROMETHANE AND CHLOROPENTAFLUOROETHANE MIXTURE with fixed boiling point, with approximately 49% chlorodifluoromethane (REFRIGERANT GAS R 502)	2.2			X	X	X	X	X	10	31	1.01		
1974	CHLORODIFLUOROBROMOMETHANE (REFRIGERANT GAS R 12B1)	2.2			X	X	X	X		10	10	1.61		
1975	NITRIC OXIDE AND DINITROGEN TETROXIDE MIXTURE (NITRIC OXIDE AND NITROGEN DIOXIDE MIXTURE)	2.3	5.1, 8	115	X		X	X		5			k, z	
1976	OCTAFLUOROCYCLOBUTANE (REFRIGERANT GAS RC 318)	2.2			X	X	X	X	X	10	11	1.32		
1978	PROPANE	2.1			X	X	X	X	X	10	23	0.43	v	
1982	TETRAFLUOROMETHANE (REFRIGERANT GAS R 14)	2.2			X	X	X	X	X	10	200 300	0.71 0.90		
1983	1-CHLORO-2,2,2-TRIFLUOROETHANE (REFRIGERANT GAS R 133a)	2.2			X	X	X	X	X	10	10	1.18		
1984	TRIFLUOROMETHANE (REFRIGERANT GAS R 23)	2.2			X	X	X	X	X	10	190 250	0.88 0.96		
2035	1,1,1-TRIFLUOROETHANE (REFRIGERANT GAS R 143a)	2.1			X	X	X	X	X	10	35	0.73		
2036	XENON	2.2			X	X	X	X	X	10	130	1.28		
2044	2,2-DIMETHYLPROPANE	2.1			X	X	X	X	X	10	10	0.53		
2073	AMMONIA SOLUTION, relative density less than 0.880 at 15°C in water, with more than 35% but not more than 40% ammonia with more than 40% but not more than 50% ammonia	2.2			X X	X X	X X	X X	X X	5 5	10 12	0.80 0.77	b b	
2188	ARSINE	2.3	2.1	178	X			X		5	42	1.10	d, k	

Part 4 – Packing and tank provisions

P200		PACKING INSTRUCTION (continued)											P200	
Table 2: LIQUEFIED GASES AND DISSOLVED GASES (continued)														
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions	
2189	DICHLOROSILANE	2.3	2.1, 8	314	X	X	X	X	X	5	10 200	0.90 1.08	a a	
2191	SULPHURYL FLUORIDE	2.3		3,020	X	X	X	X	X	5	50	1.10	u	
2192	GERMANE	2.3	2.1	620	X	X	X	X	X	5	250	0.064	d, q, r	
2193	HEXAFLUOROETHANE (REFRIGERANT GAS R 116)	2.2			X	X	X	X	X	10	200	1.13		
2194	SELENIUM HEXAFLUORIDE	2.3	8	50	X			X		5	36	1.46	k	
2195	TELLURIUM HEXAFLUORIDE	2.3	8	25	X			X		5	20	1.00	k	
2196	TUNGSTEN HEXAFLUORIDE	2.3	8	218	X	X	X	X	X	5	10	3.08	a	
2197	HYDROGEN IODIDE, ANHYDROUS	2.3	8	2,860	X	X	X	X	X	5	23	2.25	a, d	
2198	PHOSPHORUS PENTAFLUORIDE	2.3	8	261	X	X	X	X	X	5	200 300	0.90 1.25		
2199	PHOSPHINE	2.3	2.1	20	X			X		5	225 250	0.30 0.45	d, k, q d, k, q	
2200	PROPADIENE, STABILIZED	2.1			X	X	X	X	X	10	22	0.50		
2202	HYDROGEN SELENIDE, ANHYDROUS	2.3	2.1	51	X			X		5	31	1.60	k	
2203	SILANE	2.1			X	X	X	X	X	10	225 250	0.32 0.36	q q	
2204	CARBONYL SULPHIDE	2.3	2.1	1,700	X	X	X	X	X	5	30	0.87	u	
2417	CARBONYL FLUORIDE	2.3	8	360	X	X	X	X	X	5	200 300	0.47 0.70		
2418	SULPHUR TETRAFLUORIDE	2.3	8	40	X			X		5	30	0.91	k, a	
2419	BROMOTRIFLUOROETHYLENE	2.1			X	X	X	X	X	10	10	1.19		
2420	HEXAFLUOROACETONE	2.3	8	470	X	X	X	X	X	5	22	1.08		
2421	NITROGEN TRIOXIDE	2.3	5.1, 8	57	X			X		5			k	
2422	OCTAFLUOROBUT-2-ENE (REFRIGERANT GAS R 1318)	2.2			X	X	X	X	X	10	12	1.34		
2424	OCTAFLUOROPROPANE (REFRIGERANT GAS R 218)	2.2			X	X	X	X	X	10	25	1.04		
2451	NITROGEN TRIFLUORIDE	2.2	5.1		X	X	X	X	X	10	200	0.50		
2452	ETHYLACETYLENE, STABILIZED	2.1			X	X	X	X	X	10	10	0.57	c	
2453	ETHYL FLUORIDE (REFRIGERANT GAS R 161)	2.1			X	X	X	X	X	10	30	0.57		
2454	METHYL FLUORIDE (REFRIGERANT GAS R 41)	2.1			X	X	X	X	X	10	300	0.63		
2455	METHYL NITRITE	2.2	(see special provision 900)											
2517	1-CHLORO-1,1-DIFLUOROETHANE (REFRIGERANT GAS R 142b)	2.1			X	X	X	X	X	10	10	0.99		
2534	METHYLCHLOROSILANE	2.3	2.1, 8	2,810	X	X	X	X	X	5			z	
2548	CHLORINE PENTAFLUORIDE	2.3	5.1, 8	122	X			X		5	13	1.49	a, k	
2599	CHLOROTRIFLUOROMETHANE AND TRIFLUOROMETHANE AZEOTROPIC MIXTURE with approximately 60% chlorotrifluoromethane (REFRIGERANT GAS R 503)	2.2			X	X	X	X	X	10	31 42 100	0.12 0.17 0.64		
2601	CYCLOBUTANE	2.1			X	X	X	X	X	10	10	0.63		
2602	DICHLORODIFLUOROMETHANE AND DIFLUOROETHANE AZEOTROPIC MIXTURE with approximately 74% dichlorodifluoromethane (REFRIGERANT GAS R 500)	2.2			X	X	X	X	X	10	22	1.01		
2676	STIBINE	2.3	2.1	178	X			X		5	200	0.49	k, r	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P200		PACKING INSTRUCTION (continued)												P200
Table 2: LIQUEFIED GASES AND DISSOLVED GASES (continued)														
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions	
2901	BROMINE CHLORIDE	2.3	5.1, 8	290	X	X	X	X	X	5	10	1.50	a	
3057	TRIFLUOROACETYL CHLORIDE	2.3	8	10	X		X	X		5	17	1.17	k	
3070	ETHYLENE OXIDE AND DICHLORODIFLUOROMETHANE MIXTURE with not more than 12.5% ethylene oxide	2.2			X	X	X	X	X	10	18	1.09		
3083	PERCHLORYL FLUORIDE	2.3	5.1	770	X	X	X	X	X	5	33	1.21	u	
3153	PERFLUORO(METHYL VINYL ETHER)	2.1			X	X	X	X	X	10	20	0.75		
3154	PERFLUORO(ETHYL VINYL ETHER)	2.1			X	X	X	X	X	10	10	0.98		
3157	LIQUEFIED GAS, OXIDIZING, N.O.S.	2.2	5.1		X	X	X	X	X	10			z	
3159	1,1,1,2-TETRAFLUOROETHANE (REFRIGERANT GAS R 134a)	2.2			X	X	X	X	X	10	18	1.05		
3160	LIQUEFIED GAS, TOXIC, FLAMMABLE, N.O.S.	2.3	2.1	≤ 5,000	X	X	X	X	X	5			z	
3161	LIQUEFIED GAS, FLAMMABLE, N.O.S.	2.1			X	X	X	X	X	10			z	
3162	LIQUEFIED GAS, TOXIC, N.O.S.	2.3		≤ 5,000	X	X	X	X	X	5			z	
3163	LIQUEFIED GAS, N.O.S.	2.2			X	X	X	X	X	10			z	
3220	PENTAFLUOROETHANE (REFRIGERANT GAS R 125)	2.2			X	X	X	X	X	10	49 35	0.95 0.87		
3252	DIFLUOROMETHANE (REFRIGERANT GAS R 32)	2.1			X	X	X	X	X	10	48	0.78		
3296	HEPTAFLUOROPROPANE (REFRIGERANT GAS R 227)	2.2			X	X	X	X	X	10	13	1.21		
3297	ETHYLENE OXIDE AND CHLOROTETRAFLUOROETHANE MIXTURE with not more than 8.8% ethylene oxide	2.2			X	X	X	X	X	10	10	1.16		
3298	ETHYLENE OXIDE AND PENTAFLUOROETHANE MIXTURE with not more than 7.9% ethylene oxide	2.2			X	X	X	X	X	10	26	1.02		
3299	ETHYLENE OXIDE AND TETRAFLUOROETHANE MIXTURE with not more than 5.6% ethylene oxide	2.2			X	X	X	X	X	10	17	1.03		
3300	ETHYLENE OXIDE AND CARBON DIOXIDE MIXTURE with more than 87% ethylene oxide	2.3	2.1	More than 2,900	X	X	X	X	X	5	28	0.73		
3307	LIQUEFIED GAS, TOXIC, OXIDIZING, N.O.S.	2.3	5.1	≤ 5,000	X	X	X	X	X	5			z	
3308	LIQUEFIED GAS, TOXIC, CORROSIVE, N.O.S.	2.3	8	≤ 5,000	X	X	X	X	X	5			z	
3309	LIQUEFIED GAS, TOXIC, FLAMMABLE, CORROSIVE, N.O.S.	2.3	2.1, 8	≤ 5,000	X	X	X	X	X	5			z	
3310	LIQUEFIED GAS, TOXIC, OXIDIZING, CORROSIVE, N.O.S.	2.3	5.1, 8	≤ 5,000	X	X	X	X	X	5			z	
3318	AMMONIA SOLUTION, relative density less than 0.880 at 15°C in water, with more than 50% ammonia	2.3	8		X	X	X	X		5			b	
3337	REFRIGERANT GAS R 404A	2.2			X	X	X	X	X	10	36	0.82		
3338	REFRIGERANT GAS R 407A	2.2			X	X	X	X	X	10	32	0.94		
3339	REFRIGERANT GAS R 407B	2.2			X	X	X	X	X	10	33	0.93		
3340	REFRIGERANT GAS R 407C	2.2			X	X	X	X	X	10	30	0.95		
3354	INSECTICIDE GAS, FLAMMABLE, N.O.S.	2.1			X	X	X	X	X	10			z	
3355	INSECTICIDE GAS, TOXIC, FLAMMABLE, N.O.S.	2.3	2.1		X	X	X	X	X	5			z	
3374	ACETYLENE, SOLVENT FREE	2.1			X			X		5	60 52		c, p c, p	
3553	DISILANE	2.1			X	X	X	X		10	225	0.39	q	

Part 4 – Packing and tank provisions

P200		PACKING INSTRUCTION <i>(continued)</i>										P200	
Table 3: SUBSTANCES NOT IN CLASS 2													
UN No.	Proper shipping name	Class	Subsidiary hazards	LC ₅₀ , mL/m ³	Cylinders	Tubes	Pressure drums	Bundles of cylinders	MEGCs	Test period, years	Test pressure, bar	Filling ratio	Special packing provisions
1051	HYDROGEN CYANIDE, STABILIZED containing less than 3% water	6.1	3	40	X			X		5	100	0.55	k
1052	HYDROGEN FLUORIDE, ANHYDROUS	8	6.1	1307	X		X	X		5	10	0.84	t, a
1745	BROMINE PENTAFLUORIDE	5.1	6.1, 8	25	X		X	X		5	10	*	k
1746	BROMINE TRIFLUORIDE	5.1	6.1, 8	50	X		X	X		5	10	*	k
2495	IODINE PENTAFLUORIDE	5.1	6.1, 8	120	X		X	X		5	10	*	k

* A minimum ullage of 8% by volume is required.

P201	PACKING INSTRUCTION	P201
This instruction applies to UN Nos. 3167, 3168 and 3169.		
The following packagings are authorized: (1) Cylinders and gas receptacles conforming to the construction, testing and filling requirements approved by the competent authority. (2) The following combination packagings provided that the general provisions of 4.1.1 and 4.1.3 are met: Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Inner packagings: (a) For non-toxic gases, hermetically sealed inner packagings of glass or metal with a maximum capacity of 5 L per package; (b) For toxic gases, hermetically sealed inner packagings of glass or metal with a maximum capacity of 1 L per package. Packagings shall conform to the packing group III performance level.		
P202	PACKING INSTRUCTION	P202
[Reserved]		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P203	PACKING INSTRUCTION	P203
	This instruction applies to class 2 refrigerated liquefied gases.	
	Requirements for closed cryogenic receptacles: (1) The general requirements of 4.1.6.1 shall be met. (2) The requirements of chapter 6.2 shall be met. (3) The closed cryogenic receptacles shall be so insulated that they do not become coated with frost. (4) Test pressure Refrigerated liquids shall be filled in closed cryogenic receptacles with the following minimum test pressures: (a) For closed cryogenic receptacles with vacuum insulation, the test pressure shall not be less than 1.3 times the sum of the maximum internal pressure of the filled receptacle, including during filling and discharge, plus 100 kPa (1 bar); (b) For other closed cryogenic receptacles, the test pressure shall be not less than 1.3 times the maximum internal pressure of the filled receptacle, taking into account the pressure developed during filling and discharge. △ (5) Filling For non-flammable, non-toxic refrigerated liquefied gases the volume of liquid phase at the filling temperature and at a pressure of 100 kPa (1 bar) shall not exceed 98% of the water capacity of the pressure receptacle. △ For flammable refrigerated liquefied gases the gas filled into the receptacle shall remain below the level at which, if the contents were raised to the temperature at which the vapour pressure equalled the opening pressure of the relief valve, the volume of the liquid phase would reach 98% of the water capacity at that temperature. (6) Pressure-relief devices Closed cryogenic receptacles shall be fitted with at least one pressure-relief device. (7) Compatibility Materials used to ensure the leakproofness of the joints or for the maintenance of the closures shall be compatible with the contents. In the case of receptacles intended for the transport of oxidizing gases (i.e. with a subsidiary hazard of 5.1), these materials shall not react with these gases in a dangerous manner. (8) Periodic inspection The periodic inspection and test frequencies of pressure relief valves in accordance with 6.2.1.6.3 shall not exceed five years.	
	Requirements for open cryogenic receptacles: △ Only the following non-oxidizing refrigerated liquefied gases of class 2.2 may be transported in open cryogenic receptacles: UN Nos. 1913, 1951, 1963, 1970, 1977, 2591, 3136 and 3158. For these gases, when used as a coolant, the requirements of 5.5.3 shall apply. Open cryogenic receptacles shall be constructed to meet the following requirements: (1) The receptacles shall be designed, manufactured, tested and equipped in such a way as to withstand all conditions, including fatigue, to which they will be subjected during their normal use and during normal conditions of transport. (2) The capacity shall be not more than 450 L. (3) The receptacle shall have a double wall construction with the space between the inner and outer wall being evacuated (vacuum insulation). The insulation shall prevent the formation of hoar frost on the exterior of the receptacle. (4) The materials of construction shall have suitable mechanical properties at the service temperature. (5) Materials which are in direct contact with the dangerous goods shall not be affected or weakened by the dangerous goods intended to be transported and shall not cause a dangerous effect, e.g. catalysing a reaction or reacting with the dangerous goods. (6) Receptacles of glass double wall construction shall have an outer packaging with suitable cushioning or absorbent materials which withstand the pressures and impacts liable to occur under normal conditions of transport. (7) The receptacle shall be designed to remain in an upright position during transport, e.g. have a base whose smaller horizontal dimension is greater than the height of the centre of gravity when filled to capacity or be mounted on gimbals. (8) The openings of the receptacles shall be fitted with devices allowing gases to escape, preventing any splashing out of liquid, and so configured that they remain in place during transport. (9) Open cryogenic receptacles shall bear the following marks permanently affixed, e.g. by stamping, engraving or etching: (a) The manufacturer's name and address; (b) The model number or name; (c) The serial or batch number; (d) The UN number and proper shipping name of gases for which the receptacle is intended; (e) The capacity of the receptacle in litres.	

Part 4 – Packing and tank provisions

P205	PACKING INSTRUCTION	P205
This instruction applies to UN 3468.		
(1) For metal hydride storage systems, the general packing requirements of 4.1.6.1 shall be met. (2) Only pressure receptacles not exceeding 150 L in water capacity and having a maximum developed pressure not exceeding 25 MPa are covered by this packing instruction. (3) Metal hydride storage systems meeting the applicable requirements for the construction and testing of pressure receptacles containing gas of chapter 6.2 are authorized for the transport of hydrogen only. (4) When steel pressure receptacles or composite pressure receptacles with steel liners are used, only those bearing the "H" mark, in accordance with 6.2.2.9.2(j), shall be used. (5) Metal hydride storage systems shall meet the service conditions, design criteria, rated capacity, type tests, batch tests, routine tests, test pressure, rated charging pressure and provisions for pressure relief devices for transportable metal hydride storage systems specified in ISO 16111:2008 or ISO 16111:2018 and their conformity and approval shall be assessed in accordance with 6.2.2.5. (6) Metal hydride storage systems shall be filled with hydrogen at a pressure not exceeding the rated charging pressure shown in the permanent mark on the system as specified by ISO 16111:2008 or ISO 16111:2018. (7) The periodic test requirements for a metal hydride storage system shall be in accordance with ISO 16111:2008 or ISO 16111:2018 and carried out in accordance with 6.2.2.6, and the interval between periodic inspections shall not exceed five years. See 6.2.2.4 to determine which standard is applicable at the time of periodic inspection and test.		
P206	PACKING INSTRUCTION	P206
This instruction applies to UN Nos. 3500, 3501, 3502, 3503, 3504 and 3505.		
Unless otherwise indicated in these provisions, cylinders and pressure drums conforming to the applicable requirements of Chapter 6.2 are authorized. (1) The general packing requirements of 4.1.6.1 shall be met. (2) The maximum test period for periodic inspection shall be 5 years. (3) Cylinders and pressure drums shall be so filled that at 50°C the non-gaseous phase does not exceed 95% of their water capacity and they are not completely filled at 60°C. When filled, the internal pressure at 65°C shall not exceed the test pressure of the cylinders and pressure drums. The vapour pressures and volumetric expansion of all substances in the cylinders and pressure drums shall be taken into account. For liquids charged with a compressed gas both components – the liquid and the compressed gas – have to be taken into consideration in the calculation of the internal pressure in the pressure receptacle. When experimental data is not available, the following steps shall be carried out: (a) calculation of the vapour pressure of the liquid and of the partial pressure of the compressed gas at 15°C (filling temperature); (b) calculation of the volumetric expansion of the liquid phase resulting from the heating from 15°C to 65°C and calculation of the remaining volume for the gaseous phase; (c) calculation of the partial pressure of the compressed gas at 65°C considering the volumetric expansion of the liquid phase; Note: The compressibility factor of the compressed gas at 15°C and 65°C shall be considered. (d) calculation of the vapour pressure of the liquid at 65°C; (e) the total pressure is the sum of the vapour pressure of the liquid and the partial pressure of the compressed gas at 65°C; (f) consideration of the solubility of the compressed gas at 65°C in the liquid phase. The test pressure of the cylinders or pressure drums shall not be less than the calculated total pressure minus 100 kPa (1bar). If the solubility of the compressed gas in the liquid phase is not known for the calculation, the test pressure can be calculated without taking the gas solubility (subparagraph (f)) into account. (4) The minimum test pressure shall be in accordance with P200 for the propellant but shall not be less than 20 bar. Additional requirement: Cylinders and pressure drums shall not be offered for transport when connected with spray application equipment such as a hose and wand assembly. Special packing provisions: △ PP89 For UN Nos. 3501, 3502, 3503, 3504 and 3505, notwithstanding 4.1.6.1.9.2, non-refillable cylinders used may have a water capacity in L not exceeding 1 000L divided by the test pressure expressed in bars provided capacity and pressure restrictions of the construction standard comply with clause 1 of ISO 11118:2015 + Amd 1:2019, which limits the maximum capacity to 50L. PP97 For fire-extinguishing agents assigned to UN 3500 the maximum test period for periodic inspection shall be 10 years. They may be transported in tubes of a maximum water capacity of 450 L conforming to the applicable requirements of chapter 6.2.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P207	PACKING INSTRUCTION	P207
This instruction applies to UN 1950.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (a) Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2). Packagings shall conform to the packing group II performance level. (b) Rigid outer packagings with a maximum net mass as follows: Fibreboard 55 kg Other than fibreboard 125 kg The provisions of 4.1.1.3 need not be met. The packagings shall be designed and constructed to prevent excessive movement of the aerosols and inadvertent discharge during normal conditions of transport.		
Special packing provision: PP87 For UN 1950, waste aerosols transported in accordance with special provision 327, the packagings shall have a means of retaining any free liquid that might escape during transport, e.g. absorbent material. The packaging shall be adequately ventilated to prevent the creation of dangerous atmospheres and the build-up of pressure.		
P208	PACKING INSTRUCTION	P208
This instruction applies to class 2 adsorbed gases.		
(1) The following packagings are authorized provided the general packing requirements of 4.1.6.1 are met: (a) cylinders constructed as specified in 6.2.2 and in accordance with ISO 11513:2011, ISO 11513:2019, ISO 9809-1:2010 or ISO 9809-1:2019; and (b) cylinders constructed before 1 January 2016 in accordance with 6.2.3 and a specification approved by the competent authorities of the countries of transport and use. (2) The pressure of each filled cylinder shall be less than 101.3 kPa at 20°C and less than 300 kPa at 50°C. (3) The minimum test pressure of the cylinder shall be 21 bar. (4) The minimum burst pressure of the cylinder shall be 94.5 bar. (5) The internal pressure at 65°C of the filled cylinder shall not exceed the test pressure of the cylinder. (6) The adsorbent material shall be compatible with the cylinder and shall not form harmful or dangerous compounds with the gas to be adsorbed. The gas in combination with the adsorbent material shall not affect or weaken the cylinder or cause a dangerous reaction (e.g. a catalyzing reaction). (7) The quality of the adsorbent material shall be verified at the time of each fill to assure the pressure and chemical stability requirements of this packing instruction are met each time an adsorbed gas package is offered for transport. (8) The adsorbent material shall not meet the criteria of any of the classes or divisions in this Code. (9) Requirements for cylinders and closures containing toxic gases with an LC₅₀ less than or equal to 200 mL/m³ (ppm) (see table 1) shall be as follows: (a) Valve outlets shall be fitted with pressure retaining gas-tight plugs or caps having threads matching those of the valve outlets. (b) Each valve shall either be of the packless type with non-perforated diaphragm, or be of a type which prevents leakage through or past the packing. (c) Each cylinder and closure shall be tested for leakage after filling. (d) Each valve shall be capable of withstanding the test pressure of the cylinder and be directly connected to the cylinder by either a taper-thread or other means which meets the requirements of ISO 10692-2:2001. (e) Cylinders and valves shall not be fitted with a pressure relief device. (10) Valve outlets for cylinders containing pyrophoric gases shall be fitted with gas-tight plugs or caps having threads matching those of the valve outlets. (11) The filling procedure shall be in accordance with annex A of ISO 11513:2011 (applicable until 31 December 2024) or annex A of ISO 11513:2019. (12) The maximum period for periodic inspections shall be 5 years. (13) Special packing provisions that are specific to a substance (see table 1). <i>Material compatibility</i> a: Aluminium alloy cylinders shall not be used. d: When steel cylinders are used, only those bearing the “H” mark in accordance with 6.2.2.7.4 (p) are permitted. <i>Gas specific provisions</i> r: The filling ratio of this gas shall be limited such that, if complete decomposition occurs, the pressure does not exceed two thirds of the test pressure of the cylinder. <i>Material compatibility for N.O.S. adsorbed gas entries</i> z: The construction materials of the cylinders and their accessories shall be compatible with the contents and shall not react to form harmful or dangerous compounds therewith.		

Part 4 – Packing and tank provisions

P208		PACKING INSTRUCTION <i>(continued)</i>			P208
Table 1: ADSORBED GASES					
UN No.	Proper shipping name	Class or Division	Subsidiary hazard	LC ₅₀ mL/m ³	Special packing provisions
3510	ADSORBED GAS, FLAMMABLE, N.O.S.	2.1			z
3511	ADSORBED GAS, N.O.S.	2.2			z
3512	ADSORBED GAS, TOXIC, N.O.S.	2.3		≤ 5,000	z
3513	ADSORBED GAS, OXIDIZING, N.O.S.	2.2	5.1		z
3514	ADSORBED GAS, TOXIC, FLAMMABLE, N.O.S.	2.3	2.1	≤ 5,000	z
3515	ADSORBED GAS, TOXIC, OXIDIZING, N.O.S.	2.3	5.1	≤ 5,000	z
3516	ADSORBED GAS, TOXIC, CORROSIVE, N.O.S.	2.3	8	≤ 5,000	z
3517	ADSORBED GAS, TOXIC, FLAMMABLE, CORROSIVE, N.O.S.	2.3	2.1 8	≤ 5,000	z
3518	ADSORBED GAS, TOXIC, OXIDIZING, CORROSIVE, N.O.S.	2.3	5.1 8	≤ 5,000	z
3519	BORON TRIFLUORIDE, ADSORBED	2.3	8	387	a
3520	CHLORINE, ADSORBED	2.3	5.1 8	293	a
3521	SILICON TETRAFLUORIDE, ADSORBED	2.3	8	450	a
3522	ARSINE, ADSORBED	2.3	2.1	20	d
3523	GERMANE, ADSORBED	2.3	2.1	620	d, r
3524	PHOSPHORUS PENTAFLUORIDE, ADSORBED	2.3	8	190	
3525	PHOSPHINE, ADSORBED	2.3	2.1	20	d
3526	HYDROGEN SELENIDE, ADSORBED	2.3	2.1	2	

P300	PACKING INSTRUCTION	P300
This instruction applies to UN 3064.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Combination packagings consisting of inner metal cans of not more than 1 L capacity each and outer wooden boxes (4C1, 4C2, 4D or 4F) containing not more than 5 L of solution.		
Additional provisions:		
1 Metal cans shall be completely surrounded with absorbent cushioning material.		
2 Wooden boxes shall be completely lined with suitable material impervious to water and nitroglycerin.		

P301	PACKING INSTRUCTION	P301
This instruction applies to UN 3165.		
△	The following packagings are authorized, provided that the general provisions of 4.1.1.1, 4.1.1.2, 4.1.1.4, 4.1.1.5, 4.1.1.6 and 4.1.3 are met:	
	(1) Aluminium pressure receptacle made from tubing and having welded heads Primary containment of the fuel within this receptacle shall consist of a welded aluminium bladder having a maximum internal volume of 46 L. The outer receptacle shall have a minimum design gauge pressure of 1,275 kPa and a minimum burst gauge pressure of 2,755 kPa. Each receptacle shall be leak-checked during manufacture and before shipment and shall be found leakproof. The complete inner unit shall be securely packed in non-combustible cushioning material, such as vermiculite, in a strong outer tightly closed metal packaging which will adequately protect all fittings. Maximum quantity of fuel per primary containment and package is 42 L.	
	(2) Aluminium pressure receptacle Primary containment of the fuel within this receptacle shall consist of a welded vapourtight fuel compartment with an elastomeric bladder having a maximum internal volume of 46 L. The pressure receptacle shall have a minimum design gauge pressure of 2,680 kPa and a minimum burst pressure of 5,170 kPa. Each receptacle shall be leak-checked during manufacture and before shipment and shall be securely packed in non-combustible cushioning material such as vermiculite, in a strong outer tightly closed metal packaging which will adequately protect all fittings. Maximum quantity of fuel per primary containment and package is 42 L.	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P302	PACKING INSTRUCTION	P302
This instruction applies to UN 3269.		
The following combination packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G). Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2). Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Inner packagings: The activator (organic peroxide) shall have a maximum quantity of 125 ml per inner packaging if liquid, and 500 g per inner packaging if solid. The base material and the activator shall be each separately packed in inner packagings. The components may be placed in the same outer packaging provided that they will not interact dangerously in the event of a leakage. Packagings shall conform to the packing group II or III performance level according to the criteria for Class 3 applied to the base material.		

■	P303	PACKING INSTRUCTION	P303
This instruction applies to UN 3555.			
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 as well as 4.1.5.12 are met: Plastics drum non-removeable head (1H1) of maximum capacity 250 L.			
Additional requirement: The packagings shall be transported in an upright position.			
Special packing provision: PP26 For UN 3555, packagings shall be lead free.			

	P400	PACKING INSTRUCTION	P400
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. They shall be made of steel and shall be subjected to an initial test and periodic tests every 10 years at a pressure of not less than 1 MPa (10 bar, gauge pressure). During carriage, the liquid shall be under a layer of inert gas with a gauge pressure of not less than 20 kPa (0.2 bar). (2) Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F or 4G), drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1D or 1G) or jerricans (3A1, 3A2, 3B1 or 3B2) enclosing hermetically sealed metal cans with inner packagings of glass or metal, with a capacity of not more than 1 L each, having closures with gaskets. Inner packagings shall have threaded closures or closures physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. Inner packagings shall be cushioned on all sides with dry, absorbent, non-combustible material in a quantity sufficient to absorb the entire contents. Inner packagings shall not be filled to more than 90% of their capacity. Outer packagings shall have a maximum net mass of 125 kg. (3) Steel, aluminium or metal drums (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2), jerricans (3A1, 3A2, 3B1 or 3B2) or boxes (4A, 4B or 4N) with a maximum net mass of 150 kg each with hermetically sealed inner metal cans of not more than 4 L capacity each, with closures fitted with gaskets. Inner packagings shall have threaded closures or closures physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. Inner packagings shall be cushioned on all sides with dry, absorbent, non-combustible material in a quantity sufficient to absorb the entire contents. Each layer of inner packagings shall be separated by a dividing partition in addition to cushioning material. Inner packagings shall not be filled to more than 90% of their capacity.			
Special packing provision: PP86 For UN Nos. 3392 and 3394, air shall be eliminated from the vapour space by nitrogen or other means.			

Part 4 – Packing and tank provisions

P401	PACKING INSTRUCTION	P401
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. They shall be made of steel and subjected to an initial test and periodic tests every 10 years at a pressure of not less than 0.6 MPa (6 bar, gauge pressure). During carriage, the liquid shall be under a layer of inert gas with a gauge pressure of not less than 20 kPa (0.2 bar). (2) Combination packagings Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Inner packagings: Glass, metal or plastics which have threaded closures with a maximum capacity of 1 L. Each inner packaging shall be surrounded by inert cushioning and absorbent material in a quantity sufficient to absorb the entire contents. The maximum net mass per outer packaging shall not exceed 30 kg. Special packing provision: PP31 For UN Nos. 1183, 1242, 1295, 2965 and 2988, packagings shall be hermetically sealed.		
P402	PACKING INSTRUCTION	P402
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. They shall be made of steel and subjected to an initial test and periodic tests every 10 years at a pressure of not less than 0.6 MPa (6 bar, gauge pressure). During carriage, the liquid shall be under a layer of inert gas with a gauge pressure of not less than 20 kPa (0.2 bar). (2) Combination packagings Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Inner packagings with a maximum net mass as follows: Glass 10 kg Metal or plastics 15 kg Each inner packaging shall be fitted with threaded closures. Each inner packaging shall be surrounded by inert cushioning and absorbent material in a quantity sufficient to absorb the entire contents. The maximum net mass per outer packaging shall not exceed 125 kg. (3) Steel drums (1A1) with a maximum capacity of 250 L. (4) Composite packagings consisting of plastics receptacle in a steel or aluminium drum (6HA1 or 6HB1) with a maximum capacity of 250 L. Special packing provision: PP31 For UN Nos. 1389, 1391, 1392, 1420, 1421, 1422, 3148, 3184 (PG II), 3185 (PG II), 3187 (PG II), 3188 (PG II), 3398 (PG I), 3399 (PG I) and 3482, packagings shall be hermetically sealed.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P403		PACKING INSTRUCTION		P403
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.				
Combination packagings			Maximum net mass	
Inner packagings		Outer packagings		
Glass	2 kg	Drums steel (1A1, 1A2) aluminium (1B1, 1B2) other metal (1N1, 1N2) plastics (1H1, 1H2) plywood (1D) fibre (1G)	400 kg	
Plastic	15 kg		400 kg	
Metal	20 kg		400 kg	
Inner packagings shall be hermetically sealed (e.g. by taping or by threaded closures).			400 kg	
			400 kg	
			400 kg	
			400 kg	
			400 kg	
			400 kg	
			400 kg	
		400 kg		
		400 kg		
		400 kg		
		Boxes steel (4A) aluminium (4B) other metal (4N) natural wood (4C1) natural wood with sift-proof walls (4C2) plywood (4D) reconstituted wood (4F) fibreboard (4G) expanded plastics (4H1) solid plastics (4H2)	400 kg	
			400 kg	
			400 kg	
			250 kg	
			250 kg	
			250 kg	
			125 kg	
			125 kg	
			60 kg	
			250 kg	
		Jerricans steel (3A1, 3A2) aluminium (3B1, 3B2) plastics (3H1, 3H2)	120 kg	
			120 kg	
			120 kg	
Single packagings				
Drums steel (1A1, 1A2) aluminium (1B1, 1B2) metal other than steel or aluminium (1N1, 1N2) plastics (1H1, 1H2)			250 kg	
			250 kg	
			250 kg	
			250 kg	
Jerricans steel (3A1, 3A2) aluminium (3B1, 3B2) plastics (3H1, 3H2)			120 kg	
			120 kg	
			120 kg	
Composite packagings				
Plastics receptacle in steel or aluminium drum (6HA1 or 6HB1)			250 kg	
Plastics receptacle in fibre, plastics or plywood drum (6HG1, 6HH1 or 6HD1)			75 kg	
Plastics receptacle in steel, aluminium, wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)			75 kg	
Pressure receptacles, provided that the general provisions of 4.1.3.6 are met				
Special packing provisions:				
PP31 For UN Nos. 1360, 1397, 1402, 1404, 1407, 1409, 1410, 1413, 1414, 1415, 1418, 1419, 1423, 1426, 1427, 1428, 1432, 1433, 1436, 1714, 1870, 2010, 2011, 2012, 2013, 2257, 2463, 2806, 2813, 3131, 3132, 3134, 3135, 3208, 3209, 3395, 3396, 3397, 3401, 3402, 3403 and 3404, packagings shall be hermetically sealed.				
PP83 Deleted				

Part 4 – Packing and tank provisions

P404	PACKING INSTRUCTION	P404
This instruction applies to pyrophoric solids: UN Nos. 1383, 1854, 1855, 2008, 2441, 2545, 2546, 2846, 2881, 3200, 3391 and 3393.		
△	The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Combination packagings: Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2). Inner packagings: Metal receptacles with a maximum net mass of 15 kg each. Inner packagings shall be hermetically sealed. Glass receptacles, with a maximum net mass of 1 kg each, having closures with gaskets, cushioned on all sides and contained in hermetically sealed metal cans. Outer packagings shall have a maximum net mass of 125 kg. Inner packagings shall have threaded closures or closures physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. (2) Metal packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2); Jerricans (3A1, 3A2, 3B1, 3B2). Maximum gross mass: 150 kg (3) Composite packagings: Plastics receptacle in a steel or aluminium drum (6HA1 or 6HB1) Maximum gross mass: 150 kg (4) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. Special packing provisions: PP31 For UN Nos. 1383, 1854, 1855, 2008, 2441, 2545, 2546, 2846, 2881 and 3200, packagings shall be hermetically sealed. PP86 For UN Nos. 3391 and 3393, air shall be eliminated from the vapour space by nitrogen or other means.	

P405	PACKING INSTRUCTION	P405
This instruction applies to UN 1381.		
△	The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) For UN 1381, wet phosphorus: .1 Combination packagings Outer packagings: Boxes (4A, 4B, 4N, 4C1, 4C2, 4D or 4F); maximum net mass: 75 kg Inner packagings: (i) hermetically sealed metal cans, with a maximum net mass of 15 kg; or (ii) glass inner packagings cushioned on all sides with dry, absorbent, non-combustible material in a quantity sufficient to absorb the entire contents with a maximum net mass of 2 kg; or .2 Drums (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2); maximum net mass: 400 kg Jerricans (3A1 or 3B1); maximum net mass: 120 kg. These packagings shall be capable of passing the leakproofness test specified in 6.1.5.4 at the packing group II performance level. (2) For UN 1381, dry phosphorus: .1 When fused, drums (1A2, 1B2 or 1N2) with a maximum net mass of 400 kg; or .2 In projectiles or hard-cased articles when transported without class 1 components, as specified by the competent authority. Special packing provision: PP31 For UN 1381, packagings shall be hermetically sealed.	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P406	PACKING INSTRUCTION	P406
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Combination packagings Outer packagings: (4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2, 1G, 1D, 1H1, 1H2, 3H1 or 3H2) Inner packagings shall be water-resistant. (2) Plastics, plywood or fibreboard drums (1H2, 1D or 1G) or boxes (4A, 4B, 4N, 4C1, 4D, 4F, 4C2, 4G and 4H2) with a water-resistant inner bag, plastics film lining or water-resistant coating. (3) Metal drums (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2), plastics drums (1H1 or 1H2), metal jerricans (3A1, 3A2, 3B1 or 3B2), plastics jerricans (3H1 or 3H2), plastics receptacle in steel or aluminium drums (6HA1 or 6HB1), plastics receptacle in fibre, plastics or plywood drums (6HG1, 6HH1 or 6HD1), plastics receptacle in steel, aluminium, wood, plywood, fibreboard or solid plastics boxes (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2). Additional provisions: 1 Packagings shall be designed and constructed to prevent the loss of water or alcohol content or the content of the phlegmatizer. 2 Packagings shall be so constructed and closed as to avoid an explosive overpressure or pressure build-up of more than 300 kPa (3 bar). 3 The type of packaging and maximum permitted quantity per packaging are limited by the provisions of 2.1.3.4. Special packing provisions: PP24 UN Nos. 2852, 3364, 3365, 3366, 3367, 3368 and 3369 shall not be transported in quantities of more than 500 g per package. PP25 UN 1347 shall not be transported in quantities of more than 15 kg per package. PP26 For UN Nos. 1310, 1320, 1321, 1322, 1344, 1347, 1348, 1349, 1517, 2907, 3317, 3344 and 3376, packagings shall be lead-free. PP31 For UN Nos. 1310, 1320, 1321, 1322, 1336, 1337, 1344, 1347, 1348, 1349, 1354, 1355, 1356, 1357, 1517, 1571, 2555, 2556, 2557, 2852, 3317, 3364, 3365, 3366, 3367, 3368, 3369, 3370 and 3376, packagings shall be hermetically sealed. PP48 For UN 3474, metal packagings shall not be used. Packagings of other material with a small amount of metal, for example metal closures or other metal fittings such as those mentioned in 6.1.4, are not considered metal packagings. PP78 UN 3370 shall not be transported in quantities of more than 11.5 kg per package. PP80 For UN Nos. 2907 and 3344, packagings shall meet the packing group II performance level. Packagings meeting the test criteria of packing group I shall not be used.		
P407	PACKING INSTRUCTION	P407
This instruction applies to UN Nos. 1331, 1944, 1945 and 2254. The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Inner packagings: Matches shall be tightly packed in securely closed inner packagings to prevent accidental ignition under normal conditions of transport. The maximum gross mass of the package shall not exceed 45 kg except for fibreboard boxes which shall not exceed 30 kg. Packagings shall conform to the packing group III performance level. Special packing provision: PP27 UN 1331, Strike-anywhere matches, shall not be packed in the same outer packaging with any other dangerous goods other than safety matches or wax Vesta matches, which shall be packed in separate inner packagings. Inner packagings shall not contain more than 700 strike-anywhere matches.		

Part 4 – Packing and tank provisions

P408	PACKING INSTRUCTION	P408
This instruction applies to UN 3292.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) For cells: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). There shall be sufficient cushioning material to prevent contact between cells and between cells and the internal surfaces of the outer packaging and to ensure that no dangerous movement of the cells within the outer packaging occurs in transport. Packagings shall conform to the packing group II performance level. (2) Batteries may be transported unpacked or in protective enclosures (e.g. fully enclosed or wooden slatted crates). The terminals shall not support the weight of other batteries or materials packed with the batteries. Packagings need not meet the requirements of 4.1.1.3. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).		
Additional requirement: Cells and batteries shall be protected against short circuit and shall be isolated in such a manner as to prevent short circuits.		
P409	PACKING INSTRUCTION	P409
This instruction applies to UN Nos. 2956, 3242 and 3251.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Fibre drum (1G) which may be fitted with a liner or coating; maximum net mass: 50 kg. (2) Combination packagings: Fibreboard box (4G) with a single inner plastic bag; maximum net mass: 50 kg. (3) Combination packagings: Fibreboard box (4G) or fibre drum (1G) with inner plastic packagings each containing a maximum of 5 kg; maximum net mass: 25 kg.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P410		PACKING INSTRUCTION		P410
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.				
Combination packagings			Maximum net mass	
Inner packagings		Outer packagings	Packing group II	Packing group III
Glass	10 kg	Drums		
Plastics ¹	30 kg	steel (1A1, 1A2)	400 kg	400 kg
Metal	40 kg	aluminium (1B1, 1B2)	400 kg	400 kg
Paper ^{1, 2}	10 kg	other metal (1N1, 1N2)	400 kg	400 kg
Fibre ^{1, 2}	10 kg	plastics (1H1, 1H2)	400 kg	400 kg
		plywood (1D)	400 kg	400 kg
		fibre (1G) ¹	400 kg	400 kg
		Boxes		
		steel (4A)	400 kg	400 kg
		aluminium (4B)	400 kg	400 kg
		other metal (4N)	400 kg	400 kg
		natural wood (4C1)	400 kg	400 kg
		natural wood with sift-proof walls (4C2)	400 kg	400 kg
		plywood (4D)	400 kg	400 kg
		reconstituted wood (4F)	400 kg	400 kg
		fibreboard (4G) ¹	400 kg	400 kg
		expanded plastics (4H1)	60 kg	60 kg
		solid plastics (4H2)	400 kg	400 kg
		Jerricans		
		steel (3A1, 3A2)	120 kg	120 kg
		aluminium (3B1, 3B2)	120 kg	120 kg
		plastics (3H1, 3H2)	120 kg	120 kg
¹ Packagings shall be sift-proof.				
² These inner packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).				
Single packagings				
Drums				
steel (1A1 or 1A2)			400 kg	400 kg
aluminium (1B1 or 1B2)			400 kg	400 kg
metal other than steel or aluminium (1N1 or 1N2)			400 kg	400 kg
plastics (1H1 or 1H2)			400 kg	400 kg
Jerricans				
steel (3A1 or 3A2)			120 kg	120 kg
aluminium (3B1 or 3B2)			120 kg	120 kg
plastics (3H1 or 3H2)			120 kg	120 kg
Boxes				
steel (4A) ³			400 kg	400 kg
aluminium (4B) ³			400 kg	400 kg
other metal (4N) ³			400 kg	400 kg
natural wood (4C1) ³			400 kg	400 kg
natural wood with sift-proof walls (4C2) ³			400 kg	400 kg
plywood (4D) ³			400 kg	400 kg
reconstituted wood (4F) ³			400 kg	400 kg
fibreboard (4G) ³			400 kg	400 kg
solid plastics (4H2) ³			400 kg	400 kg
Bags				
Bags (5H3, 5H4, 5L3, 5M2) ^{3, 4}			50 kg	50 kg
Composite packagings				
Plastics receptacle in steel, aluminium, plywood, fibre or plastics drum (6HA1, 6HB1, 6HG1, 6HD1 or 6HH1)			400 kg	400 kg
Plastics receptacle in steel or aluminium crate or box, wooden box, plywood box, fibreboard box or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)			75 kg	75 kg
Glass receptacle in steel, aluminium, plywood or fibre drum (6PA1, 6PB1, 6PD1 or 6PG1) or in steel, aluminium, wooden, wickerwork hamper or fibreboard box (6PA2, 6PB2, 6PC, 6PD2 or 6PG2) or in expanded or solid plastics packaging (6PH1 or 6PH2)			75 kg	75 kg
³ These packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).				
⁴ For packing group II substances, these packagings may only be used when transported in a closed cargo transport unit.				
Pressure receptacles , provided that the general provisions of 4.1.3.6 are met				

Part 4 – Packing and tank provisions

P410	PACKING INSTRUCTION (continued)	P410
<i>Special packing provisions:</i>		
PP31	For UN Nos. 1326, 1339, 1340, 1341, 1343, 1352, 1358, 1373, 1374, 1378, 1379, 1382, 1384, 1385, 1390, 1393, 1394, 1395, 1396, 1398, 1400, 1401, 1402, 1405, 1409, 1417, 1418, 1431, 1436, 1437, 1871, 1923, 1929, 2004, 2008, 2318, 2545, 2546, 2624, 2805, 2813, 2830, 2835, 2844, 2881, 2940, 3078, 3088, 3131, 3132, 3134, 3135, 3170, 3182, 3189, 3190, 3205, 3206, 3208, 3209, 3395, 3396 and 3397, packagings shall be hermetically sealed.	
PP39	For UN 1378, for metal packagings a venting device is required.	
PP40	For the following UN Nos., falling in PG II, bags are not allowed: 1326, 1340, 1352, 1358, 1374, 1378, 1382, 1390, 1393, 1394, 1395, 1396, 1400, 1401, 1402, 1405, 1409, 1417, 1418, 1436, 1437, 1871, 2624, 2805, 2813, 2830, 2835, 3078, 3131, 3132, 3134, 3170, 3182, 3208 and 3209.	
PP83	<i>Deleted</i>	
PP100	For UN 2950 flexible, fibreboard or wooden packagings shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.	

P411	PACKING INSTRUCTION	P411
This instruction applies to UN 3270.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2), provided that explosion is not possible by reason of increased internal pressure. The maximum net mass shall not exceed 30 kg.		

P412	PACKING INSTRUCTION	P412
This instruction applies to UN No. 3527.		
The following combination packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) Outer packagings: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2) Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2); (2) Inner packagings: (a) The activator (organic peroxide) shall have a maximum quantity of 125 ml per inner packaging if liquid, and 500 g per inner packaging if solid. (b) The base material and the activator shall be each separately packed in inner packagings. The components may be placed in the same outer packaging provided that they will not interact dangerously in the event of a leakage. Packagings shall conform to the packing group II or III performance level according to the criteria for class 4.1 applied to the base material.		

P500	PACKING INSTRUCTION	P500
This instruction applies to UN 3356.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group II performance level. The generator(s) shall be transported in a package which meets the following requirements when one generator in the package is actuated: (1) Other generators in the package will not be actuated; (2) Packaging material will not ignite; and (3) The outside surface temperature of the completed package shall not exceed 100°C.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P501		PACKING INSTRUCTION		P501
This instruction applies to UN 2015.				
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.				
	Combination packagings	Inner packagings maximum capacity	Outer packagings maximum net mass	
△	Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4H2) or drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D) or jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2) with glass, plastics or metal inner packagings	5 L	125 kg	
△	Fibreboard box (4G) or fibre drum (1G), with plastics or metal inner packagings each in a plastics bag	2 L	50 kg	
Single packagings			Maximum capacity	
Drums steel (1A1) aluminium (1B1) metal other than steel or aluminium (1N1) plastics (1H1)			250 L 250 L 250 L 250 L	
Jerricans steel (3A1) aluminium (3B1) plastics (3H1)			60 L 60 L 60 L	
Composite packagings Plastics receptacle in steel or aluminium drum (6HA1, 6HB1) Plastics receptacle in fibre, plastics or plywood drum (6HG1, 6HH1, 6HD1) Plastics receptacle in steel or aluminium crate or box or plastics receptacle in wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2) Glass receptacle in steel, aluminium, fibre or plywood drum (6PA1, 6PB1, 6PD1 or 6PG1) or in a steel, aluminium, wood or fibreboard box or in wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 or 6PD2) or in expanded or solid plastics packaging (6PH1 or 6PH2)			250 L 250 L 60 L 60 L	
Additional provisions:				
1 Packagings shall have a minimum ullage of 10%.				
2 Packagings shall be vented.				

Part 4 – Packing and tank provisions

P502	PACKING INSTRUCTION		P502
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.			
Combination packagings			Maximum net mass
Inner packagings		Outer packagings	
Glass	5 L	Drums	
Metal	5 L	steel (1A1, 1A2)	125 kg
Plastic	5 L	aluminium (1B1, 1B2)	125 kg
		other metal (1N1, 1N2)	125 kg
		plywood (1D)	125 kg
		fibre (1G)	125 kg
		plastics (1H1, 1H2)	125 kg
		Boxes	
		steel (4A)	125 kg
		aluminium (4B)	125 kg
		other metal (4N)	125 kg
		natural wood (4C1)	125 kg
		natural wood with sift-proof walls (4C2)	125 kg
		plywood (4D)	125 kg
		reconstituted wood (4F)	125 kg
		fibreboard (4G)	125 kg
		expanded plastics (4H1)	60 kg
		solid plastics (4H2)	125 kg
Single packagings			Maximum capacity
Drums			
steel (1A1)			250 L
aluminium (1B1)			250 L
plastics (1H1)			250 L
Jerricans			
steel (3A1)			60 L
aluminium (3B1)			60 L
plastics (3H1)			60 L
Composite packagings			
Plastics receptacle in steel or aluminium drum (6HA1, 6HB1)			250 L
Plastics receptacle in fibre, plastics or plywood drum (6HG1, 6HH1, 6HD1)			250 L
Plastics receptacle in steel or aluminium crate or box or plastics receptacle in wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)			60 L
Glass receptacle in steel, aluminium, fibre or plywood drum (6PA1, 6PB1, 6PD1 or 6PG1) or in a steel, aluminium, wood or fibreboard box or in wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 or 6PD2) or in expanded or solid plastics packaging (6PH1 or 6PH2)			60 L
Special packing provision:			
PP28 For UN No. 1873, parts of packagings which are in direct contact with perchloric acid shall be constructed of glass or plastics.			

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P503 PACKING INSTRUCTION P503		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.		
Combination packagings		Maximum net mass
Inner packagings	Outer packagings	
Glass 5 kg	Drums	
Metal 5 kg	steel (1A1, 1A2)	125 kg
Plastic 5 kg	aluminium (1B1, 1B2)	125 kg
	other metal (1N1, 1N2)	125 kg
	fibre (1G)	125 kg
	plywood (1D)	125 kg
	plastics (1H1, 1H2)	125 kg
	Boxes	
	steel (4A)	125 kg
	aluminium (4B)	125 kg
	other metal (4N)	125 kg
	natural wood (4C1)	125 kg
	natural wood with sift-proof walls (4C2)	125 kg
	plywood (4D)	125 kg
	reconstituted wood (4F)	125 kg
	fibreboard (4G)	40 kg
	expanded plastics (4H1)	60 kg
	solid plastics (4H2)	125 kg
Single packagings		Maximum net mass
Metal drums (1A1, 1A2, 1B1, 1B2, 1N1 or 1N2)		250 kg
Fibreboard (1G) or plywood drums (1D) fitted with inner liners		200 kg

P504 PACKING INSTRUCTION P504		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.		
Combination packagings		Maximum net mass
(1) Outer packagings: (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2) Inner packagings: Glass receptacles with a maximum capacity of 5 L		75 kg
(2) Outer packagings: (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2) Inner packagings: Plastics receptacles with a maximum capacity of 30 L		75 kg
(3) Outer packagings: (1G, 4F or 4G) Inner packagings: Metal receptacles with a maximum capacity of 40 L		125 kg
(4) Outer packagings: (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 4A, 4B, 4N, 4C1, 4C2, 4D, 4H2) Inner packagings: Metal receptacles with a maximum capacity of 40 L		225 kg
Single packagings		Maximum capacity
Drums		
steel, non-removable head (1A1)		250 L
aluminium, non-removable head (1B1)		250 L
metal, other than steel or aluminium, non-removable head (1N1)		250 L
plastics, non-removable head (1H1)		250 L
Jerricans		
steel, non-removable head (3A1)		60 L
aluminium, non-removable head (3B1)		60 L
plastics, non-removable head (3H1)		60 L
Composite packagings		
Plastics receptacle in steel or aluminium drum (6HA1, 6HB1)		250 L
Plastics receptacle in fibre, plastics or plywood drum (6HG1, 6HH1, 6HD1)		120 L
Plastics receptacle in steel or aluminium crate or box or plastics receptacle in wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)		60 L
Glass receptacle in steel, aluminium, fibre or plywood drum (6PA1, 6PB1, 6PD1 or 6PG1) or in a steel, aluminium, wood or fibreboard box or in wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 or 6PD2) or in expanded or solid plastics packaging (6PH1 or 6PH2)		60 L
Special packing provisions:		
PP10 For UN Nos. 2014 and 3149, the packaging shall be vented.		
PP31 For UN 2626, packagings shall be hermetically sealed.		

Part 4 – Packing and tank provisions

P505 PACKING INSTRUCTION P505		
This instruction applies to UN 3375.		
The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met.		
△ △ △ △	Combination packagings	
	Inner packagings	Outer packagings
	Glass 5 L	Boxes
	Plastics 5 L	aluminium (4B)
	Metal 5 L	natural wood, ordinary (4C1)
		natural wood, sift-proof walls (4C2)
		plywood (4D)
		fibreboard (4G)
		plastics, solid (4H2)
		Drums
		aluminium, removable head (1B2)
		fibre (1G)
		other metal, removable head (1N2)
		plastics, removable head (1H2)
		plywood (1D)
		Jerricans
		aluminium, removable head (3B2)
		plastics, removable head (3H2)
Single packagings		
Drums		
aluminium (1B1, 1B2)		250 L
plastics (1H1, 1H2)		250 L
Jerricans		
aluminium (3B1, 3B2),		60 L
plastics (3H1, 3H2)		60 L
Composite packagings		
plastics receptacle with outer aluminium drum (6HB1)		250 L
plastics receptacle with outer fibre, plastics or plywood drum (6HG1, 6HH1, 6HD1)		250 L
plastics receptacle with outer aluminium crate or box or plastics receptacle with outer wooden, plywood, fibreboard or solid plastics box (6HB2, 6HC, 6HD2, 6HG2, 6HH2)		60 L
glass receptacle with outer aluminium, fibre or plywood drum (6PB1, 6PG1, 6PD1) or with outer expanded plastics or solid plastics receptacles (6PH1, 6PH2) or with outer aluminium crate or box or with outer wooden or fibreboard box or with outer wickerwork hamper (6PB2, 6PC, 6PG2, 6PD2)		60 L

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P520	PACKING INSTRUCTION								P520
This instruction applies to organic peroxides of class 5.2 and self-reactive substances of class 4.1.									
The packagings listed below are authorized provided the general provisions of 4.1.1 and 4.1.3 and special provisions of 4.1.7 are met.									
The packing methods are designated OP1 to OP8. The packing methods appropriate for the individual currently assigned organic peroxides and self-reactive substances are listed in 2.4.2.3.2.3 and 2.5.3.2.4. The quantities specified for each packing method are the maximum quantities authorized per package. The following packagings are authorized:									
△	(1) Combination packagings with outer packagings comprising boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1 and 4H2), drums (1A1, 1A2, 1B1, 1B2, 1G, 1H1, 1H2 and 1D) and jerricans (3A1, 3A2, 3B1, 3B2, 3H1 and 3H2);								
	(2) Single packagings consisting of drums (1A1, 1A2, 1B1, 1B2, 1G, 1H1, 1H2 and 1D) and jerricans (3A1, 3A2, 3B1, 3B2, 3H1 and 3H2);								
	(3) Composite packagings with plastics inner receptacles (6HA1, 6HA2, 6HB1, 6HB2, 6HC, 6HD1, 6HD2, 6HG1, 6HG2, 6HH1 and 6HH2).								
△	The maximum quantities per packaging/package for packing methods OP1 to OP8 are:								
△		OP1	OP2 ¹	OP3	OP4 ¹	OP5	OP6	OP7	OP8
△	Maximum net mass (kg) for solids and for combination packagings (liquid and solid)	0.5	0.5/10	5	5/25	25	50	50	400 ²
	Maximum contents in litres for liquids ³	0.5	–	5	–	30	60	60	225 ⁴
¹ If two values are given, the first applies to the maximum net mass per inner packaging and the second to the maximum net mass of the complete package.									
² 60 kg for jerricans/200 kg for boxes and, for solids, 400 kg in combination packagings with outer packagings comprising boxes (4C1, 4C2, 4D, 4F, 4G, 4H1 and 4H2) and with inner packagings of plastics or fibre with a maximum net mass of 25 kg.									
³ Viscous liquids shall be treated as solids when they do not meet the criteria provided in the definition for liquids presented in 1.2.1.									
⁴ 60 L for jerricans.									
Additional provisions:									
1	Metal packagings, including inner packagings of combination packagings and outer packagings of combination or composite packagings, may only be used for packing methods OP7 and OP8.								
2	In combination packagings, glass receptacles may only be used as inner packagings with a maximum content of 0.5 kg for solids or 0.5 L for liquids.								
3	In combination packagings, cushioning materials shall not be readily combustible.								
4	The packaging of an organic peroxide or self-reactive substance required to bear an EXPLOSIVE subsidiary hazard label (Model No. 1, see 5.2.2.2.2) shall also comply with the provisions given in 4.1.5.10 and 4.1.5.11.								

Part 4 – Packing and tank provisions

P520	PACKING INSTRUCTION (continued)	P520
Special packing provisions:		
PP21	For certain self-reactive substances of types B or C, UN Nos. 3221, 3222, 3223, 3224, 3231, 3232, 3233 and 3234, a smaller packaging than that allowed by packing methods OP5 or OP6 respectively shall be used (see 4.1.7 and 2.4.2.3.2.3).	
PP22	UN 3241, 2-bromo-2-nitropropane-1,3-diol, shall be packed in accordance with packing method OP6.	
PP94	Very small amounts of energetic samples of section 2.0.4.3 may be carried under UN 3223 or UN 3224, as appropriate, provided that: .1 only combination packaging with outer packaging comprising boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1 and 4H2) are used; .2 the samples are carried in microtiter plates or multi-titer plates made of plastics, glass, porcelain or stoneware as inner packaging; .3 the maximum amount per individual inner cavity does not exceed 0.01 g for solids or 0.01 mL for liquids; .4 the maximum net quantity per outer packaging is 20 g for solids or 20 mL for liquids, or in the case of mixed packing the sum of grams and millilitres does not exceed 20; and .5 when dry ice or liquid nitrogen is optionally used as a coolant for quality control measures, the requirements of 5.5.3 are complied with. Interior supports shall be provided to secure the inner packagings in their original position. The inner and outer packagings shall maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost.	
PP95	Small amounts of energetic samples of section 2.0.4.3 may be carried under UN 3223 or UN 3224, as appropriate, provided that: .1 the outer packaging consist only of corrugated fibreboard of type 4G having minimum dimensions of 60 cm (length) by 40.5 cm (width) by 30 cm (height) and minimum wall thickness of 1.3 cm; .2 the individual substance is contained in an inner packaging of glass or plastics of maximum capacity 30 mL placed in an expandable polyethylene foam matrix of at least 130 mm thickness having a density of 18 ± 1 g/L; .3 within the foam carrier, inner packagings are segregated from each other by a minimum distance of 40 mm and from the wall of the outer packaging by a minimum distance of 70 mm. The package may contain up to two layers of such foam matrices, each carrying up to 28 inner packagings; .4 the maximum content of each inner packaging does not exceed 1 g for solids or 1 mL for liquids; .5 the maximum net quantity per outer packaging is 56 g for solids or 56 mL for liquids, or in the case of mixed packing the sum of grams and millilitres does not exceed 56; and .6 when dry ice or liquid nitrogen is optionally used as a coolant for quality control measures, the requirements of 5.5.3 are complied with. Interior supports shall be provided to secure the inner packagings in their original position. The inner and outer packagings shall maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost.	
P600	PACKING INSTRUCTION	P600
This instruction applies to UN Nos. 1700, 2016 and 2017.		
△	The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2). Outer packagings shall meet the packing group II performance level. Articles shall be individually packaged and separated from each other using partitions, dividers, inner packagings or cushioning material to prevent inadvertent discharge during normal conditions of transport. Maximum net mass: 75 kg	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P601	PACKING INSTRUCTION	P601
	The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met and the packagings are hermetically sealed: (1) Combination packagings with a maximum gross mass of 15 kg, consisting of: (a) one or more glass inner packaging(s) with a maximum net quantity of 1 L each and filled to not more than 90% of their capacity; the closure(s) of which shall be physically held in place by any means capable of preventing back-off or loosening by impact or vibration during transport, individually placed in (b) metal receptacles together with cushioning and absorbent material sufficient to absorb the entire contents of the glass inner packaging(s), further packed in (c) 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings. (2) Combination packagings consisting of metal or plastics inner packagings not exceeding 5 L in capacity individually packed with absorbent material sufficient to absorb the contents and inert cushioning material in 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings with a maximum gross mass of 75 kg. Inner packagings shall not be filled to more than 90% of their capacity. The closure of each inner packaging shall be physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. (3) Packagings consisting of: Outer packagings: Steel or plastics drums, (1A1, 1A2, 1H1 or 1H2), tested in accordance with the test provisions in 6.1.5 at a mass corresponding to the mass of the assembled package either as a packaging intended to contain inner packagings, or as a single packaging intended to contain solids or liquids, and marked accordingly. Inner packagings: Drums and composite packagings (1A1, 1B1, 1N1, 1H1 or 6HA1), meeting the provisions of chapter 6.1 for single packagings, subject to the following conditions: .1 the hydraulic pressure test shall be conducted at a pressure of at least 3 bar (gauge pressure); .2 the design and production leakproofness tests shall be conducted at a test pressure of 0.30 bar; .3 they shall be isolated from the outer drum by the use of inert shock-mitigating cushioning material which surrounds the inner packaging on all sides; .4 their capacity shall not exceed 125 L; .5 closures shall be of a screw-cap type that are: (i) physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport; and (ii) provided with a cap seal. .6 The outer and inner packagings shall be subjected periodically to a leakproofness test according to .2 at intervals of not more than two and a half years; and .7 The outer and inner packagings shall bear in clearly legible and durable characters: (i) the date (month, year) of the initial testing and the latest periodic test; (ii) the name or authorized symbol of the party performing the tests and inspections. (4) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. They shall be subjected to an initial test and periodic tests every 10 years at a pressure of not less than 1 MPa (10 bar) (gauge pressure). Pressure receptacles may not be equipped with any pressure relief device. Each pressure receptacle containing a toxic by inhalation liquid with an LC₅₀ less than or equal to 200 mL/m³ (ppm) shall be closed with a plug or valve conforming to the following: (a) Each plug or valve shall have a taper-threaded connection directly to the pressure receptacle and be capable of withstanding the test pressure of the pressure receptacle without damage or leakage; (b) Each valve shall be of the packless type with non-perforated diaphragm, except that, for corrosive materials, a valve may be of the packed type with an assembly made gas-tight by means of a seal cap with gasket joint attached to the valve body or the pressure receptacle to prevent loss of material through or past the packing; (c) Each valve outlet shall be sealed by a threaded cap or threaded solid plug and inert gasket material; (d) The materials of construction for the pressure receptacle, valves, plugs, outlet caps, luting and gaskets shall be compatible with each other and with the lading. Each pressure receptacle with a wall thickness at any point of less than 2.0 mm and each pressure receptacle that does not have fitted valve protection shall be transported in an outer packaging. Pressure receptacles shall not be manifolded or interconnected.	

Part 4 – Packing and tank provisions

P602	PACKING INSTRUCTION	P602
	The following packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met and the packagings are hermetically sealed: (1) Combination packagings with a maximum gross mass of 15 kg, consisting of: (a) one or more glass inner packaging(s) with a maximum net quantity of 1 L each and filled to not more than 90% of their capacity, the closure(s) of which shall be physically held in place by any means capable of preventing back-off or loosening by impact or vibration during transport, individually placed in (b) metal receptacles together with cushioning and absorbent material sufficient to absorb the entire contents of the glass inner packaging(s), further packed in (c) 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings. (2) Combination packagings consisting of metal or plastics inner packagings individually packed with absorbent material sufficient to absorb the contents and inert cushioning material in 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings with a maximum gross mass of 75 kg. Inner packagings shall not be filled to more than 90% of their capacity. The closure of each inner packaging shall be physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. Inner packagings shall not exceed 5 L in capacity. (3) Drums and composite packagings (1A1, 1B1, 1N1, 1H1, 6HA1 or 6HH1), subject to the following conditions: .1 the hydraulic pressure test shall be conducted at a pressure of at least 3 bar (gauge pressure); .2 the design and production leakproofness tests shall be conducted at a test pressure of 0.30 bar; and .3 closures shall be of a screw-cap type that are: (i) physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport; and (ii) provided with a cap seal. (4) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. They shall be subjected to an initial test and periodic tests every 10 years at a pressure of not less than 1 MPa (10 bar) (gauge pressure). Pressure receptacles may not be equipped with any pressure relief device. Each pressure receptacle containing a toxic by inhalation liquid with an LC₅₀ less than or equal to 200 mL/m³ (ppm) shall be closed with a plug or valve conforming to the following: (a) Each plug or valve shall have a taper-threaded connection directly to the pressure receptacle and be capable of withstanding the test pressure of the pressure receptacle without damage or leakage; (b) Each valve shall be of the packless type with non-perforated diaphragm, except that, for corrosive materials, a valve may be of the packed type with an assembly made gas-tight by means of a seal cap with gasket joint attached to the valve body or the pressure receptacle to prevent loss of material through or past the packing; (c) Each valve outlet shall be sealed by a threaded cap or threaded solid plug and inert gasket material; (d) The materials of construction for the pressure receptacle, valves, plugs, outlet caps, luting and gaskets shall be compatible with each other and with the lading. Each pressure receptacle with a wall thickness at any point of less than 2.0 mm and each pressure receptacle that does not have fitted valve protection shall be transported in an outer packaging. Pressure receptacles shall not be manifolded or interconnected.	

P603	PACKING INSTRUCTION	P603
	This instruction applies to UN 3507. The following packagings are authorized provided that the general provisions of 4.1.1 and 4.1.3 and the special packing provisions of 4.1.9.1.2, 4.1.9.1.4 and 4.1.9.1.7 are met: Packagings consisting of: (a) Metal or plastics primary receptacle(s); in (b) Leakproof rigid secondary packaging(s); in (c) A rigid outer packaging: - Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); - Boxes (4A, 4B, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); - Jerricans (3A2, 3B2, 3H2). Additional requirements: 1 Primary inner receptacles shall be packed in secondary packagings in a way that, under normal conditions of transport, they cannot break, be punctured or leak their contents into the secondary packaging. Secondary packagings shall be secured in outer packagings with suitable cushioning material to prevent movement. If multiple primary receptacles are placed in a single secondary packaging, they shall be either individually wrapped or separated so as to prevent contact between them. 2 The contents shall comply with the provisions of 2.7.2.4.5.2. 3 The provisions of 6.4.4 shall be met. 4 In the case of fissile-excepted material, limits specified in 2.7.2.3.5 shall be met.	

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Chapter 4.1 – Use of packagings, including IBCs and large packagings

P620	PACKING INSTRUCTION	P620
This instruction applies to UN Nos. 2814 and 2900.		
The following packagings are authorized, provided the special packing provisions of 4.1.8 are met: Packagings meeting the provisions of chapter 6.3 and approved accordingly consisting of: .1 Inner packagings comprising: (i) leakproof primary receptacle(s); (ii) a leakproof secondary packaging; (iii) other than for solid infectious substances, an absorbent material in sufficient quantity to absorb the entire contents placed between the primary receptacle(s) and the secondary packaging; if multiple primary receptacles are placed in a single secondary packaging, they shall be either individually wrapped or separated so as to prevent contact between them; .2 A rigid outer packaging: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). The smallest external dimension shall be not less than 100 mm.		
Additional provisions: △ 1 Inner packagings containing infectious substances shall not be consolidated with inner packagings containing unrelated types of goods. Complete packages may be overpacked in accordance with the provisions of 1.2.1 and 5.1.2: such an overpack may contain dry ice. When dry ice or other refrigerants presenting a risk of asphyxiation are used as a coolant, the requirements of 5.5.3 shall apply. 2 Other than for exceptional consignments, such as whole organs which require special packaging, the following additional provisions shall apply: (a) <i>Substances consigned at ambient temperatures or at a higher temperature.</i> Primary receptacles shall be of glass, metal or plastics. Positive means of ensuring a leakproof seal shall be provided, e.g. a heat seal, a skirted stopper or a metal crimp seal. If screw caps are used, they shall be secured by positive means, e.g. tape, paraffin sealing tape or a manufactured locking closure; △ (b) <i>Substances consigned refrigerated or frozen.</i> Ice, dry ice or other refrigerant shall be placed around the secondary packaging(s) or alternatively in an overpack with one or more complete packages marked in accordance with 6.3.3. Interior supports shall be provided to secure secondary packaging(s) or packages in position after the ice or dry ice has dissipated. When dry ice or other refrigerants presenting a risk of asphyxiation are used as a coolant, the requirements of 5.5.3 shall apply. If ice is used, the outer packaging or overpack shall be leakproof. If dry ice is used, the outer packaging or overpack shall permit the release of carbon dioxide gas. The primary receptacle and the secondary packaging shall maintain their integrity at the temperature of the refrigerant used; △ (c) <i>Substances consigned in liquid nitrogen.</i> When liquid nitrogen is used as a coolant, the requirements of 5.5.3 shall apply. Plastics primary receptacles capable of withstanding very low temperature shall be used. The secondary packaging shall also be capable of withstanding very low temperatures, and in most cases will need to be fitted over the primary receptacle individually. Provisions for the consignment of liquid nitrogen shall also be fulfilled. The primary receptacle and the secondary packaging shall maintain their integrity at the temperature of the liquid nitrogen. (d) Lyophilized substances may also be transported in primary receptacles that are flame-sealed glass ampoules or rubber-stoppered glass vials fitted with metal seals. 3 Whatever the intended temperature of the consignment, the primary receptacle or the secondary packaging shall be capable of withstanding, without leakage, an internal pressure producing a pressure differential of not less than 95 kPa. This primary receptacle or secondary packaging shall also be capable of withstanding temperatures in the range –40°C to +55°C. 4 Other dangerous goods shall not be packed in the same packaging as class 6.2 infectious substances unless they are necessary for maintaining the viability, stabilizing or preventing degradation or neutralizing the hazards of the infectious substances. A quantity of 30 mL or less of dangerous goods included in classes 3, 8 or 9 may be packed in each primary receptacle containing infectious substances. These small quantities of dangerous goods of classes 3, 8 or 9 are not subject to any additional provisions of this Code when packed in accordance with this packing instruction. 5 Alternative packagings for the transport of animal material may be authorized by the competent authority in accordance with the provisions of 4.1.3.7.		

Part 4 – Packing and tank provisions

P621	PACKING INSTRUCTION		P621
This instruction applies to UN 3291.			
The following packagings are authorized provided that the general provisions of 4.1.1 except 4.1.1.15 and 4.1.3 are met:			
(1) Provided that there is sufficient absorbent material to absorb the entire amount of liquid present and the packaging is capable of retaining liquids:			
Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G);			
Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2);			
Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2).			
Packagings shall conform to the packing group II performance level for solids.			
(2) For packages containing larger quantities of liquid:			
Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G);			
Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2);			
Composite packagings (6HA1, 6HB1, 6HG1, 6HH1, 6HD1, 6HA2, 6HB2, 6HC, 6HD2, 6HG2, 6HH2, 6PA1, 6PB1, 6PG1, 6PD1, 6PH1, 6PH2, 6PA2, 6PB2, 6PC, 6PG2 or 6PD2).			
Packagings shall conform to the packing group II performance level for liquids.			
Additional requirement:			
Packagings intended to contain sharp objects such as broken glass and needles shall be resistant to puncture and retain liquids under the performance test conditions in Chapter 6.1.			

P622	PACKING INSTRUCTION		P622
This instruction applies to waste of UN 3549 transported for disposal.			
The following large packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met:			
Inner packagings		Intermediate packagings	Outer packagings
metal plastics		metal plastics	Boxes steel (4A) aluminium (4B) plywood (4D) fibreboard (4G) other metal (4N) plastics, solid (4H2) Drums steel (1A2) aluminium (1B2) plywood (1D) fibre (1G) other metal (1N2) plastics (1H2) Jerricans steel (3A2) aluminium (3B2) plastics (3H2)
The outer packaging shall conform to the packing group I performance level for solids.			
Additional requirements:			
1. Fragile articles shall be contained in either a rigid inner packaging or a rigid intermediate packaging.			
2. Inner packagings containing sharp objects such as broken glass and needles shall be rigid and resistant to puncture.			
3. The inner packaging, the intermediate packaging and the outer packaging shall be capable of retaining liquids. Outer packagings that are not capable of retaining liquids by design shall be fitted with a liner or suitable measure of retaining liquids.			
4. The inner packaging and/or the intermediate packaging may be flexible. When flexible packagings are used, they shall be capable of passing the impact resistance test of at least 165 g according to ISO 7765-1:1988 <i>Plastics film and sheeting – Determination of impact resistance by the free-falling dart method – Part 1: Staircase methods</i> and the tear resistance test of at least 480 g in both parallel and perpendicular planes with respect to the length of the bag in accordance with ISO 6383-2:1983 <i>Plastics – Film and sheeting – Determination of tear resistance – Part 2: Elmendorf method</i> . The maximum net mass of each flexible inner packaging shall be 30 kg.			
5. Each flexible intermediate packaging shall contain only one inner packaging.			
6. Inner packagings containing a small amount of free liquid may be included in intermediate packaging provided that there is sufficient absorbent or solidifying material in the inner or intermediate packaging to absorb or solidify all the liquid content present. Suitable absorbent material which may withstand the temperatures and vibrations liable to occur under normal conditions of transport shall be used.			
7. Intermediate packagings shall be secured in outer packagings with suitable cushioning and/or absorbent material.			

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P650	PACKING INSTRUCTION	P650
	This instruction applies to UN 3373.	
	(1) The packaging shall be of good quality, strong enough to withstand the shocks and loadings normally encountered during transport, including transshipment between cargo transport units and between cargo transport units and warehouses as well as any removal from a pallet or overpack for subsequent manual or mechanical handling. Packagings shall be constructed and closed to prevent any loss of contents that might be caused under normal conditions of transport by vibration or by changes in temperature, humidity or pressure. (2) The packaging shall consist of at least three components: (a) a primary receptacle; (b) a secondary packaging; and (c) an outer packaging of which either the secondary or the outer packaging shall be rigid. (3) Primary receptacles shall be packed in secondary packagings in such a way that, under normal conditions of transport, they cannot break, be punctured or leak their contents into the secondary packaging. Secondary packagings shall be secured in outer packagings with suitable cushioning material. Any leakage of the contents shall not compromise the integrity of the cushioning material or of the outer packaging. (4) For transport, the mark illustrated below shall be displayed on the external surface of the outer packaging on a background of a contrasting colour and shall be clearly visible and legible. The mark shall be in the form of a square set at an angle of 45° (diamond-shaped) with each side having a length of at least 50 mm, the width of the line shall be at least 2 mm and the letters and numbers shall be at least 6 mm high. The proper shipping name "BIOLOGICAL SUBSTANCE, CATEGORY B" in letters at least 6 mm high shall be marked on the outer packaging adjacent to the diamond-shaped mark. (5) At least one surface of the outer packaging shall have a minimum dimension of 100 mm × 100 mm. △ (6) The completed package shall be capable of withstanding a 1.2 m drop in any orientation without leakage from the primary receptacle(s), which shall remain protected by absorbent material, when required, in the secondary packaging. Note: Capability may be demonstrated by testing, assessment or experience. (7) For liquid substances (a) The primary receptacle(s) shall be leakproof; (b) The secondary packaging shall be leakproof; (c) If multiple fragile primary receptacles are placed in a single secondary packaging, they shall either be individually wrapped or separated to prevent contact between them; △ (d) Absorbent material shall be placed between the primary receptacle(s) and the secondary packaging. The absorbent material shall be in a quantity sufficient to absorb the entire contents of the primary receptacle(s) so that any release of the liquid substance will not compromise the integrity of the cushioning material or of the outer packaging; and (e) The primary receptacle or the secondary packaging shall be capable of withstanding, without leakage, an internal pressure of 95 kPa (0.95 bar). △ Note: Capability may be demonstrated by testing, assessment or experience. (8) For solid substances (a) The primary receptacle(s) shall be siftproof; (b) The secondary packaging shall be siftproof; △ (c) If multiple fragile primary receptacles are placed in a single secondary packaging, they shall either be individually wrapped or separated to prevent contact between them; and (d) If there is any doubt as to whether or not residual liquid may be present in the primary receptacle during transport then a packaging suitable for liquids, including absorbent materials, shall be used.	

Part 4 – Packing and tank provisions

P650	PACKING INSTRUCTION (continued)	P650
△	(9) Refrigerated or frozen specimens: ice, dry ice and liquid nitrogen (a) When dry ice or liquid nitrogen is used as a coolant, the requirements of 5.5.3 shall apply. When used, ice shall be placed outside the secondary packagings or in the outer packaging or an overpack. Interior supports shall be provided to secure the secondary packagings in the original position. If ice is used, the outside packaging or overpack shall be leakproof; and (b) The primary receptacle and the secondary packaging shall maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost. (10) When packages are placed in an overpack, the package marks required by this packing instruction shall either be clearly visible or be reproduced on the outside of the overpack. (11) Infectious substances assigned to UN 3373 which are packed and marked in accordance with this packing instruction are not subject to any other provisions of this Code. (12) Clear instructions on filling and closing such packages shall be provided by packaging manufacturers and subsequent distributors to the consignor or to the person who prepares the package (e.g. patient) to enable the package to be correctly prepared for transport. (13) Other dangerous goods shall not be packed in the same packaging as class 6.2 infectious substances unless they are necessary for maintaining the viability, stabilizing or preventing degradation or neutralizing the hazards of the infectious substances. A quantity of 30 mL or less of dangerous goods included in classes 3, 8 or 9 may be packed in each primary receptacle containing infectious substances. When these small quantities of dangerous goods are packed with infectious substances in accordance with this packing instruction, no other provisions of the Code need be met. Additional provision: Alternative packagings for the transport of animal material may be authorized by the competent authority in accordance with the provisions of 4.1.3.7.	

P800	PACKING INSTRUCTION	P800
This instruction applies to UN Nos. 2803 and 2809.		
The following packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met:		
(1) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met.		
(2) Steel flasks or bottles with threaded closures with a capacity not exceeding 3.0 L; or		
(3) Combination packagings which conform to the following provisions:		
(a) Inner packagings shall comprise glass, metal or rigid plastics intended to contain liquids with a maximum net mass of 15 kg each.		
(b) The inner packagings shall be packed with sufficient cushioning material to prevent breakage.		
(c) Either the inner packagings or the outer packagings shall have inner liners or bags of strong leakproof and puncture-resistant material impervious to the contents and completely surrounding the contents to prevent it from escaping from the package irrespective of its position or orientation.		
(d) The following outer packagings and maximum net masses are authorized:		
Outer packaging		Maximum net mass
Drums		
steel (1A1, 1A2)		400 kg
metal, other than steel or aluminium (1N1, 1N2)		400 kg
plastics (1H1, 1H2)		400 kg
plywood (1D)		400 kg
fibre (1G)		400 kg
Boxes		
steel (4A)		400 kg
metal, other than steel or aluminium (4N)		400 kg
natural wood (4C1)		250 kg
natural wood with sift-proof walls (4C2)		250 kg
plywood (4D)		250 kg
reconstituted wood (4F)		125 kg
fibreboard (4G)		125 kg
expanded plastics (4H1)		60 kg
solid plastics (4H2)		125 kg
Special packing provision:		
PP41 For UN 2803, when it is necessary to transport gallium at low temperatures in order to maintain it in a completely solid state, the above packagings may be overpacked in a strong, water-resistant outer packaging which contains dry ice or other means of refrigeration. When dry ice or other means of refrigeration presenting a risk of asphyxiation are used as a coolant, the requirements of 5.5.3 shall apply. If a refrigerant is used, all of the above materials used in the packaging of gallium shall be chemically and physically resistant to the refrigerant and shall have impact resistance at the low temperatures of the refrigerant employed. If dry ice is used, the outer packaging shall permit the release of carbon dioxide gas. Interior supports shall be provided to prevent movement after the dissipation of the refrigerant.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P801	PACKING INSTRUCTION	P801
This instruction applies to UN Nos. 2794, 2795 and 3028.		
The following packagings are authorized, provided that the provisions of 4.1.1.1, 4.1.1.2, 4.1.1.6, and 4.1.3 are met: (1) Rigid outer packagings, wooden slatted crates or pallets. Additionally, the following conditions shall be met: (a) Batteries' stacks shall be in tiers separated by a layer of electrically non-conductive material; (b) Battery terminals shall not support the weight of other superimposed elements; (c) Batteries shall be packaged or secured to prevent inadvertent movement; (d) Batteries shall not leak under normal conditions of transport or appropriate measures shall be taken to prevent the release of electrolyte from the package (e.g. individually packaging batteries or other equally effective methods); and (e) Batteries shall be protected against short circuits. (2) Stainless steel or plastics bins may also be used to transport used batteries. Additionally, the following conditions shall be met: (a) The bins shall be resistant to the electrolyte that was contained in the batteries; (b) The bins shall not be filled to a height greater than the height of their sides; (c) The outside of the bins shall be free of residues of electrolyte contained in the batteries; (d) Under normal conditions of transport, no electrolyte shall leak from the bins; (e) Measures shall be taken to ensure that filled bins cannot lose their content; and (f) Measures shall be taken to prevent short circuits (e.g. batteries are discharged, individual protection of the battery terminals, etc.). Note: The packagings authorized in (1) and (2) may exceed a net mass of 400 kg (see 4.1.3.3).		
P802	PACKING INSTRUCTION	P802
The following packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met: (1) Combination packagings Outer packagings: 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2; maximum net mass: 75 kg. Inner packagings: glass or plastics; maximum capacity: 10 L. (2) Combination packagings Outer packagings: 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2; maximum net mass: 125 kg Inner packagings: metal; maximum capacity: 40 L (3) Composite packagings: Glass receptacle in steel, aluminium or plywood drum (6PA1, 6PB1 or 6PD1) or in a steel, aluminium or wood box or in wickerwork hamper (6PA2, 6PB2, 6PC or 6PD2) or in solid plastics packaging (6PH2); maximum capacity: 60 L. (4) Steel drums (1A1) with a maximum capacity of 250 L. (5) Pressure receptacles may be used provided that the general provisions of 4.1.3.6 are met. Special packing provisions: PP79 For UN 1790 with more than 60% but not more than 85% hydrogen fluoride, see P001. PP81 For UN 1790 with not more than 85% hydrogen fluoride and UN 2031 with more than 55% nitric acid, the permitted use of plastics drums and jerricans as single packagings shall be two years from their date of manufacture.		
P803	PACKING INSTRUCTION	P803
This instruction applies to UN 2028.		
△	The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2). Packagings shall conform to the packing group II performance level. Articles shall be individually packaged and separated from each other using partitions, dividers, inner packagings or cushioning material to prevent inadvertent discharge during normal conditions of transport. Maximum net mass: 75 kg.	

Part 4 – Packing and tank provisions

P804	PACKING INSTRUCTION	P804
This instruction applies to UN 1744.		
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met and the packagings are hermetically sealed: (1) Combination packagings with a maximum gross mass of 25 kg, consisting of one or more glass inner packaging(s) with a maximum capacity of 1.3 L each and filled to no more than 90% of their capacity; the closure(s) of which shall be physically held in place by any means capable of preventing back-off or loosening by impact or vibration during transport, individually placed in: (a) metal or rigid plastics receptacles together with cushioning and absorbent material sufficient to absorb the entire contents of the glass inner packaging(s), further packed in: (b) 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings. (2) Combination packagings consisting of metal or polyvinylidene fluoride (PVDF) inner packagings, not exceeding 5 L in capacity individually packed with absorbent material sufficient to absorb the contents and inert cushioning material in 1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G, 4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G or 4H2 outer packagings with a maximum gross mass of 75 kg. Inner packagings shall not be filled to more than 90% of their capacity. The closure of each inner packaging shall be physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport. (3) Packagings consisting of: Outer packagings: Steel or plastics drums (1A1, 1A2, 1H1 or 1H2) tested in accordance with the test requirements in 6.1.5 at a mass corresponding to the mass of the assembled package either as a packaging intended to contain inner packagings, or as a single packaging intended to contain solids or liquids, and marked accordingly; Inner packagings: Drums and composite packagings (1A1, 1B1, 1N1, 1H1 or 6HA1) meeting the requirements of chapter 6.1 for single packagings, subject to the following conditions: (a) The hydraulic pressure test shall be conducted at a pressure of at least 300 kPa (3 bar) (gauge pressure); (b) The design and production leakproofness tests shall be conducted at a test pressure of 30 kPa (0.3 bar); (c) They shall be isolated from the outer drum by the use of inert shock-mitigating cushioning material which surrounds the inner packaging on all sides; (d) Their capacity shall not exceed 125 L; (e) Closures shall be of a screw type that are: (i) Physically held in place by any means capable of preventing back-off or loosening of the closure by impact or vibration during transport; (ii) Provided with a cap seal; (f) The outer and inner packagings shall be subjected periodically to an internal inspection and leakproofness test according to (b) at intervals of not more than two and a half years; and (g) The outer and inner packagings shall bear in clearly legible and durable characters: (i) the date (month, year) of the initial test and the latest periodic test and inspection of the inner packaging; and (ii) the name or authorized symbol of the expert performing the tests and inspections. (4) Pressure receptacles, provided that the general provisions of 4.1.3.6 are met. (a) They shall be subjected to an initial test and periodic tests every 10 years at a pressure of not less than 1 MPa (10 bar) (gauge pressure); (b) They shall be subjected periodically to an internal inspection and leakproofness test at intervals of not more than two and a half years; (c) They may not be equipped with any pressure relief device; (d) Each pressure receptacle shall be closed with a plug or valve(s) fitted with a secondary closure device; and (e) The materials of construction for the pressure receptacle, valves, plugs, outlet caps, luting and gaskets shall be compatible with each other and with the contents.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P900	PACKING INSTRUCTION	P900
	This instruction applies to UN 2216.	
	The following packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met: (1) Packagings according to P002; or (2) Bags (5H1, 5H2, 5H3, 5H4, 5L1, 5L2, 5L3, 5M1 or 5M2) with a maximum net mass of 50 kg. Fish meal may also be transported unpackaged when it is packed in closed cargo transport units and the free air space has been restricted to a minimum.	
P901	PACKING INSTRUCTION	P901
	This instruction applies to UN 3316.	
	The following combination packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A1, 1A2, 1B1, 1B2, 1N1, 1N2, 1H1, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A1, 3A2, 3B1, 3B2, 3H1, 3H2). Packagings shall conform to the performance level consistent with the packing group assigned to the kit as a whole (see 3.3.1, special provision 251). Where the kit contains only dangerous goods to which no packing group is assigned, packagings shall meet Packing Group II performance level. Maximum quantity of dangerous goods per outer packaging: 10 kg excluding the mass of any carbon dioxide, solid (dry ice) used as a refrigerant. ■ If dry ice is used as a coolant, the requirements of 5.5.3 shall apply. Additional requirement: Dangerous goods in kits shall be packed in inner packagings which shall be protected from other materials in the kit.	
P902	PACKING INSTRUCTION	P902
△	This instruction applies to UN Nos. 3268 and 3559.	
△	(1) Packaged articles: The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group III performance level. The packagings shall be designed and constructed so as to prevent movement of the articles and inadvertent operation during normal conditions of transport. △ (2) Unpackaged articles: △ Except for UN 3559, the articles may also be transported unpackaged in dedicated handling devices, vehicles or containers when moved to, from, or between where they are manufactured and an assembly plant including intermediate handling locations. Additional requirement: Any pressure receptacle shall be in accordance with the requirements of the competent authority for the substance(s) contained therein.	

Part 4 – Packing and tank provisions

P903	PACKING INSTRUCTION	P903
△	This instruction applies to UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552.	
△	For the purpose of this packing instruction, “equipment” means apparatus for which the cells or batteries will provide electrical power for its operation. The following packagings are authorized provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) For cells and batteries: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Cells or batteries shall be packed in packagings so that the cells or batteries are protected against damage that may be caused by the movement or placement of the cells or batteries within the packaging. Packagings shall conform to the packing group II performance level. (2) In addition for a cell or a battery with a gross mass of 12 kg or more employing a strong, impact resistant outer casing: (a) Strong outer packagings; (b) Protective enclosures (e.g. fully enclosed or wooden slatted crates); or (c) Pallets or other handling devices. Cells or batteries shall be secured to prevent inadvertent movement, and the terminals shall not support the weight of other superimposed elements. Packagings need not meet the requirements of 4.1.1.3. (3) For cells or batteries packed with equipment: Packagings conforming to the requirements in paragraph (1) of this packing instruction, then placed with the equipment in an outer packaging; or Packagings that completely enclose the cells or batteries, then placed with equipment in a packaging conforming to the requirements in paragraph (1) of this packing instruction. The equipment shall be secured against movement within the outer packaging. (4) For cells or batteries contained in equipment: Strong outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use. They shall be constructed in such a manner as to prevent accidental operation during transport. Packagings need not meet the requirements of 4.1.1.3. Large equipment can be offered for transport unpackaged or on pallets when the cells or batteries are afforded equivalent protection by the equipment in which they are contained. When intentionally active, devices such as radio frequency identification (RFID) tags, watches and temperature loggers, which are not capable of generating a dangerous evolution of heat, may be transported in strong outer packagings. (5) For packaging containing both cells or batteries packed with equipment and contained in equipment: (a) For cells and batteries, packagings that completely enclose the cells or batteries, then placed with equipment in a packaging conforming to the requirements in paragraph (1) of this packing instruction; or (b) Packagings conforming to the requirements in paragraph (1) of this packing instruction, then placed with the equipment in a strong outer packaging constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use. The outer packaging shall be constructed in such a manner as to prevent accidental operation during transport and need not meet the requirements of 4.1.1.3. The equipment shall be secured against movement within the outer packaging. When intentionally active, devices such as radio frequency identification (RFID) tags, watches and temperature loggers, which are not capable of generating a dangerous evolution of heat, may be transported in strong outer packagings. Note: The packagings authorized in (2), (4) and (5) may exceed a net mass of 400 kg (see 4.1.3.3).	
△	Additional requirement: Cells or batteries shall be protected against short circuit.	

P904	PACKING INSTRUCTION	P904
This instruction applies to UN 3245.		
The following packagings are authorized: (1) Packagings meeting the provisions of 4.1.1.1, 4.1.1.2, 4.1.1.4, 4.1.1.8 and 4.1.3 and so designed that they meet the construction requirements of 6.1.4. Outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use, shall be used. Where this packing instruction is used for the transport of inner packagings of combination packagings the packaging shall be designed and constructed to prevent inadvertent discharge during normal conditions of transport. (2) Packagings, which need not conform to the packaging test requirements of part 6, but conforming to the following: (a) An inner packaging comprising: (i) primary receptacle(s) and a secondary packaging, the primary receptacle(s) or the secondary packaging shall be leakproof for liquids or siftproof for solids; (ii) for liquids, absorbent material placed between the primary receptacle(s) and the secondary packaging. The absorbent material shall be in a quantity sufficient to absorb the entire contents of the primary receptacle(s) so that any release of the liquid substance will not compromise the integrity of the cushioning material or of the outer packaging; (iii) if multiple fragile primary receptacles are placed in a single secondary packaging they shall be individually wrapped or separated to prevent contact between them; (b) An outer packaging shall be strong enough for its capacity, mass and intended use, and with a smallest external dimension of at least 100 mm. For transport, the mark illustrated below shall be displayed on the external surface of the outer packaging on a background of a contrasting colour and shall be clearly visible and legible. The mark shall be in the form of a square set at an angle of 45° (diamond-shaped) with each side having a length of at least 50 mm; the width of the line shall be at least 2 mm and the letters and numbers shall be at least 6 mm high.		
Additional requirement: When dry ice or liquid nitrogen is used as a coolant, the requirements of 5.5.3 shall apply. When used, ice shall be placed outside the secondary packagings or in the outer packaging or an overpack. Interior supports shall be provided to secure the secondary packaging in the original position. If ice is used, the outside packaging or overpack shall be leakproof.		

Part 4 – Packing and tank provisions

P905	PACKING INSTRUCTION	P905
This instruction applies to UN Nos. 2990 and 3072.		
Any suitable packaging is authorized, provided the general provisions of 4.1.1 and 4.1.3 are met, except that packagings need not conform to the provisions of part 6. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3). When the life-saving appliances are constructed to incorporate or are contained in rigid outer weatherproof casings (such as for lifeboats), they may be transported unpackaged.		
Additional provisions:		
1 All dangerous substances and articles contained as equipment within the appliances shall be secured to prevent inadvertent movement and in addition: (a) signal devices of class 1 shall be packed in plastics or fibreboard inner packagings; (b) gases (class 2.2) shall be contained in cylinders as specified by the competent authority, which may be connected to the appliance; (c) electric storage batteries (class 8) and lithium batteries and sodium ion batteries (class 9) shall be disconnected or electrically isolated and secured to prevent any spillage of liquid; and (d) small quantities of other dangerous substances (for example in classes 3, 4.1 and 5.2) shall be packed in strong inner packagings. 2 Preparation for transport and packaging shall include provisions to prevent any accidental inflation of the appliance.		
P906	PACKING INSTRUCTION	P906
This instruction applies to UN Nos. 2315, 3151, 3152 and 3432.		
The following packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met:		
(1) For liquids and solids containing or contaminated with PCBs, polyhalogenated biphenyls, polyhalogenated terphenyls or halogenated monomethyldiphenylmethanes: Packagings in accordance with P001 or P002, as appropriate. (2) For transformers and condensers and other articles: (a) Packagings in accordance with packing instructions P001 or P002. The articles shall be secured with suitable cushioning material to prevent inadvertent movement during normal conditions of transport; or (b) Leakproof packagings which are capable of containing, in addition to the articles, at least 1.25 times the volume of the liquid PCBs, polyhalogenated biphenyls, polyhalogenated terphenyls or halogenated monomethyldiphenylmethanes present in them. There shall be sufficient absorbent material in the packagings to absorb at least 1.1 times the volume of liquid which is contained in the articles. In general, transformers and condensers shall be carried in leakproof metal packagings which are capable of holding, in addition to the transformers and condensers, at least 1.25 times the volume of the liquid present in them. Note 1: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3). Notwithstanding the above, liquids and solids not packaged in accordance with P001 and P002 and unpackaged transformers and condensers may be transported in cargo transport units fitted with a leakproof metal tray to a height of at least 800 mm, containing sufficient inert absorbent material to absorb at least 1.1 times the volume of any free liquid. Note 2: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).		
Additional provision: Adequate provisions shall be taken to seal the transformers and condensers to prevent leakage during normal conditions of transport.		
P907	PACKING INSTRUCTION	P907
This instruction applies to articles, such as machinery, apparatus or devices of UN No. 3363.		
If the article is constructed and designed so that the receptacles containing the dangerous goods are afforded adequate protection, an outer packaging is not required. Dangerous goods in an article shall otherwise be packed in outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use, and meeting the applicable requirements of 4.1.1.1. Receptacles containing dangerous goods shall conform to the general provisions in 4.1.1, except that 4.1.1.3, 4.1.1.4, 4.1.1.12 and 4.1.1.14 do not apply. For class 2.2 gases, the inner cylinder or receptacle, its contents and filling ratio shall be to the satisfaction of the competent authority of the country in which the cylinder or receptacle is filled. In addition, the manner in which receptacles are contained within the article shall be such that, under normal conditions of transport, damage to receptacles containing the dangerous goods is unlikely; and in the event of damage to the receptacles containing solid or liquid dangerous goods, no leakage of the dangerous goods from the article is possible (a leakproof liner may be used to satisfy this requirement). Receptacles containing dangerous goods shall be so installed, secured or cushioned as to prevent their breakage or leakage and so as to control their movement within the article during normal conditions of transport. Cushioning material shall not react dangerously with the content of the receptacles. Any leakage of the contents shall not substantially impair the protective properties of the cushioning material. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3).		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P908	PACKING INSTRUCTION	P908
△	This instruction applies to damaged or defective cells and batteries, including those contained in equipment, of UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552. The following packagings are authorized provided the general provisions for 4.1.1 and 4.1.3 are met: For cells and batteries and equipment containing cells and batteries: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G) Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2) Jerricans (3A2, 3B2, 3H2) Packagings shall conform to the packing group II performance level. ■ Packagings shall also meet the following requirements: (a) Each damaged or defective cell or battery or equipment containing such cells or batteries shall be individually packed in inner packaging and placed inside of an outer packaging. The inner packaging or outer packaging shall be leakproof to prevent the potential release of electrolyte. (b) Each inner packaging shall be surrounded by sufficient non-combustible and electrically non-conductive thermal insulation material to protect against a dangerous evolution of heat. (c) Sealed packagings shall be fitted with a venting device when appropriate. (d) Appropriate measures shall be taken to minimize the effects of vibrations and shocks, prevent movement of the cells or batteries within the package that may lead to further damage and a dangerous condition during transport. Cushioning material that is non-combustible and electrically non-conductive may also be used to meet this requirement. △ (e) The non-combustibility of the thermal insulation material and the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured. For leaking cells or batteries, sufficient inert absorbent material shall be added to the inner or outer packaging to absorb any release of electrolyte. A cell or battery with a net mass of more than 30 kg shall be limited to one cell or battery per outer packaging. Additional requirement: Cells or batteries shall be protected against short circuit.	

Part 4 – Packing and tank provisions

P909	PACKING INSTRUCTION	P909
△	This instruction applies to UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552 transported for disposal or recycling, either packed together with or packed without non-lithium batteries.	
△	(1) Cells and batteries shall be packed in accordance with the following: (a) The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3, are met: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H2); and Jerricans (3A2, 3B2, 3H2). (b) Packagings shall conform to the packing group II performance level. (c) Metal packagings shall be fitted with an electrically non-conductive lining material (e.g. plastics) of adequate strength for the intended use. (2) However, lithium ion or sodium ion cells with a Watt-hour rating of not more than 20 Wh, lithium ion or sodium ion batteries with a Watt-hour rating of not more than 100 Wh, lithium metal cells with a lithium content of not more than 1 g and lithium metal batteries with an aggregate lithium content of not more than 2 g may be packed in accordance with the following: (a) In strong outer packaging up to 30 kg gross mass meeting the general provisions of 4.1.1, except 4.1.1.3, and 4.1.3. (b) Metal packagings shall be fitted with an electrically non-conductive lining material (e.g. plastics) of adequate strength for the intended use. (3) For cells or batteries contained in equipment, strong outer packagings constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use, may be used. Packagings need not meet the requirements of 4.1.1.3. Equipment may also be offered for transport unpackaged or on pallets when the cells or batteries are afforded equivalent protection by the equipment in which they are contained. (4) In addition, for cells or batteries with a gross mass of 12 kg or more employing a strong, impact resistant outer casing, strong outer packagings constructed of suitable material and of adequate strength and design in relation to the packagings capacity and its intended use, may be used. Packagings need not meet the requirements of 4.1.1.3. Note: The packagings authorized in (3) and (4) may exceed a net mass of 400 kg (see 4.1.3.3). Additional requirements: 1 Cells and batteries shall be designed or packed to prevent short circuits and the dangerous evolution of heat. 2 Protection against short circuits and the dangerous evolution of heat includes, but is not limited to: (a) individual protection of the battery terminals, (b) inner packaging to prevent contact between cells and batteries, (c) batteries with recessed terminals designed to protect against short circuits, or (d) the use of an electrically non-conductive and non-combustible cushioning material to fill empty space between the cells or batteries in the packaging. 3 Cells and batteries shall be secured within the outer packaging to prevent excessive movement during transport (e.g. by using an electrically non-conductive and non-combustible cushioning material or through the use of a tightly closed plastics bag).	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

P910	PACKING INSTRUCTION	P910
△	This instruction applies to UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552 production runs consisting of not more than 100 cells or batteries and to pre-production prototypes of cells or batteries when these prototypes are transported for testing. The following packagings are authorized provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) For cells and batteries, including when packed with equipment: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group II performance level and shall meet the following requirements: (a) Batteries and cells, including equipment, of different sizes, shapes or masses shall be packaged in an outer packaging of a tested design type listed above provided the total gross mass of the package does not exceed the gross mass for which the design type has been tested; (b) Each cell or battery shall be individually packed in an inner packaging and placed inside an outer packaging; (c) Each inner packaging shall be completely surrounded by sufficient non-combustible and electrically non-conductive thermal insulation material to protect against a dangerous evolution of heat; (d) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the cells or batteries within the package that may lead to damage and a dangerous condition during transport. Cushioning material that is non-combustible and electrically non-conductive may be used to meet this requirement; (e) The non-combustibility of the thermal insulation material and the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured; (f) A cell or battery with a net mass of more than 30 kg shall be limited to one cell or battery per outer packaging. (2) For cells and batteries contained in equipment: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). Packagings shall conform to the packing group II performance level and shall meet the following requirements: (a) Equipment of different sizes, shapes or masses shall be packaged in an outer packaging of a tested design type listed above provided the total gross mass of the package does not exceed the gross mass for which the design type has been tested; (b) The equipment shall be constructed or packaged in such a manner as to prevent accidental operation during transport; (c) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the equipment within the package that may lead to damage and a dangerous condition during transport. When cushioning material is used to meet this requirement it shall be non-combustible and electrically non-conductive; and (d) The non-combustibility of the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured. (3) The equipment or the batteries may be transported unpackaged under conditions specified by the competent authority. Additional conditions that may be considered in the approval process include, but are not limited to: (a) The equipment or the battery shall be strong enough to withstand the shocks and loadings normally encountered during transport, including transshipment between cargo transport units and between cargo transport units and warehouses as well as any removal from a pallet for subsequent manual or mechanical handling; and (b) The equipment or the battery shall be fixed in cradles or crates or other handling devices in such a way that it will not become loose during normal conditions of transport. Note: The packagings authorized may exceed a net mass of 400 kg (see 4.1.3.3). Additional requirements:	
△	The cells and batteries shall be protected against short circuit. Protection against short circuits includes, but is not limited to, (a) individual protection of the battery terminals, (b) inner packaging to prevent contact between cells and batteries, (c) batteries with recessed terminals designed to protect against short circuits, or (d) the use of an electrically non-conductive and non-combustible cushioning material to fill empty space between the cells or batteries in the packaging.	

Part 4 – Packing and tank provisions

P911	PACKING INSTRUCTION	P911
△	This instruction applies to damaged or defective cells and batteries of UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552 liable to rapidly disassemble, dangerously react, produce a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours under normal conditions of transport. The following packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: For cells and batteries and equipment containing cells and batteries: Drums (1A2, 1B2, 1N2, 1H2, 1D, 1G); Boxes (4A, 4B, 4N, 4C1, 4C2, 4D, 4F, 4G, 4H1, 4H2); Jerricans (3A2, 3B2, 3H2). The packagings shall conform to the packing group I performance level. (1) The packaging shall be capable of meeting the following additional performance requirements in case of rapid disassembly, dangerous reaction, production of a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours of the cells or batteries: (a) The outside surface temperature of the completed package shall not have a temperature of more than 100°C. A momentary spike in temperature up to 200°C is acceptable; (b) No flame shall occur outside the package; (c) No projectiles shall exit the package; (d) The structural integrity of the package shall be maintained; (e) The packagings shall have a gas management system (e.g. filter system, air circulation, containment for gas, gas tight packaging, etc.), as appropriate. (2) The additional packaging performance requirements shall be verified by a test as specified by the competent authority.* A verification report shall be available on request. As a minimum requirement, the cell or battery name, the cell or battery number, the mass, type, energy content of the cells or batteries, the packaging identification and the test data according to the verification method as specified by the competent authority shall be listed in the verification report. (3) When dry ice or liquid nitrogen is used as a coolant, the requirements of section 5.5.3 shall apply. The inner packaging and outer packaging shall maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost. Additional requirement: Cells or batteries shall be protected against short circuit. * The following criteria, as relevant, may be considered to assess the performance of the packaging: (a) The assessment shall be done under a quality management system (as described, e.g. in section 2.9.4.5) allowing for the traceability of tests results, reference data and characterization models used; (b) The list of hazards expected in case of thermal runaway for the cell or battery type, in the condition it is transported (e.g. usage of an inner packaging, state of charge (SOC), use of sufficient non-combustible, electrically non-conductive and absorbent cushioning material, etc.), shall be clearly identified and quantified; the reference list of possible hazards for cells or batteries (e.g. rapidly disassemble, dangerously react, produce a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours) can be used for this purpose. The quantification of these hazards shall rely on available scientific literature; (c) The mitigating effects of the packaging shall be identified and characterized, based on the nature of the protections provided and the construction material properties. A list of technical characteristics and drawings shall be used to support this assessment (density [kg·m⁻³], specific heat capacity [J·kg⁻¹·K⁻¹], heating value [kJ·kg⁻¹], thermal conductivity [W·m⁻¹·K⁻¹], melting temperature and flammability temperature [K], heat transfer coefficient of the outer packaging [W·m⁻²·K⁻¹], ...); (d) The test and any supporting calculations shall assess the result of a thermal runaway of the cell or battery inside the packaging in the normal conditions of transport; (e) In case the SOC of the cell or battery is not known, the assessment used shall be done with the highest possible SOC corresponding to the cell or battery use conditions; (f) The surrounding conditions in which the packaging may be used and transported shall be described (including for possible consequences of gas or smoke emissions on the environment, such as ventilation or other methods) according to the gas management system of the packaging; (g) The tests or the model calculation shall consider the worst case scenario for the thermal runaway triggering and propagation inside the cell or battery: this scenario includes the worst possible failure in the normal transport condition, the maximum heat and flame emissions for the possible propagation of the reaction; (h) These scenarios shall be assessed over a period long enough to allow all the possible consequences to occur (e.g. 24 hours). (i) In the case of multiple batteries and multiple items of equipment containing batteries, additional requirements, such as the maximum number of batteries and items of equipment, the total maximum energy content of the batteries, and the configuration inside the package, including separations and protections of the parts, shall be considered.	
△		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

■	P912 PACKING INSTRUCTION P912
	This instruction applies to UN Nos. 3556, 3557 and 3558. The vehicle shall be secured in a strong, rigid outer packaging constructed of suitable material, and of adequate strength and design in relation to the packaging capacity and its intended use. It shall be constructed in such a manner as to prevent accidental operation during transport. Packagings need not meet the requirements of 4.1.1.3. The vehicle shall be secured by means capable of restraining the vehicle in the outer packaging to prevent any movement during transport which would change the orientation or cause the battery in the vehicle to be damaged. Vehicles transported in a packaging may have some parts of the vehicle, other than the battery, detached from its frame to fit into the packaging. Note: The packagings may exceed a net mass of 400 kg (see 4.1.3.3). Vehicles with an individual net mass of 30 kg or more: (a) may be loaded into crates or secured to pallets; (b) may be transported unpackaged providing that the vehicle is capable of remaining upright during transport without additional support and the vehicle provides adequate protection to the battery so that no damage to the battery can occur; or (c) where the vehicles have the potential to topple over during transport (e.g. motor cycles), may be transported unpackaged in a cargo transport unit fitted out with the means to prevent toppling in transport, such as by the use of bracing, frames or racking.

Part 4 – Packing and tank provisions

4.1.4.2 Packing instructions concerning the use of IBCs

IBC01	PACKING INSTRUCTION	IBC01
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (31A, 31B and 31N).		

IBC02	PACKING INSTRUCTION	IBC02
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (31A, 31B and 31N); Rigid plastics (31H1 and 31H2); Composite (31HZ1).		
Special packing provisions:		
B5	For UN Nos. 1791, 2014, 2984 and 3149, IBCs shall be provided with a device to allow venting during transport. The inlet to the venting device shall be sited in the vapour space of the IBC under maximum filling conditions during transport.	
B8	The pure form of this substance shall not be transported in IBCs since it is known to have a vapour pressure of more than 110 kPa at 50°C or 130 kPa at 55°C.	
B15	For UN 2031 with more than 55% nitric acid, the permitted use of rigid plastics IBCs and of rigid plastics inner receptacles of composite IBCs shall be two years from their date of manufacture.	
B16	For UN 3375, IBCs of type 31A and 31N are not allowed without competent authority approval.	
B20	For UN Nos. 1716, 1717, 1736, 1737, 1738, 1742, 1743, 1755, 1764, 1768, 1776, 1778, 1782, 1789, 1790, 1796, 1826, 1830, 1832, 2031, 2308, 2353, 2513, 2584, 2796 and 2817 coming under PG II, IBCs shall be fitted with two shut-off devices.	

IBC03	PACKING INSTRUCTION	IBC03
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (31A, 31B and 31N); Rigid plastics (31H1 and 31H2); Composite (31HZ1 and 31HA2, 31HB2, 31HN2, 31HD2 and 31HH2).		
Special packing provisions:		
B8	The pure form of this substance shall not be transported in IBCs since it is known to have a vapour pressure of more than 110 kPa at 50°C or 130 kPa at 55°C.	
B11	Notwithstanding the provisions of the second paragraph of 4.1.1.10, UN 2672 ammonia solution in concentrations not exceeding 25% may be transported in IBCs.	
B19	For UN Nos. 3532 and 3534, IBCs shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the IBCs in the event of loss of stabilization.	

IBC04	PACKING INSTRUCTION	IBC04
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N).		
Special packing provision:		
B1	For packing group I substances, IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	

IBC05	PACKING INSTRUCTION	IBC05
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N); Rigid plastics (11H1, 11H2, 21H1, 21H2, 31H1 and 31H2); Composite (11HZ1, 21HZ1 and 31HZ1).		
Special packing provisions:		
B1	For packing group I substances, IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	
B21	For solid substances in IBCs other than metal or rigid plastics IBCs, the IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

IBC06	PACKING INSTRUCTION	IBC06
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N); Rigid plastics (11H1, 11H2, 21H1, 21H2, 31H1 and 31H2); Composite (11HZ1, 11HZ2, 21HZ1, 21HZ2 and 31HZ1).		
Additional provision: Where the solid may become liquid during transport see 4.1.3.4.		
Special packing provisions:		
B1	For packing group I substances, IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	
B12	For UN 2907, IBCs shall meet the packing group II performance level. IBCs meeting the test criteria of packing group I shall not be used.	
B21	For solid substances in IBCs other than metal or rigid plastics IBCs, the IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	

IBC07	PACKING INSTRUCTION	IBC07
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N); Rigid plastics (11H1, 11H2, 21H1, 21H2, 31H1 and 31H2); Composite (11HZ1, 11HZ2, 21HZ1, 21HZ2 and 31HZ1); Wooden (11C, 11D and 11F).		
Additional provisions: 1 Where the solid may become liquid during transport see 4.1.3.4. 2 Liners of wooden IBCs shall be sift-proof.		
Special packing provisions:		
B1	For packing group I substances, IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	
B4	Flexible, fibreboard or wooden IBCs shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.	
B18	For UN Nos. 3531 and 3533, IBCs shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the IBCs in the event of loss of stabilization.	
B21	For solid substances in IBCs other than metal or rigid plastics IBCs, the IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	
B40	UN 3550 may be transported in flexible IBCs (13H3 or 13H4) with siftproof liners to prevent any egress of dust during transport.	

IBC08	PACKING INSTRUCTION	IBC08
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N); Rigid plastics (11H1, 11H2, 21H1, 21H2, 31H1 and 31H2); Composite (11HZ1, 11HZ2, 21HZ1, 21HZ2 and 31HZ1); Fibreboard (11G); Wooden (11C, 11D and 11F); Flexible (13H1, 13H2, 13H3, 13H4, 13H5, 13L1, 13L2, 13L3, 13L4, 13M1 or 13M2).		
Additional provisions: Where the solid may become liquid during transport see 4.1.3.4.		
Special packing provisions:		
B3	Flexible IBCs shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.	
B4	Flexible, fibreboard or wooden IBCs shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.	
B6	For UN Nos. 1327, 1363, 1364, 1365, 1386, 1408, 1841, 2211, 2217, 2793 and 3314, IBCs are not required to meet the IBC testing provisions of chapter 6.5.	
B21	For solid substances in IBCs other than metal or rigid plastics IBCs, the IBCs shall be carried in closed cargo transport units or in freight containers/vehicles, which shall have rigid sides or fences at least to the height of the IBC.	

IBC99	PACKING INSTRUCTION	IBC99
Only IBCs which are approved for these goods by the competent authority may be used (see 4.1.3.7). A copy of the competent authority approval shall accompany each consignment or the transport document shall include an indication that the packaging was approved by the competent authority.		

Part 4 – Packing and tank provisions

IBC100	PACKING INSTRUCTION	IBC100
This instruction applies to UN Nos. 0082, 0222, 0241, 0331 and 0332.		
The following IBCs are authorized, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 and special provisions of 4.1.5 are met: Metal (11A, 11B, 11N, 21A, 21B, 21N, 31A, 31B and 31N); Flexible (13H2, 13H3, 13H4, 13L2, 13L3, 13L4 and 13M2); Rigid plastics (11H1, 11H2, 21H1, 21H2, 31H1 and 31H2); Composite (11HZ1, 11HZ2, 21HZ1, 21HZ2, 31HZ1 and 31HZ2).		
Additional provisions:		
1 IBCs shall only be used for free-flowing substances.		
2 Flexible IBCs shall only be used for solids.		
Special packing provisions:		
B2	For UN 0222 in IBCs other than metal or rigid plastics IBCs, the IBCs shall be transported in closed cargo transport units.	
B3	For UN 0222, flexible IBCs shall be sift-proof and water resistant or shall be fitted with a sift-proof and water resistant liner.	
B9	For UN 0082, this packing instruction may only be used when the substances are mixtures of ammonium nitrate or other inorganic nitrates with other combustible substances which are not explosive ingredients. Such explosives shall not contain nitroglycerin, similar liquid organic nitrates, or chlorates. Metal IBCs are not authorized.	
B10	For UN 0241, this packing instruction may only be used for substances which consist of water as an essential ingredient and high proportions of ammonium nitrate or other oxidizing substances, some or all of which are in solution. The other constituents may include hydrocarbons or aluminium powder, but shall not include nitro-derivatives such as trinitrotoluene. Metal IBCs are not authorized.	
B17	For UN 0222, metal IBCs are not authorized.	

IBC520		PACKING INSTRUCTION		IBC520	
This instruction applies to organic peroxides and self-reactive substances of type F.					
The IBCs listed below are authorized for the formulations listed, provided the general provisions of 4.1.1, 4.1.2 and 4.1.3 and special provisions of 4.1.7.2 are met. The formulations not listed in 2.4.2.3.2.3 or 2.5.3.2.4 but listed below may also be transported packed in accordance with packing method OP8 of packing instruction P520 of 4.1.4.1, with the same control and emergency temperatures, if applicable. For formulations not listed below, only IBCs which are approved by the competent authority may be used (see 4.1.7.2.2).					
UN No.	Organic peroxide	Type of IBC	Maximum quantity (litres)	Control temperature	Emergency temperature
3109	ORGANIC PEROXIDE TYPE F, LIQUID				
	<i>tert</i> -Butyl cumyl peroxide	31HA1	1,000		
	<i>tert</i> -Butyl hydroperoxide, not more than 72% with water	31A 31HA1	1,250 1,000		
	<i>tert</i> -Butyl peroxyacetate, not more than 32% in diluent type A	31A 31HA1	1,250 1,000		
	<i>tert</i> -Butyl peroxybenzoate, not more than 32% in diluent type A	31A	1,250		
	<i>tert</i> -Butyl peroxy-3,5,5-trimethylhexanoate, not more than 37% in diluent type A	31A 31HA1	1,250 1,000		
	Cumyl hydroperoxide, not more than 90% in diluent type A	31HA1	1,250		
	Dibenzoyl peroxide, not more than 42% as a stable dispersion	31H1	1,000		
	Di- <i>tert</i> -butyl peroxide, not more than 52% in diluent type A	31A 31HA1	1,250 1,000		
	1,1-Di-(<i>tert</i> -butylperoxy)cyclohexane, not more than 37% in diluent type A	31A	1,250		
	1,1-Di-(<i>tert</i> -butylperoxy)cyclohexane, not more than 42% in diluent type A	31H1	1,000		
	Dilauroyl peroxide, not more than 42%, stable dispersion, in water	31HA1	1,000		
	2,5-Dimethyl-2,5-di(<i>tert</i> -butylperoxy)hexane, not more than 52% in diluent type A	31HA1	1,000		
	Isopropylcumyl hydroperoxide, not more than 72% in diluent type A	31HA1	1,250		
	<i>p</i> -Menthyl hydroperoxide, not more than 72% in diluent type A	31HA1	1,250		
	Peroxyacetic acid, stabilized, not more than 17%	31H1 31H2 31HA1 31A	1,500 1,500 1,500 1,500		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

	3,6,9-Triethyl-3,6,9-trimethyl-1,4,7-triperoxonane, not more than 27% in diluent type A	31HA1	1,000		
IBC520 PACKING INSTRUCTION (continued) IBC520					
UN No.	Organic peroxide	Type of IBC	Maximum quantity (litres)	Control temperature	Emergency temperature
3110	ORGANIC PEROXIDE TYPE F, SOLID Dicumyl peroxide	31A 31H1 31HA1	2,000		
3119	ORGANIC PEROXIDE TYPE F, LIQUID, TEMPERATURE CONTROLLED <i>tert</i> -Amyl peroxy-2-ethylhexanoate, not more than 62% in diluent type A	31HA1	1,000	+15°C	+20°C
	<i>tert</i> -Amyl peroxy-pivalate, not more than 32% in diluent type A	31A	1,250	+10°C	+15°C
	<i>tert</i> -Amyl peroxy-pivalate, not more than 42%, stable dispersion, in water	31HA1	1,000	0°C	+10°C
	<i>tert</i> -Butyl peroxy-2-ethylhexanoate, not more than 32% in diluent type B	31HA1 31A	1,000 1,250	+30°C +30°C	+35°C +35°C
	<i>tert</i> -Butyl peroxyneodecanoate, not more than 32% in diluent type A	31A	1,250	0°C	+10°C
	<i>tert</i> -Butyl peroxyneodecanoate, not more than 42%, stable dispersion, in water	31A	1,250	–5°C	+5°C
	<i>tert</i> -Butyl peroxyneodecanoate, not more than 52%, stable dispersion, in water	31A	1,250	–5°C	+5°C
	<i>tert</i> -Butyl peroxy-pivalate, not more than 27% in diluent type B	31HA1 31A	1,000 1,250	+10°C +10°C	+15°C +15°C
	<i>tert</i> -Butyl peroxy-pivalate, not more than 42% in diluent type A	31HA1 31A	1,000 1,250	+10°C +10°C	+15°C +15°C
	Di-(2-neodecanoylperoxyisopropyl)benzene, not more than 42%, stable dispersion, in water	31A	1,250	–15°C	–5°C
	3-Hydroxy-1,1-dimethylbutyl peroxyneodecanoate, not more than 52%, stable dispersion, in water	31A	1,250	–15°C	–5°C
	Cumyl peroxyneodecanoate, not more than 52%, stable dispersion, in water	31A	1,250	–15°C	–5°C
	Di-(4- <i>tert</i> -butylcyclohexyl) peroxydicarbonate, not more than 42%, stable dispersion, in water	31HA1	1,000	+30°C	+35°C
	Dicetyl peroxydicarbonate, not more than 42%, stable dispersion, in water	31HA1	1,000	+30°C	+35°C
	Dicyclohexyl peroxydicarbonate, not more than 42% as a stable dispersion, in water	31A	1,250	+10°C	+15°C
	Di-(2-ethylhexyl) peroxydicarbonate, not more than 62%, stable dispersion, in water	31A 31HA1	1,250 1,000	–20°C –20°C	–10°C –10°C
	Dimyristyl peroxydicarbonate, not more than 42%, stable dispersion, in water	31HA1	1,000	+15°C	+20°C
	Di-(3,5,5-trimethylhexanoyl) peroxide, not more than 52% in diluent type A	31HA1 31A	1,000 1,250	+10°C +10°C	+15°C +15°C
	Di-(3,5,5-trimethylhexanoyl) peroxide, not more than 52%, stable dispersion, in water	31HA1 31A	1,000 1,250	+10°C +10°C	+15°C +15°C
	Diisobutyl peroxide, not more than 28%, stable dispersion, in water	31HA1 31A	1,000 1,250	–20°C –20°C	–10°C –10°C
	Diisobutyl peroxide, not more than 42%, stable dispersion, in water	31HA1 31A	1,000 1,250	–25°C –25°C	–15°C –15°C
	1,1,3,3-Tetramethylbutyl peroxy-2-ethylhexanoate, not more than 67%, in diluent type A	31HA1	1,000	+15°C	+20°C
	1,1,3,3-Tetramethylbutyl peroxyneodecanoate, not more than 52%, stable dispersion, in water	31A 31HA1	1,250 1,000	–5°C –5°C	+5°C +5°C
3120	ORGANIC PEROXIDE, TYPE F, SOLID, TEMPERATURE CONTROLLED				

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Part 4 – Packing and tank provisions

Additional provisions:

- 1 IBCs shall be provided with a device to allow venting during transport. The inlet to the pressure relief device shall be sited in the vapour space of the IBC under maximum filling conditions during transport.
- 2 To prevent explosive rupture of metal IBCs or composite IBCs with complete metal casing, the emergency relief devices shall be designed to vent all the decomposition products and vapours evolved during self-accelerating decomposition or during a period of not less than one hour of fire-engulfment as calculated by the formula in 4.2.1.13.8. The control and emergency temperatures specified in this packing instruction are based on a non-insulated IBC. When consigning an organic peroxide in an IBC in accordance with this instruction, it is the responsibility of the consignor to ensure that:
 - (a) the pressure and emergency relief devices installed on the IBC are designed to take appropriate account of the self-accelerating decomposition of the organic peroxide and of fire engulfment; and
 - (b) when applicable, the control and emergency temperatures indicated are appropriate, taking into account the design (such as insulation) of the IBC to be used.

Chapter 4.1 – Use of packagings, including IBCs and large packagings

IBC620	PACKING INSTRUCTION	IBC620
This instruction applies to UN 3291.		
The following IBCs are authorized, provided that the general provisions of 4.1.1, except 4.1.1.15, 4.1.2 and 4.1.3 are met: Rigid, leakproof IBCs conforming to the packing group II performance level.		
Additional provisions: 1 There shall be sufficient absorbent material to absorb the entire amount of liquid present in the IBC. 2 IBCs shall be capable of retaining liquids. 3 IBCs intended to contain sharp objects such as broken glass and needles shall be resistant to puncture.		

4.1.4.3 Packing instructions concerning the use of large packagings

LP01		PACKING INSTRUCTION (LIQUIDS)				LP01
The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met.						
Inner packagings		Large outer packagings	Packing group I	Packing group II	Packing group III	
Glass	10 L	Steel (50A)	Not allowed	Not allowed	3 m³	
Plastics	30 L	Aluminium (50B)				
Metal	40 L	Metal other than steel or aluminium (50N)				
		Rigid plastics (50H)				
		Natural wood (50C)				
		Plywood (50D)				
		Reconstituted wood (50F)				
		Rigid fibreboard (50G)				

LP02		PACKING INSTRUCTION (SOLIDS)			LP02
The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met.					
Inner packagings		Large outer packagings	Packing group I	Packing group II	Packing group III
Glass	10 kg	Steel (50A)	Not allowed	Not allowed	3 m ³
Plastics ^b	50 kg	Aluminium (50B)			
Metal	50 kg	Metal other than steel or			
Paper ^{a, b}	50 kg	aluminium (50N)			
Fibre ^{a, b}	50 kg	Rigid plastics (50H)			
		Natural wood (50C)			
		Plywood (50D)			
		Reconstituted wood (50F)			
		Rigid fibreboard (50G)			
		Flexible plastics (51H) ^c			

^a These packagings shall not be used when the substances being transported may become liquid during transport.^b Packagings shall be sift-proof.^c To be used with flexible inner packagings only.**Special packing provision:****L2** Deleted**L3** For UN Nos.1309, 1376, 1483, 1869, 2793, 2858 and 2878, flexible or fibre inner packagings shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.**L4** For UN Nos.1932, 2008, 2009, 2545, 2546, 2881 and 3189 flexible or fibre inner packagings shall be hermetically sealed.

Part 4 – Packing and tank provisions

LP03	PACKING INSTRUCTION	LP03
This instruction applies to UN Nos. 3537, 3538, 3540, 3541, 3546, 3547 and 3548.		
(1) The following large packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: Rigid large packagings conforming to the packing group II performance level, made of: - steel (50A); - aluminium (50B); - metal other than steel or aluminium (50N); - rigid plastics (50H); - natural wood (50C); - plywood (50D); - reconstituted wood (50F); - rigid fibreboard (50G). (2) Additionally, the following conditions shall be met: (a) Receptacles within articles containing liquids or solids shall be constructed of suitable materials and secured in the article in such a way that, under normal conditions of transport, they cannot break, be punctured or leak their contents into the article itself or the outer packaging; (b) Receptacles containing liquids with closures shall be packed with their closures correctly oriented. The receptacles shall in addition conform to the internal pressure test provisions of 6.1.5.5; (c) Receptacles that are liable to break or be punctured easily, such as those made of glass, porcelain or stoneware or of certain plastics materials shall be properly secured. Any leakage of the contents shall not substantially impair the protective properties of the article or of the outer packaging; (d) Receptacles within articles containing gases shall meet the requirements of section 4.1.6 and chapter 6.2 as appropriate or be capable of providing an equivalent level of protection as packing instructions P200 or P208; and (e) Where there is no receptacle within the article, the article shall fully enclose the dangerous substances and prevent their release under normal conditions of transport. (3) Articles shall be packed to prevent movement and inadvertent operation during normal conditions of transport. ■ (4) Articles containing pre-production prototype lithium cells or batteries when these prototypes are transported for testing or production runs of not more than 100 lithium cells or batteries that are of a type that have not met the testing requirements of the Manual of Tests and Criteria, part III, subsection 38.3 shall in addition meet the following: (a) Packagings shall conform to the requirements in paragraph (1) of this packing instruction. (b) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the article within the package that may lead to damage and a dangerous condition during transport. When cushioning material is used to meet this requirement it shall be non-combustible and electrically non-conductive. (c) Non-combustibility of the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured.		
LP99	PACKING INSTRUCTION	LP99
Only large packagings which are approved for these goods by the competent authority may be used (see 4.1.3.7). A copy of the competent authority approval shall accompany each consignment or the transport document shall include an indication that the packaging was approved by the competent authority.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

LP101 PACKING INSTRUCTION LP101		
The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 and special provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Large packagings
<i>Not necessary</i>	<i>Not necessary</i>	Steel (50A) Aluminium (50B) Metal other than steel or aluminium (50N) Rigid plastics (50H) Natural wood (50C) Plywood (50D) Reconstituted wood (50F) Rigid fibreboard (50G)
Special packing provision: L1 For UN Nos. 0006, 0009, 0010, 0015, 0016, 0018, 0019, 0034, 0035, 0038, 0039, 0048, 0056, 0137, 0138, 0168, 0169, 0171, 0181, 0182, 0183, 0186, 0221, 0243, 0244, 0245, 0246, 0254, 0280, 0281, 0286, 0287, 0297, 0299, 0300, 0301, 0303, 0321, 0328, 0329, 0344, 0345, 0346, 0347, 0362, 0363, 0370, 0412, 0424, 0425, 0434, 0435, 0436, 0437, 0438, 0451, 0488, 0502 and 0510: Large and robust explosives articles, normally intended for military use, without their means of initiation or with their means of initiation containing at least two effective protective features, may be transported unpackaged. When such articles have propelling charges or are self-propelled, their ignition systems shall be protected against stimuli encountered during normal conditions of transport. A negative result in Test Series 4 on an unpackaged article indicates that the article can be considered for transport unpackaged. Such unpackaged articles may be fixed to cradles or contained in crates or other suitable handling devices.		

LP102 PACKING INSTRUCTION LP102		
The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 and special provisions of 4.1.5 are met.		
Inner packagings	Intermediate packagings	Outer packagings
Bags water-resistant Receptacles fibreboard metal plastics wood Sheets fibreboard, corrugated Tubes fibreboard	<i>Not necessary</i>	Steel (50A) Aluminium (50B) Metal other than steel or aluminium (50N) Rigid plastics (50H) Natural wood (50C) Plywood (50D) Reconstituted wood (50F) Rigid fibreboard (50G)

LP200 PACKING INSTRUCTION LP200		
This instruction applies to UN 1950 and UN 2037.		
The following large packagings are authorized for aerosols and gas cartridges, provided that the general provisions of 4.1.1 and 4.1.3 are met: Rigid large packagings conforming to the packing group II performance level, made of: steel (50A); aluminium (50B); metal other than steel or aluminium (50N); rigid plastics (50H); natural wood (50C); plywood (50D); reconstituted wood (50F); rigid fibreboard (50G).		
Special packing provision: L2 The large packagings shall be designed and constructed to prevent dangerous movement and inadvertent discharge during normal conditions of transport. For waste aerosols transported in accordance with special provision 327, the large packagings shall have a means of retaining any free liquid that might escape during transport, e.g. absorbent material. For waste aerosols and waste gas cartridges carried in accordance with special provision 327, the large packagings shall be adequately ventilated to prevent the creation of dangerous atmospheres and the build-up of pressure.		

Part 4 – Packing and tank provisions

LP621 PACKING INSTRUCTION LP621		
This instruction applies to UN 3291.		
The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met: (1) For clinical waste placed in inner packagings: Rigid, leakproof large packagings conforming to the provisions of chapter 6.6 for solids, at the packing group II performance level, provided there is sufficient absorbent material to absorb the entire amount of liquid present and the large packaging is capable of retaining liquids. (2) For packages containing larger quantities of liquid: Large rigid packagings conforming to the provisions of chapter 6.6, at the packing group II performance level, for liquids.		
Additional provision: Large packagings intended to contain sharp objects such as broken glass and needles shall be resistant to puncture and retain liquids under the performance test conditions in chapter 6.6.		

LP622 PACKING INSTRUCTION LP622		
This instruction applies to waste of UN 3549 transported for disposal.		
The following large packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met:		
Inner packagings	Intermediate packagings	Outer packagings
metal plastics	metal plastics	Steel (50A) Aluminium (50B) Metal other than steel or aluminium (50N) Plywood (50D) Rigid fibreboard (50G) Rigid plastics (50H)
The outer packaging shall conform to the packing group I performance level for solids.		
Additional requirement: 1. Fragile articles shall be contained in either a rigid inner packaging or a rigid intermediate packaging. 2. Inner packagings containing sharp objects such as broken glass and needles shall be rigid and resistant to puncture. 3. The inner packaging, the intermediate packaging and the outer packaging shall be capable of retaining liquids. Outer packagings that are not capable of retaining liquids by design shall be fitted with a liner or suitable measure of retaining liquids. 4. The inner packaging and/or the intermediate packaging may be flexible. When flexible packagings are used, they shall be capable of passing the impact resistance test of at least 165 g according to ISO 7765-1:1988 "Plastics film and sheeting – Determination of impact resistance by the free-falling dart method – Part 1: Staircase methods" and the tear resistance test of at least 480 g in both parallel and perpendicular planes with respect to the length of the bag in accordance with ISO 6383-2:1983 <i>Plastics – Film and sheeting – Determination of tear resistance – Part 2: Elmendorf method</i>. The maximum net mass of each flexible inner packaging shall be 30 kg. 5. Each flexible intermediate packaging shall contain only one inner packaging. 6. Inner packagings containing a small amount of free liquid may be included in intermediate packaging provided that there is sufficient absorbent or solidifying material in the inner or intermediate packaging to absorb or solidify all the liquid content present. Suitable absorbent material which withstands the temperatures and vibrations liable to occur under normal conditions of transport shall be used. 7. Intermediate packagings shall be secured in outer packagings with suitable cushioning and/or absorbent material.		

Chapter 4.1 – Use of packagings, including IBCs and large packagings

LP902	PACKING INSTRUCTION	LP902
This instruction applies to UN 3268.		
(1) Packaged articles: The following large packagings are authorized, provided the general provisions of 4.1.1 and 4.1.3 are met: Rigid large packagings conforming to the packing group III performance level, made of: - steel (50A); - aluminium (50B); - metal other than steel or aluminium (50N); - rigid plastics (50H); - natural wood (50C); - plywood (50D); - reconstituted wood (50F); - rigid fibreboard (50G). The packagings shall be designed and constructed to prevent movement of the articles and inadvertent operation during normal conditions of transport. (2) Unpackaged articles: The articles may also be transported unpackaged in dedicated handling devices, vehicles, containers or wagons when moved to, from, or between where they are manufactured and an assembly plant including intermediate handling locations.		
Additional provision: Any pressure receptacle shall be in accordance with the requirements of the competent authority for the substance(s) contained in the pressure receptacle(s).		

LP903	PACKING INSTRUCTION	LP903
This instruction applies to large cells with a gross mass of more than 500 g, large batteries with a gross mass of more than 12 kg, and equipment containing large cells or large batteries of UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552.		
The following large packagings are authorized for cells, batteries and equipment containing cells or batteries, provided that the general provisions of 4.1.1 and 4.1.3 are met: Rigid large packagings conforming to the packing group II performance level, made of: - steel (50A); - aluminium (50B); - metal other than steel or aluminium (50N); - rigid plastics (50H); - natural wood (50C); - plywood (50D); - reconstituted wood (50F); - rigid fibreboard (50G). Cells, batteries or equipment shall be placed in inner packagings or separated by other suitable means, such as placement in trays or by dividers, to ensure protection against damage that may be caused under normal conditions of transport by: (a) its movement or placement within the large packaging; (b) contact with other cells, batteries or equipment within the large packaging; and (c) any loads arising from the superimposed weight of cells, batteries, equipment and packaging components above the cell, battery or equipment within the large packaging. When multiple cells, batteries or items of equipment, are packed in the large packaging, bags (e.g. plastics) alone shall not be used to satisfy these requirements.		
Additional requirement: Batteries shall be protected against short circuit.		

Part 4 – Packing and tank provisions

LP904	PACKING INSTRUCTION	LP904
△	This instruction applies to single damaged or defective batteries and to single items of equipment containing damaged or defective cells or batteries of UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552.	
	The following large packagings are authorized for a single damaged or defective battery and for a single item of equipment containing damaged or defective cells or batteries, provided the general provisions of 4.1.1 and 4.1.3 are met. For batteries and equipment containing cells and batteries: Rigid large packagings conforming to the packing group II performance level, made of: - steel (50A) - aluminium (50B) - metal other than steel or aluminium (50N) - rigid plastics (50H) - plywood (50D) ■ Large packagings shall also meet the following requirements: (a) The damaged or defective battery or equipment containing such cells or batteries shall be individually packed in an inner packaging and placed inside of an outer packaging. The inner packaging or outer packaging shall be leakproof to prevent the potential release of electrolyte. (b) The inner packaging shall be surrounded by sufficient non-combustible and electrically non-conductive thermal insulation material to protect against a dangerous evolution of heat. (c) Sealed packagings shall be fitted with a venting device when appropriate. (d) Appropriate measures shall be taken to minimize the effects of vibrations and shocks, prevent movement of the battery or the equipment within the package that may lead to further damage and a dangerous condition during transport. Cushioning material that is non-combustible and electrically non-conductive may also be used to meet this requirement. △ (e) The non-combustibility of the thermal insulation material and the cushioning material shall be assessed according to a standard recognized in the country where the packaging is designed or manufactured. For leaking batteries and cells, sufficient inert absorbent material shall be added to the inner or outer packaging to absorb any release of electrolyte.	
	Additional requirement: Batteries and cells shall be protected against short circuit.	

Chapter 4.1 – Use of packagings, including IBCs and large packagings

LP905	PACKING INSTRUCTION	LP905
△	This instruction applies to UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552 production runs consisting of not more than 100 cells or batteries and to pre-production prototypes of cells or batteries when these prototypes are transported for testing. The following large packagings are authorized for a single battery and for a single item of equipment containing cells or batteries, provided that the general provisions of 4.1.1 and 4.1.3 are met: (1) For a single battery: Rigid large packagings conforming to the packing group II performance level, made of: steel (50A); aluminium (50B); metal other than steel or aluminium (50N); rigid plastics (50H); natural wood (50C); plywood (50D); reconstituted wood (50F); rigid fibreboard (50G). Large packagings shall also meet the following requirements: (a) A battery of different size, shape or mass may be packed in an outer packaging of a tested design type listed above provided the total gross mass of the package does not exceed the gross mass for which the design type has been tested; (b) The battery shall be packed in an inner packaging and placed inside the outer packaging; (c) The inner packaging shall be completely surrounded by sufficient non-combustible and electrically non-conductive thermal insulation material to protect against a dangerous evolution of heat; (d) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the battery within the package that may lead to damage and a dangerous condition during transport. When cushioning material is used to meet this requirement it shall be non-combustible and electrically non-conductive; and (e) The non-combustibility of the thermal insulation material and the cushioning material shall be assessed according to a standard recognized in the country where the large packaging is designed or manufactured. (2) For a single item of equipment containing cells or batteries: Rigid large packagings conforming to the packing group II performance level, made of: steel (50A); aluminium (50B); metal other than steel or aluminium (50N); rigid plastics (50H); natural wood (50C); plywood (50D); reconstituted wood (50F); rigid fibreboard (50G). Large packagings shall also meet the following requirements: (a) A single item of equipment of different size, shape or mass may be packed in an outer packaging of a tested design type listed above provided the total gross mass of the package does not exceed the gross mass for which the design type has been tested; (b) The equipment shall be constructed or packed in such a manner as to prevent accidental operation during transport; (c) Appropriate measures shall be taken to minimize the effects of vibration and shocks and prevent movement of the equipment within the package that may lead to damage and a dangerous condition during transport. When cushioning material is used to meet this requirement, it shall be non-combustible and electrically non-conductive; and (d) The non-combustibility of the cushioning material shall be assessed according to a standard recognized in the country where the large packaging is designed or manufactured. Additional requirement: Cells and batteries shall be protected against short circuit.	

Part 4 – Packing and tank provisions

LP906	PACKING INSTRUCTION	LP906
This instruction applies to damaged or defective batteries of UN Nos. 3090, 3091, 3480, 3481, 3551 and 3552 liable to rapidly disassemble, dangerously react, produce a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours under normal conditions of transport.		
The following large packagings are authorized, provided that the general provisions of 4.1.1 and 4.1.3 are met: For batteries and items of equipment containing batteries: Rigid large packagings conforming to the packing group I performance level, made of: - steel (50A); - aluminium (50B); - metal other than steel or aluminium (50N); - rigid plastics (50H); - plywood (50D); - rigid fibreboard (50G).		
(1) The large packaging shall be capable of meeting the following additional performance requirements in case of rapid disassembly, dangerous reaction, production of a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours of the battery: (a) The outside surface temperature of the completed package shall not have a temperature of more than 100°C. A momentary spike in temperature up to 200°C is acceptable; (b) No flame shall occur outside the package; (c) No projectiles shall exit the package; (d) The structural integrity of the package shall be maintained; and (e) The large packagings shall have a gas management system (e.g. filter system, air circulation, containment for gas, gas tight packaging etc.), as appropriate.		
(2) The additional large packaging performance requirements shall be verified by a test as specified by the competent authority.* A verification report shall be made available on request. As a minimum requirement, the name of the batteries, their type as defined in section 38.3.2.3 of the <i>Manual of Tests and Criteria</i>, the maximum number of batteries, the total mass of batteries, the total energy content of the batteries, the large packaging identification and the test data according to the verification method as specified by the competent authority shall be listed in the verification report. A set of specific instructions describing the way to use the package shall also be part of the verification report.		
(3) When dry ice or liquid nitrogen is used as a coolant, the requirements of section 5.5.3 shall apply. The inner packaging and outer packaging shall maintain their integrity at the temperature of the refrigerant used as well as the temperatures and the pressures which could result if refrigeration were lost.		
(4) The specific instructions for use of the package shall be made available by the packaging manufacturers and subsequent distributors to the consignor. They shall include at least the identification of the batteries and items of equipment that may be contained inside the packaging, the maximum number of batteries contained in the package and the maximum total of the batteries' energy content, as well as the configuration inside the package, including the separations and protections used during the performance verification test.		
Additional requirement: Batteries shall be protected against short circuit.		
* The following criteria, as relevant, may be considered to assess the performance of the large packaging: (a) The assessment shall be done under a quality management system (as described, e.g. in section 2.9.4.5) allowing for the traceability of tests results, reference data and characterization models used; (b) The list of hazards expected in case of thermal runaway for the battery type, in the condition it is transported (e.g. usage of an inner packaging, state of charge (SOC), use of sufficient non-combustible, electrically non-conductive and absorbent cushioning material etc.), shall be clearly identified and quantified; the reference list of possible hazards for batteries (e.g. rapidly disassemble, dangerously react, produce a flame or a dangerous evolution of heat or a dangerous emission of toxic, corrosive or flammable gases or vapours) can be used for this purpose. The quantification of these hazards shall rely on available scientific literature; (c) The mitigating effects of the large packaging shall be identified and characterized, based on the nature of the protections provided and the construction material properties. A list of technical characteristics and drawings shall be used to support this assessment (density [kg·m³], specific heat capacity [J·kg⁻¹·K⁻¹], heating value [kJ·kg⁻¹], thermal conductivity [W·m⁻¹·K⁻¹], melting temperature and flammability temperature [K], heat transfer coefficient of the outer packaging [W·m⁻²·K⁻¹], ...); (d) The test and any supporting calculations shall assess the result of a thermal run-away of the battery inside the large packaging in the normal conditions of transport; (e) In case the SOC of the battery is not known, the assessment used shall be done with the highest possible SOC corresponding to the battery use conditions; (f) The surrounding conditions in which the large packaging may be used and transported shall be described (including for possible consequences of gas or smoke emissions on the environment, such as ventilation or other methods) according to the gas management system of the large packaging; (g) The tests or the model calculation shall consider the worst case scenario for the thermal runaway triggering and propagation inside the battery: this scenario includes the worst possible failure in the normal transport condition, the maximum heat and flame emissions for the possible propagation of the reaction; (h) These scenarios shall be assessed over a period long enough to allow all the possible consequences to occur (e.g. 24 hours). (i) In the case of multiple batteries and multiple items of equipment containing batteries, additional requirements such as the maximum number of batteries and items of equipment, the total maximum energy content of the batteries, and the configuration inside the package, including separations and protections of the parts, shall be considered.		

4.1.5 Special packing provisions for goods of class 1

- 4.1.5.1 The general provisions of 4.1.1 shall be met.
- 4.1.5.2 All packagings for class 1 goods shall be so designed and constructed that:
- .1 they will protect the explosives, prevent them escaping and cause no increase in the risk of unintended ignition or initiation when subjected to normal conditions of transport, including foreseeable changes in temperature, humidity and pressure;
 - .2 the complete package can be handled safely in normal conditions of transport; and
 - .3 the packages will withstand any loading imposed on them by foreseeable stacking to which they will be subject during transport so that they do not add to the risk presented by the explosives, the containment function of the packagings is not harmed, and they are not distorted in a way or to an extent which will reduce their strength or cause instability of a stack.
- 4.1.5.3 All explosive substances and articles, as prepared for transport, shall have been classified in accordance with the procedures detailed in 2.1.3.
- 4.1.5.4 Class 1 goods shall be packed in accordance with the appropriate packing instruction shown in columns 8 and 9 of the Dangerous Goods List, as detailed in 4.1.4.
- 4.1.5.5 Unless otherwise specified in this Code, packagings, including IBCs and large packagings, shall conform to the requirements of chapters 6.1, 6.5 or 6.6, as appropriate, and shall meet their test provisions for packing group II.
- 4.1.5.6 The closure device of packagings containing liquid explosives shall ensure a double protection against leakage.
- 4.1.5.7 The closure device of metal drums shall include a suitable gasket; if a closure device includes a screw-thread, the ingress of explosive substances into the screw-thread shall be prevented.
- 4.1.5.8 Packagings for water-soluble substances shall be water-resistant. Packagings for desensitized or phlegmatized substances shall be closed to prevent changes in concentration during transport.
- 4.1.5.9 When the packaging includes a double envelope filled with water which may freeze during transport, a sufficient quantity of an anti-freeze agent shall be added to the water to prevent freezing. Anti-freeze that could create a fire hazard because of its inherent flammability shall not be used.
- 4.1.5.10 Nails, staples and other closure devices made of metal without protective covering shall not penetrate to the inside of the outer packaging unless the inner packaging adequately protects the explosives against contact with the metal.
- 4.1.5.11 Inner packagings, fittings and cushioning materials and the placing of explosive substances or articles in packages shall be accomplished in a manner which prevents the explosive substances or articles from becoming loose in the outer packaging under normal conditions of transport. Metallic components of articles shall be prevented from making contact with metal packagings. Articles containing explosive substances not enclosed in an outer casing shall be separated from each other in order to prevent friction and impact. Padding, trays, partitioning in the inner or outer packaging, mouldings or receptacles may be used for this purpose.
- 4.1.5.12 Packagings shall be made of materials compatible with, and impermeable to, the explosives contained in the package, so that neither interaction between the explosives and the packaging materials nor leakage causes the explosive to become unsafe to transport, or the hazard division or compatibility group to change.
- 4.1.5.13 The ingress of explosive substances into the recesses of seamed metal packagings shall be prevented.
- 4.1.5.14 Plastics packagings shall not be liable to generate or accumulate sufficient static electricity so that a discharge could cause the packaged explosive substances or articles to initiate, ignite or function.
- 4.1.5.15 Large and robust explosives articles, normally intended for military use, without their means of initiation or with their means of initiation containing at least two effective protective features may be transported unpackaged. When such articles have propelling charges or are self-propelled, their ignition systems shall be protected against stimuli encountered during normal conditions of transport. A negative result in Test Series 4 on an unpackaged article indicates that the article can be considered for transport unpackaged. Such unpackaged articles may be fixed to cradles or contained in crates or other suitable handling, storage or launching devices in such a way that they will not become loose during normal conditions of transport. Where such large explosive articles are, as part of their operational safety and suitability tests, subjected to test regimes that meet the provisions of this Code and such tests have been successfully undertaken, the competent authority may approve such articles to be transported under this Code.
- 4.1.5.16 Explosive substances shall not be packed in inner or outer packagings where the differences in internal and external pressures, due to thermal or other effects, could cause an explosion or rupture of the package.

Part 4 – Packing and tank provisions

- 4.1.5.17 Whenever loose explosive substances or the explosive substance of an uncased or partly cased article may come into contact with the inner surface of metal packagings (1A1, 1A2, 1B1, 1B2, 4A, 4B and metal receptacles), the metal packaging shall be provided with an inner liner or coating (see 4.1.1.2).
- 4.1.5.18 Packing instruction P101 may be used for any explosive provided the package has been approved by a competent authority regardless of whether the packaging complies with the packing instruction assignment in the Dangerous Goods List.
- 4.1.5.19 Government-owned military dangerous goods, packaged prior to 1 January 1990 in accordance with the provisions of the IMDG Code in effect at that time, may be transported provided the packagings maintain their integrity and the goods are declared as government-owned goods packaged prior to 1 January 1990.

4.1.6 Special packing provisions for goods of class 2**4.1.6.1 General provisions**

- 4.1.6.1.1 This section provides general requirements applicable to the use of pressure receptacles for the transport of class 2 gases and other dangerous goods in pressure receptacles (e.g. UN 1051 Hydrogen cyanide, stabilized). Pressure receptacles shall be constructed and closed so as to prevent any loss of contents which might be caused under normal conditions of transport, including by vibration, or by changes in temperature, humidity or pressure (resulting from change in altitude, for example).
- △ 4.1.6.1.2 Parts of pressure receptacles which are in direct contact with dangerous goods shall not be affected or weakened by those dangerous goods and shall not cause a dangerous effect (e.g. catalysing a reaction or reacting with the dangerous goods). The provisions of ISO 11114-1:2020 and ISO 11114-2:2021 shall be met as applicable.
- 4.1.6.1.3 Pressure receptacles, including their closures, shall be selected to contain a gas or a mixture of gases according to the requirements of 6.2.1.2 and the requirements of the specific packing instructions of 4.1.4.1. This section also applies to pressure receptacles which are elements of MEGCs.
- 4.1.6.1.4 Refillable pressure receptacles shall not be filled with a gas or gas mixture different from that previously contained unless the necessary operations for change of gas service have been performed. The change of service for compressed and liquefied gases shall be in accordance with ISO 11621:1997, as applicable. In addition, a pressure receptacle that previously contained a class 8 corrosive substance or a substance of another class with a corrosive subsidiary hazard shall not be authorized for the transport of a class 2 substance unless the necessary inspection and testing as specified in 6.2.1.6 have been performed.
- 4.1.6.1.5 Prior to filling, the filler shall perform an inspection of the pressure receptacle and ensure that the pressure receptacle is authorized for the gas and, in case of a chemical under pressure, for the propellant to be transported and that the provisions of this Code have been met. Shut-off valves shall be closed after filling and remain closed during transport. The consignor shall verify that the closures and equipment are not leaking.
- 4.1.6.1.6 Pressure receptacles shall be filled according to the working pressures, filling ratios and provisions specified in the appropriate packing instruction for the specific substance being filled and taking into account the lowest design pressure of any component. Service equipment having a design pressure lower than other components shall nevertheless comply with 6.2.1.3.1. Reactive gases and gas mixtures shall be filled to a pressure such that if complete decomposition of the gas occurs, the working pressure of the pressure receptacle shall not be exceeded.
- 4.1.6.1.7 Pressure receptacles, including their closures, shall conform to the design, construction, inspection and testing requirements detailed in chapter 6.2. When outer packagings are prescribed, the pressure receptacles shall be firmly secured therein. Unless otherwise specified in the detailed packing instructions, one or more inner packagings may be enclosed in an outer packaging.
- 4.1.6.1.8 Valves shall be designed and constructed in such a way that they are inherently able to withstand damage without release of the contents or shall be protected from damage which could cause inadvertent release of the contents of the pressure receptacle, by one of the following methods:
- .1 Valves are placed inside the neck of the pressure receptacle and protected by a threaded plug or cap;
 - △ .2 Valves are protected by caps or guards. Caps shall possess vent-holes of sufficient cross-sectional area to evacuate the gas if leakage occurs at the valves;
 - .3 Valves are protected by shrouds or permanent protective attachments;
 - .4 Pressure receptacles are transported in frames (e.g. bundles); or
 - .5 Pressure receptacles are transported in an outer packaging. The packaging as prepared for transport shall be capable of meeting the drop test specified in 6.1.5.3 at the packing group I performance level.
- △ For pressure receptacles with valves as described in .2, the requirements of ISO 11117:1998, ISO 11117:2008 + Cor 1:2009 or ISO 11117:2019 shall be met. Requirements for shrouds and permanent protective attachments used as valve protection under .3, are given in the relevant pressure receptacle shell design standards, see 6.2.2.1. Valves with inherent protection used for refillable pressure receptacles shall meet the requirements of clause 4.6.2 of ISO 10297:2006 or clause 5.5.2 of ISO 10297:2014 or clause 5.5.2 of ISO

10297:2014 + Amd 1:2017, or in case of self-closing valves, of clause 5.4.2 of ISO 17879:2017. For valves with inherent protection used for non-refillable cylinders, the requirements of clause 9.2.5 of ISO 11118:2015 or of clause 9.2.5 of ISO 11118:2015 + Amd 1:2019 shall be met.

For metal hydride storage systems, the valve protection requirements specified in ISO 16111:2008 or ISO 16111:2018 shall be met.

4.1.6.1.9 Non-refillable pressure receptacles shall:

- .1 be transported in an outer packaging, such as a box, or crate, or in shrink-wrapped trays or stretch-wrapped trays;
- .2 be of a water capacity less than or equal to 1.25 L when filled with flammable or toxic gas;
- .3 not be used for toxic gases with an LC_{50} less than or equal to 200 mL/m³; and
- .4 not be repaired after being put into service.

4.1.6.1.10 Refillable pressure receptacles, other than closed cryogenic receptacles, shall be periodically inspected in accordance with 6.2.1.6 and packing instruction P200, P205, P206 or P208, as applicable. Pressure relief valves for closed cryogenic receptacles shall be subject to periodic inspections and tests according to the provisions of 6.2.1.6.3 and packing instruction P203. Pressure receptacles shall not be filled after they become due for periodic inspection but may be transported after the expiry of the time limit.

4.1.6.1.11 Repairs shall be consistent with the manufacture and testing requirements of the applicable design and construction standards and are only permitted as indicated in the relevant periodic inspection standards specified in 6.2.2.4. Pressure receptacles, other than the jacket of closed cryogenic receptacles, shall not be subjected to repairs of any of the following:

- .1 weld cracks or other weld defects;
- .2 cracks in walls;
- .3 leaks or defects in the material of the wall, head or bottom.

4.1.6.1.12 Pressure receptacles shall not be offered for filling:

- .1 when damaged to such an extent that the integrity of the pressure receptacle or its service equipment may be affected;
- .2 unless the pressure receptacle and its service equipment has been examined and found to be in good working order; or
- .3 unless the required certification, retest, and filling marks are legible.

4.1.6.1.13 Filled pressure receptacles shall not be offered for transport:

- .1 when leaking;
- .2 when damaged to such an extent that the integrity of the pressure receptacle or its service equipment may be affected;
- .3 unless the pressure receptacle and its service equipment has been examined and found to be in good working order; or
- .4 unless the required certification, retest, and filling marks are legible.

4.1.6.1.14 Where in packing instruction P200 cylinders and other pressure receptacles for gases conforming to the requirements of this subsection and chapter 6.2 are authorized, use is also authorized of cylinders and pressure receptacles which conform to the requirements of the competent authority of the country in which the cylinder or pressure receptacle is filled. Valves shall be suitably protected. Pressure receptacles with capacities of 1 L or less shall be packed in outer packagings constructed of suitable material of adequate strength and design in relation to the capacity of the packaging and its intended use and secured or cushioned so as to prevent significant movement within the outer packaging during normal conditions of transport.

4.1.7 Special packing provisions for organic peroxides (class 5.2) and self-reactive substances of class 4.1

4.1.7.0 General

4.1.7.0.1 For organic peroxides, all receptacles shall be “effectively closed”. Where significant internal pressure may develop in a package by the evolution of gas, a vent may be fitted, provided the gas emitted will not cause danger, otherwise the degree of filling shall be limited. Any venting device shall be so constructed that liquid will not escape when the package is in an upright position and it shall be able to prevent ingress of impurities. The outer packaging, if any, shall be so designed as not to interfere with the operation of the venting device.

Part 4 – Packing and tank provisions**4.1.7.1 Use of packagings (except IBCs)**

- 4.1.7.1.1** Packagings for organic peroxides and self-reactive substances shall conform to the provisions of chapter 6.1 and shall meet its test provisions for packing group II.
- 4.1.7.1.2** The packing methods for organic peroxides and self-reactive substances are listed in packing instruction P520 and are designated OP1 to OP8. The quantities specified for each packing method are the maximum quantities authorized per package.
- 4.1.7.1.3** The packing methods appropriate for the individual currently assigned self-reactive substances and organic peroxides are listed in 2.4.2.3.2.3 and 2.5.3.2.4.
- 4.1.7.1.4** For new organic peroxides, new self-reactive substances or new formulations of currently assigned organic peroxides or self-reactive substances, the following procedure shall be used to assign the appropriate packing method:
- .1 ORGANIC PEROXIDE TYPE B or SELF-REACTIVE SUBSTANCE TYPE B:**
Packing method OP5 shall be assigned, provided that the organic peroxide (or self-reactive substance) satisfies the criteria of 2.5.3.3.2.2 (resp. 2.4.2.3.3.2.2) in a packaging authorized by the packing method. If the organic peroxide (or self-reactive substance) can only satisfy these criteria in a smaller packaging than those authorized by packing method OP5 (viz. one of the packagings listed for OP1 to OP4), then the corresponding packing method with the lower OP number is assigned;
 - .2 ORGANIC PEROXIDE TYPE C or SELF-REACTIVE SUBSTANCE TYPE C:**
Packing method OP6 shall be assigned, provided that the organic peroxide (or self-reactive substance) satisfies the criteria of 2.5.3.3.2.3 (resp. 2.4.2.3.3.2.3) in packaging authorized by the packing method. If the organic peroxide (or self-reactive substance) can only satisfy these criteria in a smaller packaging than those authorized by packing method OP6, then the corresponding packing method with the lower OP number is assigned;
 - .3 ORGANIC PEROXIDE TYPE D or SELF-REACTIVE SUBSTANCE TYPE D:**
Packing method OP7 shall be assigned to this type of organic peroxide or self-reactive substance;
 - .4 ORGANIC PEROXIDE TYPE E or SELF-REACTIVE SUBSTANCE TYPE E:**
Packing method OP8 shall be assigned to this type of organic peroxide or self-reactive substance;
 - .5 ORGANIC PEROXIDE TYPE F or SELF-REACTIVE SUBSTANCE TYPE F:**
Packing method OP8 shall be assigned to this type of organic peroxide or self-reactive substance.

4.1.7.2 Use of intermediate bulk containers

- 4.1.7.2.1** The currently assigned organic peroxides specifically listed in packing instruction IBC520 may be transported in IBCs in accordance with this packing instruction. IBCs shall conform to the requirements of chapter 6.5 and shall meet its test provisions for packing group II.
- 4.1.7.2.2** Other organic peroxides and self-reactive substances of type F may be transported in IBCs under conditions established by the competent authority of the country of origin when, on the basis of the appropriate tests, that competent authority is satisfied that such transport may be safely conducted. The tests undertaken shall include those necessary:
- .1** to prove that the organic peroxide (or self-reactive substance) complies with the principles for classification;
 - .2** to prove the compatibility of all materials normally in contact with the substance during the transport;
 - .3** to determine, when applicable, the control and emergency temperatures associated with the transport of the product in the IBC concerned as derived from the SADT;
 - .4** to design, when applicable, pressure and emergency relief devices; and
 - .5** to determine if any special provisions are necessary for safe transport of the substance.
- 4.1.7.2.3** For self-reactive substances, temperature control is required according to 2.4.2.3.4. For organic peroxides, temperature control is required according to 2.5.3.4.1. Temperature control provisions are given in 7.3.7.
- 4.1.7.2.4** Emergencies to be taken into account are self-accelerating decomposition and fire engulfment. To prevent explosive rupture of metal or composite IBCs with a complete metal casing, the emergency relief devices shall be designed to vent all the decomposition products and vapours evolved during self-accelerating decomposition or during a period of not less than one hour of complete fire engulfment calculated by the equations given in 4.2.1.13.8.

4.1.8 Special packing provisions for infectious substances of category A (class 6.2, UN 2814 and UN 2900)

- 4.1.8.1** Consignors of infectious substances shall ensure that packages are prepared in such a manner that they arrive at their destination in good condition and present no hazard to persons or animals during transport.

- 4.1.8.2** The definitions in 1.2.1 and the general packing provisions of 4.1.1.1 to 4.1.1.14, except 4.1.1.10 to 4.1.1.12, apply to infectious substances packages. However, liquids shall only be filled into packagings which have an appropriate resistance to the internal pressure that may develop under normal conditions of transport.
- 4.1.8.3** An itemized list of contents shall be enclosed between the secondary packaging and the outer packaging. When the infectious substances to be transported are unknown, but suspected of meeting the criteria for inclusion in category A, the words “suspected category A infectious substance” shall be shown, in parentheses, following the proper shipping name on the document inside the outer packaging.
- 4.1.8.4** Before an empty packaging is returned to the consignor, or sent elsewhere, it shall be disinfected or sterilized to nullify any hazard and any label or mark indicating that it had contained an infectious substance shall be removed or obliterated.
- 4.1.8.5** Provided an equivalent level of performance is maintained, the following variations in the primary receptacles placed within an intermediate packaging are allowed without further testing of the completed package:
- .1** Primary receptacles of equivalent or smaller size as compared to the tested primary receptacles may be used provided:
 - (a) the primary receptacles are of similar design to the tested primary receptacle (such as shape: round, rectangular, etc.);
 - (b) the material of construction of the primary receptacle (glass, plastics, metal, etc.) offers resistance to impact and stacking forces equal to or greater than that of the originally tested primary receptacle;
 - (c) the primary receptacles have the same or smaller openings and the closure is of similar design (such as screw cap, friction lid, etc.);
 - (d) sufficient additional cushioning material is used to take up void spaces and to prevent significant movement of the primary receptacles; and
 - (e) primary receptacles are oriented within the intermediate packaging in the same manner as in the tested package.
 - .2** A lesser number of the tested primary receptacles, or of the alternative types of primary receptacles identified in .1 above, may be used provided sufficient cushioning is added to fill the void space(s) and to prevent significant movement of the primary receptacles.

4.1.9 Special packing provisions for radioactive material

4.1.9.1 General

- 4.1.9.1.1** Radioactive material, packagings and packages shall meet the provisions of chapter 6.4. The quantity of radioactive material in a package shall not exceed the limits specified in 2.7.2.2, 2.7.2.4.1, 2.7.2.4.4, 2.7.2.4.5, 2.7.2.4.6 and 4.1.9.3.

The types of packages for radioactive materials covered by the provisions of this Code are:

- .1** Excepted package (see 1.5.1.5);
- .2** Industrial package Type 1 (Type IP-1 package);
- .3** Industrial package Type 2 (Type IP-2 package);
- .4** Industrial package Type 3 (Type IP-3 package);
- .5** Type A package;
- .6** Type B(U) package;
- .7** Type B(M) package;
- .8** Type C package.

Packages containing fissile material or uranium hexafluoride are subject to additional requirements.

- 4.1.9.1.2** The non-fixed contamination on the external surfaces of any package shall be kept as low as practicable and, under routine conditions of transport, shall not exceed the following limits:
- (a) 4 Bq/cm² for beta and gamma emitters and low-toxicity alpha emitters, and
 - (b) 0.4 Bq/cm² for all other alpha emitters.

These limits are applicable when averaged over any area of 300 cm² of any part of the surface.

- 4.1.9.1.3** A package shall not contain any items other than those that are necessary for the use of the radioactive material. The interaction between these items and the package under the conditions of transport applicable to the design shall not reduce the safety of the package.

Part 4 – Packing and tank provisions

- 4.1.9.1.4** Except as provided in 7.1.4.5.11, the level of non-fixed contamination on the external and internal surfaces of overpacks, cargo transport units and conveyances shall not exceed the limits specified in 4.1.9.1.2. This requirement does not apply to the internal surfaces of freight containers being used as packagings, either loaded or empty.
- 4.1.9.1.5** For radioactive material having other dangerous properties the package design shall take into account those properties. Radioactive material with a subsidiary hazard, packaged in packages that do not require competent authority approval, shall be transported in packagings, IBCs, tanks or bulk containers fully complying with the provisions of the relevant chapters of part 6 as appropriate, as well as applicable provisions of chapters 4.1, 4.2 or 4.3 for that subsidiary hazard.
- 4.1.9.1.6** Before a packaging is first used to transport radioactive material, it shall be confirmed that it has been manufactured in conformity with the design specifications to ensure compliance with the relevant provisions of this Code and any applicable certificate of approval. The following requirements shall also be fulfilled, if applicable:
- .1 if the design pressure of the containment system exceeds 35 kPa (gauge), it shall be ensured that the containment system of each packaging conforms to the approved design requirements relating to the capability of that system to maintain its integrity under that pressure;
 - .2 for each packaging intended for use as a Type B(U), Type B(M) or Type C package and for each packaging intended to contain fissile material, it shall be ensured that the effectiveness of its shielding and containment and, where necessary, the heat transfer characteristics and the effectiveness of the confinement system are within the limits applicable to or specified for the approved design;
 - .3 for each packaging intended to contain fissile material, it shall be ensured that the effectiveness of the criticality safety features is within the limits applicable to or specified for the design and in particular where, in order to comply with the requirements of 6.4.11.1 neutron poisons are specifically included, checks shall be performed to confirm the presence and distribution of those neutron poisons.
- 4.1.9.1.7** Before each shipment of any package, it shall be ensured that the package contains neither:
- .1 radionuclides different from those specified for the package design; nor
 - .2 contents in a form, or physical or chemical state different from those specified for the package design.
- 4.1.9.1.8** Before each shipment of any package, it shall be ensured that all the requirements specified in the relevant provisions of this Code and in the applicable certificates of approval have been fulfilled. The following requirements shall also be fulfilled, if applicable:
- .1 it shall be ensured that lifting attachments which do not meet the requirements of 6.4.2.2 have been removed or otherwise rendered incapable of being used for lifting the package, in accordance with 6.4.2.3;
 - .2 each Type B(U), Type B(M) and Type C package shall be held until equilibrium conditions have been approached closely enough to demonstrate compliance with the requirements for temperature and pressure unless an exemption from these requirements has received unilateral approval;
 - .3 for each Type B(U), Type B(M) and Type C package, it shall be ensured by inspection and/or appropriate tests that all closures, valves and other openings of the containment system through which the radioactive contents might escape are properly closed and, where appropriate, sealed in the manner for which the demonstrations of compliance with the requirements of 6.4.8.8 and 6.4.10.3 were made;
 - .4 for packages containing fissile material the measurement specified in 6.4.11.5 (b) and the tests to demonstrate closure of each package as specified in 6.4.11.8 shall be performed.
 - .5 for packages intended to be used for shipment after storage, it shall be ensured that all packaging components and radioactive contents have been maintained during storage in a manner such that all the requirements specified in the relevant provisions of this Code and in the applicable certificates of approval have been fulfilled.
- 4.1.9.1.9** The consignor shall also have a copy of any instructions with regard to the proper closing of the package and any preparation for shipment before making any shipment under the terms of the certificates.
- 4.1.9.1.10** Except for consignments under exclusive use, the transport index of any package or overpack shall not exceed 10, nor shall the criticality safety index of any package or overpack exceed 50.
- 4.1.9.1.11** Except for packages or overpacks transported under exclusive use by rail or by road under the conditions specified in 7.1.4.5.5.1, or under exclusive use and special arrangement by ship under the conditions specified in 7.1.4.5.7, the maximum dose rate at any point on any external surface of a package or overpack shall not exceed 2 mSv/h.
- 4.1.9.1.12** The maximum dose rate at any point on any external surface of a package or overpack under exclusive use shall not exceed 10 mSv/h.
- 4.1.9.1.13** Pyrophoric radioactive material shall be packaged in Type A, Type B(U), Type B(M) or Type C packages and shall also be suitably inerted.

4.1.9.2 Provisions and controls for transport of LSA material and SCO

- 4.1.9.2.1** The quantity of LSA material or SCO in a single Type IP-1 package, Type IP-2 package, Type IP-3 package, or object or collection of objects, whichever is appropriate, shall be so restricted that the external dose rate at 3 m from the unshielded material or object or collection of objects does not exceed 10 mSv/h.
- 4.1.9.2.2** For LSA material and SCO which are or contain fissile material, which is not excepted under 2.7.2.3.5, the applicable requirements of 7.1.4.5.15 and 7.1.4.5.16 shall be met.
- 4.1.9.2.3** For LSA material and SCO which are or contain fissile material, the applicable requirements of 6.4.11.1 shall be met.
- 4.1.9.2.4** LSA material and SCO in groups LSA-I, SCO-I and SCO-III may be transported unpackaged under the following conditions:
- .1 all unpackaged material other than ores containing only naturally occurring radionuclides shall be transported in such a manner that, under routine conditions of transport, there will be no escape of the radioactive contents from the conveyance nor will there be any loss of shielding;
 - .2 each conveyance shall be under exclusive use, except when only transporting SCO-I on which the contamination on the accessible and the inaccessible surfaces is not greater than ten times the applicable level specified in 2.7.1.2;
 - .3 for SCO-I where it is suspected that non-fixed contamination exists on inaccessible surfaces in excess of the values specified in 2.7.2.3.2.1.1, measures shall be taken to ensure that the radioactive material is not released into the conveyance; and
 - .4 unpackaged fissile material shall meet the requirements of 2.7.2.3.5.5.
 - .5 for SCO-III;
 - .1 transport shall be under exclusive use by road, rail, inland waterway or sea;
 - .2 stacking shall not be permitted;
 - .3 all activities associated with the shipment, including radiation protection, emergency response and any special precautions or special administrative or operational controls that are to be employed during transport shall be described in a transport plan. The transport plan shall demonstrate that the overall level of safety in transport is at least equivalent to that which would be provided if the requirements of 6.4.7.14 (only for the test specified in 6.4.15.6, preceded by the tests specified in 6.4.15.2 and 6.4.15.3) had been met.
 - .4 the requirements of 6.4.5.1 and 6.4.5.2 for a Type IP-2 package shall be satisfied, except that the maximum damage referred to in 6.4.15.4 may be determined based on provisions in the transport plan, and the requirements of 6.4.15.5 are not applicable.
 - .5 the object and any shielding are secured to the conveyance in accordance with 6.4.2.1.
 - .6 the shipment shall be subject to multilateral approval.
- 4.1.9.2.5** LSA material and SCO, except as otherwise specified in 4.1.9.2.4, shall be packaged in accordance with table 4.1.9.2.5.

Table 4.1.9.2.5 – Industrial package provisions for LSA material and SCO

Radioactive contents	Industrial package type	
	Exclusive use	Not under exclusive use
LSA-I		
Solid ^a	Type IP-1	Type IP-1
Liquid	Type IP-1	Type IP-2
LSA-II		
Solid	Type IP-2	Type IP-2
Liquid and gas	Type IP-2	Type IP-3
LSA-III	Type IP-2	Type IP-3
SCO-I ^a	Type IP-1	Type IP-1
SCO-II	Type IP-2	Type IP-2

^a Under the conditions specified in 4.1.9.2.4, LSA-I material and SCO-I may be transported unpackaged.

4.1.9.3 Packages containing fissile material

The contents of packages containing fissile material shall be as specified for the package design either directly in the provisions of this Code or in the certificate of approval.